1

Introduction to Algebra

Carson James

Contents

Notation **ix**

Preface **1**

I Sets and Order **3**

1 Logic **5**

2 Set Theory **7**
2.1 Basic Structures . . . 7
2.2 Relations . . . 10
2.2.1 Introduction . . . 10
2.2.2 Subrelations . . . 13
2.2.3 Products of Relations . . . 14
2.3 Equivalence Relations . . . 16
2.4 Products . . . 21

3 Ordered Sets **23**
3.1 To Do . . . 23
3.1.1 To Do . . . 23
3.2 Prosets . . . 23
3.2.1 Introduction . . . 23
3.2.2 Products . . . 24
3.2.3 Quotients . . . 24
3.2.4 Functions . . . 24
3.2.5 Upper and Lower Sets . . . 24
3.2.6 Monotone Maps . . . 34
3.2.7 Galois Connections . . . 36
3.2.8 Suprema and Infima . . . 41
3.3 Posets . . . 43
3.3.1 Introduction . . . 43
3.3.2 Galois Connections . . . 44
3.3.3 Suprema and Infima . . . 46
3.3.4 Monotone Maps . . . 49
3.4 Directed Sets . . . 50

II Algebraic Structures **51**

4 Lattices **53**
4.0.1 To Do . . . 53
4.1 Introduction . . . 53

4.1.1 Skew Semilattices . . . 53
4.1.2 Skew Semilattice Homomorphisms . . . 55
4.1.3 Skew Lattices . . . 56
4.1.4 Semilattices . . . 58
4.1.5 Semilattice Homomorphisms . . . 62
4.1.6 Lattices . . . 63
4.1.7 Lattice Homomorphisms . . . 65
4.2 Basic Properties . . . 66
4.2.1 Bounded Lattices . . . 66
4.2.2 Bounded Lattice Homomorphisms . . . 67
4.2.3 Irreducibility and Primality . . . 68
4.3 Lattice Ideals and Filters . . . 70
4.3.1 Introduction . . . 70
4.3.2 Properties of Ideals and Filters . . . 75
4.4 Contructions with Lattices . . . 77
4.4.1 Sublattices . . . 77
4.4.2 Product Lattices . . . 79
4.4.3 Coproduct Lattices . . . 79
4.4.4 Quotient Lattices . . . 79
4.5 Modular and Distributive Lattices . . . 79
4.5.1 Introduction . . . 79

5 Complete Lattices **85**
5.1 Introduction . . . 85
5.2 Complete Lattice Homomorphisms . . . 86
5.3 Properties of Ideals and Filters . . . 86
5.3.1 Prime Ideals and Filters in Complete Lattices . . . 86
5.4 Constructions with Complete Lattices . . . 89
5.4.1 Complete Sublattices . . . 89
5.4.2 Products of Complete Lattices . . . 89
5.4.3 Coproducts of Complete Lattices . . . 89
5.4.4 Quotients of Complete Lattices . . . 89
5.5 Fundamental Examples . . . 90
5.5.1 Introduction . . . 90
5.5.2 Sets . . . 90
5.5.3 Equivalence Relations . . . 90
5.5.4 Ideal and Filter Lattices . . . 90
5.5.5 Topological Spaces . . . 90
5.6 Least Upper Bound Property . . . 90
5.6.1 Complete Lattice Homomorphisms . . . 90

6 Frames and Locales **91**
6.1 Introduction . . . 91
6.1.1 Frames . . . 91
6.1.2 Locales . . . 91
6.2 More Lattice Stuff to Come . . . 92

7 Some Chapter **93**
7.1 Closure Operators . . . 93

8 Universal Algebra **95**
8.1 Introduction . . . 95
8.2 Subalgebras . . . 97
8.3 Congruences . . . 99

9 Groups **101**
9.0.1 Direct Products . . . 101
9.1 Rings . . . 103
9.2 Modules . . . 104
9.2.1 Introduction . . . 104
9.3 Fields . . . 107
9.4 Vector Spaces . . . 108
9.5 Appendix . . . 108
9.5.1 Monoids . . . 108

A Summation **109**

B Asymptotic Notation **111**

C Categories **113**

D Vector Spaces **115**
D.1 Introduction . . . 115
D.2 Bases . . . 116
D.3 Multilinear Maps . . . 117
D.4 Tensor Products . . . 118

Notation

$\mathcal{M}_+(X, \mathcal{A})$	finite measures on $(X, \mathcal{A})$
v	velocity

Preface

cc-by-nc-sa

Part I

Sets and Order

Chapter 1

Logic

Chapter 2

Set Theory

2.1 Basic Structures

move to notes on intro to math and make these set theory notes assuming familiarity with producsts of two sets, similar to the set theory in the analysis notes.

In this section, we provide an exposure to a basic framework, called set theory. This framework provides a unifying language which allows us to discuss many objects of interest in mathematics by providing the ability to

1. more precisely define objects of interest,
2. more precisely formulate and prove statements about those objects
3. perform (1) and (2) in a way which is aligned with our intuition.

At an intuitive level, the main idea of a **set** is that of a collection of distinct objects, which we refer to as **elements** or **members**. When denoting a set, it is customary to list the elements of the set, separated by a comma and inside of a pair of curly brackets. Some examples of sets include the following:

- the set of letters of the English alphabet: $\{a, b, \ldots, z\}$, a set with 26 elements
- the set of inhabitants of a specific household: $\{$"Marielle", "Fred", "Dexter"$\}$, a set with 3 elements
- the set of nonnegative even integers: $\{0, 2, \ldots\}$, a set with infintely many elements

Note 2.1.0.1. Since the elements of a set must be distinct, we would consider, for instance, the sets $\{a, b, c\}$ and $\{a, a, b, c\}$ to represent the same set.

When expressing that a given object is a member of set, it is customary to use the **membership symbol**, denoted $\in$. Some examples include the following:

- $2 \in \{1, 2, 3, 4\}$
- "Earth" $\in \{$"Mercury", "Venus", "Earth", "Mars"$\}$.

Note 2.1.0.2. We note that this introduction to set theory is not formal, but intuitve. As such, we make no attempt to precisely define the notion of a set or the notion of membership to a set.

At this point in our discussion, it might seem that any advantage conferred by the language of sets in studying mathematics remains to be seen. Indeed, more time and experience may be required to fully appreciate the power of this framework, but for now we summarize why set theory is relevant in mathematics with the following mantra:

In mathematics, everything is a set.

There is a similar perspective adopted in the context of Linux operating systems where every object is a file or in the context of object oriented programming where every object is a class.

If one already has familiarity with various objects in mathematics such as various numbers, functions, and shapes, they might wonder how these objects could be considered as a set, i.e. a list of elements. For example, which elements $a_1, a_2, \ldots$, which are themselves sets (because everything is a set), would allow one to express the integer 2 as the set $\{a_1, a_2, \ldots\}$? At first glance, we might attempt to represent 2 with an arbitrary set containing two elements, say {"cow", "chicken"}. The intuition here might be that the number of elements reflects our notion of the "quantity" or "magnitude" we associate to the number 2. However, upon further reflection, such a set might not satisfy other notions we associate to the number 2.

For example we typically think of the number 2 as having a unique "ranking" or "place" in the number line. That is, the number 2 is the integer that preceeds the number 3. So we would need to not only define a set representing the number 3, but also a set representing the notion of subsequence that would align with our intuition that 3 is subsequent to 2.

In a different vein, one might also wish for our representation of the number 2 to agree with the idea that the sum of 2 and 4 is 6. The issue here is how to describe the notion of integer addition. After some thought, one might realize that the idea of adding two integers is basically that of a function, i.e. an object which eats two integer inputs and spits out an integer output.

It might seem tricky to be able to represent all the nonnegative integers, the notion of integer subsequence and the notion of integer addition by sets in a way that aligns with our intution of those objects, relations and operations. With our limited toolset, this would indeed be quite challenging, but after a few more ideas, this will become a relatively simple goal. For now, we focus on providing the necessary calculus of sets to be able to do so.

The first two relations between sets that we define pertain to when one set is a subset of another and when one set is equal to another. To motivate this, we observe that each element of $\{a, b\}$ is also an element of $\{a, b, c\}$. Our intuition dictates that the former is in some sense "less than or equal to" the latter. However, the same could not be said of the converse. Therefore, the two sets should not be considered equal.

Definition 2.1.0.3. Let A and B be sets. Then

- A is said to be a **subset of** B if for each set x, $x \in A$ implies that $x \in B$.
  (A is a subset of $B \iff \forall x(x \in A \implies x \in B)$)
- A is said to be **equal to** B if $A \subset B$ and $B \subset A$.
- A is said to be **empty** if for each set x, $x \notin A$.
  (A is empty $\iff \forall x(\neg(x \in A))$)

Note 2.1.0.4.

- When defining a term, it is conventional to use an "if", but really the relationship is an "iff".
- For convenience, we will typically write $x \notin A$ in place of $\neg(x \in A)$.

The following exercise shows us that there is at most one set that is empty.

Exercise 2.1.0.5. Let A, B be sets. If A is empty and B is empty, then $A = B$.

Proof. Suppose that A is empty and B is empty. Let x be a set.

- Since A is empty, $x \notin A$. Therefore $x \in A$ implies that $x \in B$. Since x is an arbitrary set, we have that for each set x, $x \in A$ implies that $x \in B$. Hence $A \subset B$.
- Similarly $B \subset A$

Since $A \subset B$ and $B \subset A$, we have that $A = B$. □

Even though there can be at most one set that is empty, we are not guaranteed the existence of such a set. Since its existence is quite useful and intuitively quite reasonable, we will assume such a set does in fact exist.

Axiom 2.1.0.6. There exists a set $\varnothing$, referred to as the **empty set**, such that for each set x, $x \notin \varnothing$.

Exercise 2.1.0.7. Let A be a set. Then $\varnothing \subset A$.

Proof. Let x be a set. Then $x \notin \varnothing$. Hence $x \in \varnothing$ implies that $x \in A$. Since x is an arbitrary set, we have that for each set x, $x \in \varnothing$ implies that $x \in A$. Thus $\varnothing \subset A$. □

give intuition for union, intersection, complement, show we can define union in terms of intersection and complement, call back to demorgans laws for logic

Definition 2.1.0.8. Let A, B be sets. We define intersection, union, empty set

discuss and introduce/make pictures of venn diagrams

discuss how definitions by convention use an if, but it should be treated as an iff. The if makes sense because if we say that a dog is "blarge" if it is black and weighs more than 15 kg, if we had put an iff in this definition, then the we would be saying that if a dog is blarge, then it is black and weighs more than 15 kgs, which wouldn really make sense in a definition because we have not defined previously and do not know what it is blarge, so it is a language thing. but really we are defining a set. $x \in \{\text{blarge dogs}\} \iff x \in \{\text{black dogs}\}\text{and}\{\text{dogs above 15 kg}\}$ i.e. $\{\text{blarge dogs}\} = \{\text{black dogs}\} \cap \{\text{dogs above 15 kg}\}$.

discuss how we can define natural numbers

discuss how we define $(a, b) = \{\{a\}, \{a, b\}\}$ or similar

Definition 2.1.0.9. products $A \times B$

Exercise 2.1.0.10. content...

Exercise 2.1.0.11.

2.2 Relations

keep this part here, after section on sets similar to analysis notes, so we can assume arbitrary unions, products, etc, not just unions/products/etc of two sets

2.2.1 Introduction

Definition 2.2.1.1. Let A, B be sets.

- Let $R \subset A \times B$. Then R is said to be a **relation between** A **and** B.
- We define $\mathcal{R}(A, B) := \mathcal{P}(A, B)$

Definition 2.2.1.2. Let A, B, C be sets, $R \in \mathcal{R}(A, B)$, $S \in \mathcal{R}(B, C)$. We define the **composition relation of** S **with** R, denoted $S \circ_{\mathbf{Rel}} R \in \mathcal{R}(A, C)$, by

$$S \circ_{\mathbf{Rel}} R := \{(a, c) \in A \times C : \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S\}.$$

Note 2.2.1.3. When the context is clear, we write $S \circ R$ in place of $S \circ_{\mathbf{Rel}} R$.

Exercise 2.2.1.4. Let A, B, C, D be sets, R a relation between A and B, S relation between B and C and T relation between C and D. Then $(T \circ S) \circ R = T \circ (S \circ R)$.

Proof.

- Let $(a, d) \in (T \circ S) \circ R$. Then there exists $b \in B$ such that $(a, b) \in R$ and $(b, d) \in T \circ S$. Since $(b, d) \in T \circ S$, there exists $c \in C$ such that $(b, c) \in S$ and $(c, d) \in T$. Since $(a, b) \in R$ and $(b, c) \in S$, $(a, c) \in S \circ R$. Since $(a, c) \in S \circ R$ and $(c, d) \in T$, we have that $(a, d) \in T \circ (S \circ R)$. Since $(a, d) \in (T \circ S) \circ R$ is arbitrary, we have that for each $(a, d) \in (T \circ S) \circ R$, $(a, d) \in T \circ (S \circ R)$. Hence $(T \circ S) \circ R \subset T \circ (S \circ R)$.
- Similarly, $T \circ (S \circ R) \subset (T \circ S) \circ R$.

Since $(T \circ S) \circ R \subset T \circ (S \circ R)$ and $T \circ (S \circ R) \subset (T \circ S) \circ R$, we have that $(T \circ S) \circ R = T \circ (S \circ R)$. □

Definition 2.2.1.5. Let A be a set. We define the **diagonal relation on** A, denoted $\Delta_A \in \mathcal{R}(A, A)$, by

$$\Delta_A := \{(a, a) : a \in A\}.$$

Exercise 2.2.1.6. Let A, B be sets and $R \in \mathcal{R}(A, B)$. Then

1. $R \circ \Delta_A = R$
2. $\Delta_B \circ R = R$

Proof.

1.
   - Let $(a, b) \in R \circ \Delta_A$. Then there exists $a' \in A$ such that $(a, a') \in \Delta_A$ and $(a', b) \in R$. By definition of Δ_A, $a = a'$. Therefore $(a, b) \in R$. Since $(a, b) \in R \circ \Delta_A$ is arbitrary, we have that for each $(a, b) \in R \circ \Delta_A$, $(a, b) \in R$. Hence $R \circ \Delta_A \subset R$.
   - Let $(a, b) \in R$. Then $a \in A$. Therefore $(a, a) \in \Delta_A$. Since $(a, a) \in \Delta_A$ and $(a, b) \in R$, we have that $(a, b) \in R \circ \Delta_A$. Since $(a, b) \in R$ is arbitrary, we have that for each $(a, b) \in R$, $(a, b) \in R \circ \Delta_A$. Hence $R \circ \Delta_A \subset R$.

   Since $R \circ \Delta_A \subset R$ and $R \circ \Delta_A \subset R$, we have that $R \circ \Delta_A = R$.
2. Similar to (1).

□

Exercise 2.2.1.7. Let A, B, C be sets.

1. Let $R \in \mathcal{R}(A, B)$ and $(S_\gamma)_{\gamma \in \Gamma} \subset \mathcal{R}(B, C)$. Then

(a) $\left(\bigcup_{\gamma\in\Gamma} S_\gamma\right)\circ R = \bigcup_{\gamma\in\Gamma}(S_\gamma\circ R)$,

(b) $\left(\bigcap_{\gamma\in\Gamma} S_\gamma\right)\circ R \subset \bigcap_{\gamma\in\Gamma}(S_\gamma\circ R)$

2. Let $(R_\gamma)_{\gamma\in\Gamma}\subset\mathcal{R}(A,B)$ and $S\in\mathcal{R}(B,C)$. Then

(a) $S\circ\left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) = \bigcup_{\gamma\in\Gamma}(S\circ R_\gamma)$.

(b) $S\circ\left(\bigcap_{\gamma\in\Gamma} R_\gamma\right) \subset \bigcap_{\gamma\in\Gamma}(S\circ R_\gamma)$.

Proof.

1. (a) We note that

$$\begin{aligned}
(a,c)\in\left(\bigcup_{\gamma\in\Gamma} S_\gamma\right)\circ R &\iff \text{there exists } b\in B \text{ such that } (a,b)\in R \text{ and } (b,c)\in\bigcup_{\gamma\in\Gamma} S_\gamma\\
&\iff \text{there exists } b\in B \text{ and } \gamma_0\in\Gamma \text{ such that } (a,b)\in R \text{ and } (b,c)\in S_{\gamma_0}\\
&\iff \text{there exists } \gamma_0\in\Gamma \text{ such that } (a,c)\in S_{\gamma_0}\circ R\\
&\iff (a,c)\in\bigcup_{\gamma\in\Gamma}(S_\gamma\circ R).
\end{aligned}$$

Hence $\left(\bigcup_{\gamma\in\Gamma} S_\gamma\right)\circ R = \bigcup_{\gamma\in\Gamma}(S_\gamma\circ R)$.

(b) We note that

$$\begin{aligned}
(a,c)\in\left(\bigcap_{\gamma\in\Gamma} S_\gamma\right)\circ R &\iff \text{there exists } b\in B \text{ such that } (a,b)\in R \text{ and } (b,c)\in\bigcap_{\gamma\in\Gamma} S_\gamma\\
&\iff \text{there exists } b\in B \text{ such that } (a,b)\in R \text{ and for each } \gamma\in\Gamma \text{ and } (b,c)\in S_\gamma\\
&\implies \text{for each } \gamma\in\Gamma, \text{ there exists } b\in B \text{ such that } (a,b)\in R \text{ and } (b,c)\in S_\gamma\\
&\iff \text{for each } \gamma\in\Gamma,\ (a,c)\in S_\gamma\circ R\\
&\iff (a,c)\in\bigcap_{\gamma\in\Gamma}(S_\gamma\circ R).
\end{aligned}$$

Hence $\left(\bigcap_{\gamma\in\Gamma} S_\gamma\right)\circ R = \bigcap_{\gamma\in\Gamma}(S_\gamma\circ R)$.

2. (a) We note that

$$\begin{aligned}
(a,c)\in S\circ\left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) &\iff \text{there exists } b\in B \text{ such that } (a,b)\in\bigcup_{\gamma\in\Gamma} R_\gamma \text{ and } (b,c)\in S\\
&\iff \text{there exists } b\in B \text{ and } \gamma_0\in\Gamma \text{ such that } (a,b)\in R_{\gamma_0} \text{ and } (b,c)\in S\\
&\iff \text{there exists } \gamma_0\in\Gamma \text{ such that } (a,c)\in S\circ R_{\gamma_0}\\
&\iff (a,c)\in\bigcup_{\gamma\in\Gamma}(S\circ R_\gamma).
\end{aligned}$$

Hence $S\circ\left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) = \bigcup_{\gamma\in\Gamma}(S_\gamma\circ R)$.

(b) We note that

$$\begin{aligned}
(a,c) \in S \circ \left(\bigcap_{\gamma \in \Gamma} R_\gamma \right) &\iff \text{there exists } b \in B \text{ such that } (a,b) \in \bigcap_{\gamma \in \Gamma} R_\gamma \text{ and } (b,c) \in S \\
&\iff \text{there exists } b \in B \text{ such that for each } \gamma \in \Gamma,\ (a,b) \in R_\gamma \text{ and } (b,c) \in S \\
&\implies \text{for each } \gamma \in \Gamma, \text{ there exists } b \in B \text{ such that } (a,b) \in R_\gamma \text{ and } (b,c) \in S \\
&\iff \text{for each } \gamma \in \Gamma,\ (a,c) \in S \circ R_\gamma \\
&\iff (a,c) \in \bigcap_{\gamma \in \Gamma} (S \circ R_\gamma).
\end{aligned}$$

Hence $S \circ \left(\bigcap_{\gamma \in \Gamma} R_\gamma \right) \subset \bigcap_{\gamma \in \Gamma} (S_\gamma \circ R)$.

□

Exercise 2.2.1.8. Let A, B, C be sets.

1. Let $R_1, R_2 \in \mathcal{R}(A,B)$ and $S \in \mathcal{R}(B,C)$. If $R_1 \subset R_2$, then $S \circ R_1 \subset S \circ R_2$.
2. Let $R \in \mathcal{R}(A,B)$ and $S_1, S_2 \in \mathcal{R}(B,C)$. If $S_1 \subset S_2$, then $S_1 \circ R \subset S_2 \circ R$.
3. Let $R_1, R_2 \in \mathcal{R}(A,B)$ and $S_1, S_2 \in \mathcal{R}(B,C)$. If $R_1 \subset R_2$ and $S_1 \subset S_2$, then $S_1 \circ R_1 \subset S_2 \circ R_2$.

Proof.

1. Suppose that $R_1 \subset R_2$. Then

$$\begin{aligned}
(a,c) \in S \circ R_1 &\iff \text{there exists } b \in B \text{ such that } (a,b) \in R_1 \text{ and } (b,c) \in S \\
&\implies \text{there exists } b \in B \text{ such that } (a,b) \in R_2 \text{ and } (b,c) \in S \\
&\iff (a,c) \in S \circ R_2.
\end{aligned}$$

   Hence $S \circ R_1 \subset S \circ R_2$.

2. Suppose that $S_1 \subset S_2$. Then

$$\begin{aligned}
(a,c) \in S_1 \circ R &\iff \text{there exists } b \in B \text{ such that } (a,b) \in R \text{ and } (b,c) \in S_1 \\
&\implies \text{there exists } b \in B \text{ such that } (a,b) \in R \text{ and } (b,c) \in S_2 \\
&\iff (a,c) \in S_2 \circ R.
\end{aligned}$$

   Hence $S_2 \circ R \subset S_1 \circ R$.

3. Suppose that $R_1 \subset R_2$ and $S_1 \subset S_2$. Then the previous parts imply that

$$\begin{aligned}
S_1 \circ R_1 &\subset S_2 \circ R_1 \\
&\subset S_2 \circ R_2.
\end{aligned}$$

□

Definition 2.2.1.9. Let A, B be sets and $R \in \mathcal{R}(A,B)$. We define the **converse of** R, denoted $R^* \in \mathcal{R}(B,A)$, by

$$R^* := \{(b,a) \in B \times A : (a,b) \in R\}.$$

Exercise 2.2.1.10. Let A, B be sets and R a relation between A and B. Then $(R^*)^* = R$.

Proof.

$$\begin{aligned}
(a,b) \in (R^*)^* &\iff (b,a) \in R^* \\
&\iff (a,b) \in R.
\end{aligned}$$

Hence $(R^*)^* = R$. □

Exercise 2.2.1.11. Let A, B, C be sets, $R \in \mathcal{R}(A, B)$ and $S \in \mathcal{R}(B, C)$. Then $(S \circ R)^* = R^* \circ S^*$.

Proof. We note that

$$\begin{aligned}
(c, a) \in (S \circ R)^* &\iff (a, c) \in S \circ R \\
&\iff \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S \\
&\iff \text{there exists } b \in B \text{ such that } (c, b) \in S^* \text{ and } (b, a) \in R^* \\
&\iff (c, a) \in R^* \circ S^*.
\end{aligned}$$

Hence $(S \circ R)^* = R^* \circ S^*$. □

Exercise 2.2.1.12. Let A and B be sets and $(R_\gamma)_{\gamma \in \Gamma} \subset \mathcal{R}(A, B)$. Then

1. $\left(\bigcup_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcup_{\gamma \in \Gamma} R_\gamma^*$
2. $\left(\bigcap_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcap_{\gamma \in \Gamma} R_\gamma^*$

Proof.

1. We note that

$$\begin{aligned}
(b, a) \in \left(\bigcup_{\gamma \in \Gamma} R_\gamma \right)^* &\iff (a, b) \in \bigcup_{\gamma \in \Gamma} R_\gamma \\
&\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a, b) \in R_{\gamma_0} \\
&\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (b, a) \in R_{\gamma_0}^* \\
&\iff (b, a) \in \bigcup_{\gamma \in \Gamma} R_\gamma^*.
\end{aligned}$$

   Hence $\left(\bigcup_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcup_{\gamma \in \Gamma} R_\gamma^*$.

2. We note that

$$\begin{aligned}
(b, a) \in \left(\bigcap_{\gamma \in \Gamma} R_\gamma \right)^* &\iff (a, b) \in \bigcap_{\gamma \in \Gamma} R_\gamma \\
&\iff \text{for each } \gamma \in \Gamma,\ (a, b) \in R_\gamma \\
&\iff \text{for each } \gamma \in \Gamma,\ (b, a) \in R_\gamma^* \\
&\iff (b, a) \in \bigcap_{\gamma \in \Gamma} R_\gamma^*.
\end{aligned}$$

   Hence $\left(\bigcap_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcap_{\gamma \in \Gamma} R_\gamma^*$.

□

2.2.2 Subrelations

Definition 2.2.2.1. Let A, B be sets, $R \in \mathcal{R}(A, B)$, $E \subset A$ and $F \subset B$. We define the **subrelation of R between E and F**, denoted $R|_{E \times F} \in \mathcal{R}(E, F)$, by $R|_{E \times F}$

maybe dont do this subsection, since relations are like morphisms, so unless you can find a categorical way to present restrictions on domain or codomain $f|_E$ or $f|^F$, just get rid of this subsection

2.2.3 Products of Relations

Definition 2.2.3.1. Let A, B, C, D be sets and $R \in \mathcal{R}(A, B), S \in \mathcal{R}(C, D)$. We define the **product relation of R and S**, denoted $R \times_{\mathbf{Rel}} S \in \mathcal{R}(A \times C, B \times D)$, by

$$R \times_{\mathbf{Rel}} S := \{((a, c), (b, d)) \in (A \times C) \times (B \times D) : (a, b) \in R \text{ and } (c, d) \in S\}.$$

Note 2.2.3.2. When the context is clear, we write $R \times S$ in place of $R \times_{\mathbf{Rel}} S$.

Exercise 2.2.3.3. Let A, B, C, D be sets.

1. Let $R \in \mathcal{R}(A, B)$ and $(S_\gamma)_{\gamma \in \Gamma} \subset \mathcal{R}(C, D)$. Then
   (a) $R \times \left(\bigcup_{\gamma \in \Gamma} S_\gamma \right) = \bigcup_{\gamma \in \Gamma} (R \times S_\gamma)$
   (b) $R \times \left(\bigcap_{\gamma \in \Gamma} S_\gamma \right) = \bigcap_{\gamma \in \Gamma} (R \times S_\gamma)$
2. Let $(R_\gamma)_{\gamma \in \Gamma} \subset\in \mathcal{R}(A, B)$ and $S \in \mathcal{R}(C, D)$. Then
   (a) $\left(\bigcup_{\gamma \in \Gamma} R_\gamma \right) \times S = \bigcup_{\gamma \in \Gamma} (R_\gamma \times S)$
   (b) $\left(\bigcap_{\gamma \in \Gamma} R_\gamma \right) \times S = \bigcap_{\gamma \in \Gamma} (R_\gamma \times S)$

Proof.

1. (a) We note that

$$\begin{aligned}
((a, c), (b, d)) \in R \times \left(\bigcup_{\gamma \in \Gamma} S_\gamma \right) &\iff (a, b) \in R \text{ and } (c, d) \in \bigcup_{\gamma \in \Gamma} S_\gamma \\
&\iff (a, b) \in R \text{ and there exists } \gamma_0 \in \Gamma \text{ such that } (c, d) \in S_{\gamma_0} \\
&\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a, b) \in R \text{ and } (c, d) \in S_{\gamma_0} \\
&\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } ((a, c), (b, d)) \in R \times S_{\gamma_0} \\
&\iff ((a, c), (b, d)) \in \bigcup_{\gamma \in \Gamma} (R \times S_\gamma).
\end{aligned}$$

   Hence $R \times \left(\bigcup_{\gamma \in \Gamma} S_\gamma \right) = \bigcup_{\gamma \in \Gamma} (R \times S_\gamma)$.

   (b) We note that

$$\begin{aligned}
((a, c), (b, d)) \in R \times \left(\bigcap_{\gamma \in \Gamma} S_\gamma \right) &\iff (a, b) \in R \text{ and } (c, d) \in \bigcap_{\gamma \in \Gamma} S_\gamma \\
&\iff (a, b) \in R \text{ for each } \gamma \in \Gamma,\ (c, d) \in S_\gamma \\
&\iff \text{for each } \gamma \in \Gamma,\ (a, b) \in R \text{ and } (c, d) \in S_\gamma \\
&\iff \text{for each } \gamma \in \Gamma,\ ((a, c), (b, d)) \in R \times S_\gamma \\
&\iff ((a, c), (b, d)) \in \bigcap_{\gamma \in \Gamma} (R \times S_\gamma).
\end{aligned}$$

   Hence $R \times \left(\bigcap_{\gamma \in \Gamma} S_\gamma \right) = \bigcap_{\gamma \in \Gamma} (R \times S_\gamma)$.

2. (a) We note that

$$\begin{aligned}((a,c),(b,d)) \in \left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) \times S &\iff (a,b) \in \left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) \text{ and } (c,d) \in S \\ &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a,b) \in R_{\gamma_0} \text{ and } (c,d) \in S \\ &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } ((a,c),(b,d)) \in R_{\gamma_0} \times S \\ &\iff ((a,c),(b,d)) \in \bigcup_{\gamma\in\Gamma} (R_\gamma \times S)\end{aligned}$$

Hence $\left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) \times S = \bigcup_{\gamma\in\Gamma} (R_\gamma \times S)$.

(b) We note that

$$\begin{aligned}((a,c),(b,d)) \in \left(\bigcap_{\gamma\in\Gamma} R_\gamma\right) \times S &\iff (a,b) \in \left(\bigcap_{\gamma\in\Gamma} R_\gamma\right) \text{ and } (c,d) \in S \\ &\iff \text{for each } \gamma \in \Gamma,\ (a,b) \in R_\gamma \text{ and } (c,d) \in S \\ &\iff \text{for each } \gamma \in \Gamma,\ ((a,c),(b,d)) \in R_\gamma \times S \\ &\iff ((a,c),(b,d)) \in \bigcap_{\gamma\in\Gamma} (R_\gamma \times S)\end{aligned}$$

Hence $\left(\bigcap_{\gamma\in\Gamma} R_\gamma\right) \times S = \bigcap_{\gamma\in\Gamma} (R_\gamma \times S)$.

$\square$

2.3 Equivalence Relations

Note 2.3.0.1. Let X be a set and $\nu \in \mathcal{R}^2(X)$. When the context is clear, we may write $a\nu b$ in place of $(a, b) \in \nu$.

Definition 2.3.0.2. Let X be a set. Suppose that $X \neq \varnothing$.

- Let $\theta \in \mathcal{R}^2(X)$. Then θ is said to be
  - **reflexive** if for each $a \in X$, $(a, a) \in \theta$
  - **symmetric** if for each $a, b \in X$, $(a, b) \in \theta$ implies that $(b, a) \in \theta$
  - **transitive** if for each $a, b, c \in X$, $(a, b) \in \theta$ and $(b, c) \in \theta$ implies that $(a, c) \in \theta$
  - an **equivalence relation on** X if θ is reflexive, symmetric and transitive
- We define $\mathrm{Equiv}(X) := \{\theta \in \mathcal{R}^2(X) : \theta \text{ is an equivalence relation on } X\}$

Exercise 2.3.0.3. Let X be a set and $\theta \in \mathcal{R}^2(X)$. Then

1. θ is reflexive iff $\Delta_X \subset \theta$
2. θ is symmetric iff $\theta^* \subset \theta$
3. θ is transitive iff $\theta \circ \theta \subset \theta$

Proof.

1.
   - $(\implies)$:
     Suppose that θ is reflexive. Let $(a, a) \in \Delta_X$. Then $a \in X$. Since θ is reflexive, $(a, a) \in \theta$. Since $(a, a) \in \Delta_X$ is arbitrary, we have that for each $(a, a) \in \Delta_X$, $(a, a) \in \theta$. Hence $\Delta_X \subset \theta$.
   - $(\impliedby)$:
     Suppose that $\Delta_X \subset \theta$. Let $a \in X$. By definition and assumption,
     $$\begin{aligned}(a, a) &\in \Delta_X \\ &\subset \theta.\end{aligned}$$
     Since $a \in X$ is arbitrary, we have that for each $a \in X$, $(a, a) \in \theta$. Hence θ is reflexive.
2.
   - $(\implies)$:
     Suppose that θ is symmetric. Let $(b, a) \in \theta^*$. By definition, $(a, b) \in \theta$. Since θ is symmetric, $(b, a) \in \theta$. Since $(b, a) \in \theta^*$ is arbitrary, we have that for each $(b, a) \in \theta^*$, $(b, a) \in \theta$. Hence $\theta^* \subset \theta$.
   - $(\impliedby)$:
     Suppose that $\theta^* \subset \theta$. Let $a, b \in X$. Suppose that $(a, b) \in \theta$. By definition and assumption,
     $$\begin{aligned}(b, a) &\in \theta^* \\ &\subset \theta.\end{aligned}$$
     Since $a, b \in X$ with $(a, b) \in \theta$ are arbitrary, we have that for each $a, b \in X$, $(a, b) \in \theta$ implies that $(b, a) \in \theta$. Hence θ is symmetric.
3.
   - $(\implies)$:
     Suppose that θ is transitive. Let $(a, c) \in \theta \circ \theta$. By definition, there exists $b \in X$ such that $(a, b) \in \theta$ and $(b, c) \in \theta$. Since θ is transitive, $(a, c) \in \theta$. Since $(a, c) \in \theta \circ \theta$ is arbitrary, we have that for each $(a, c) \in \theta \circ \theta$, $(a, c) \in \theta$. Hence $\theta \circ \theta \subset \theta$.
   - $(\impliedby)$:
     Suppose that $\theta \circ \theta \subset \theta$. Let $a, b, c \in X$. Suppose that $(a, b) \in \theta$ and $(b, c) \in \theta$. By definition and assumption,
     $$\begin{aligned}(a, c) &\in \theta \circ \theta \\ &\subset \theta.\end{aligned}$$
     Since $a, b, c \in X$ with $(a, b) \in \theta$ and $(b, c) \in \theta$ are arbitrary, we have that for each $a, b \in X$, $(a, b) \in \theta$ and $(b, c) \in \theta$ implies that $(a, c) \in \theta$. Hence θ is transitive.

□

Definition 2.3.0.4. Let X be a set. Suppose that $X \neq \varnothing$. We define the **universal relation on** X, denoted $\nabla_X \in \mathcal{R}^2(X)$, by $\nabla_X := X^2$.

Exercise 2.3.0.5. Let X be a set. Suppose that $X \neq \varnothing$. Then $\nabla_X \in \mathrm{Equiv}(X)$.

Proof. Let $a, b, c \in X$.

1. Since $a \in X$, $(a, a) \in \nabla_X$.
2. Suppose that $(a, b) \in \nabla_X$. Since $a, b \in X$, $(b, a) \in X^2$. Therefore $(a, b) \in \nabla_X$ implies that $(b, a) \in \nabla_X$.
3. Suppose that $(a, b), (b, c) \in \nabla_X$. Since $a, c \in X$, $(a, c) \in X^2$. Therefore $(a, b), (b, c) \in X^2$ implies that $(a, c) \in \nabla_X$.

□

Exercise 2.3.0.6. Let X be a set. Suppose that $X \neq \varnothing$. Then $\Delta_X \in \mathrm{Equiv}(X)$.

Proof. Let $a, b, c \in X$.

1. Since $a \in X$, $(a, a) \in \Delta_X$.
2. Suppose that $(a, b) \in \Delta_X$. Then $a = b$ and
$$\begin{aligned}(b, a) &= (a, a)\\ &\in \Delta_X.\end{aligned}$$
Therefore $(a, b) \in \Delta_X$ implies that $(b, a) \in \Delta_X$.
3. Suppose that $(a, b), (b, c) \in \Delta_X$. Then $a = b$ and $b = c$. Hence $a = c$ and
$$\begin{aligned}(a, c) &= (a, a)\\ &\in \Delta_X.\end{aligned}$$
Therefore $(a, b), (b, c) \in \Delta_X$ implies that $(a, c) \in \Delta_X$.

□

Exercise 2.3.0.7. Let X be a set. Suppose that $X \neq \varnothing$. Then for each $(\theta_\alpha)_{\alpha \in A} \subset \mathrm{Equiv}(X)$, $\bigcap_{\alpha \in A} \theta_\alpha \in \mathrm{Equiv}(X)$.

Proof. Let $(\theta_\alpha)_{\alpha \in A} \subset \mathrm{Equiv}(X)$. Set $\theta := \bigcap_{\alpha \in A} \theta_\alpha$. Let $a, b, c \in X$.

1. Let $\alpha \in A$. Since $\theta_\alpha \in \mathrm{Equiv}(X)$, $(a, a) \in \theta_\alpha$. Since $\alpha \in A$ is arbitrary, we have that for each $\alpha \in A$, $(a, a) \in \theta_\alpha$. Hence $(a, a) \in \theta$.
2. Suppose that $(a, b) \in \theta$. Let $\alpha \in A$. Then $(a, b) \in \theta_\alpha$. Since $\theta_\alpha \in \mathrm{Equiv}(X)$, we have that $(b, a) \in \theta_{a\langle}$. Since $\alpha \in A$ is arbitrary, we have that for each $\alpha \in A$, $(b, a) \in \theta_\alpha$. Hence $(b, a) \in \theta$. Therefore $(a, b) \in \theta$ implies that $(b, a) \in \theta$.
3. Suppose that $(a, b) \in \theta$ and $(b, c) \in \theta$. Let $\alpha \in A$. Then $(a, b), (b, c) \in \theta_\alpha$. Since $\theta_\alpha \in \mathrm{Equiv}(X)$, $(a, c) \in \theta_\alpha$. Since $\alpha \in A$ is arbitrary, we have that for each $\alpha \in A$, $(a, c) \in \theta_\alpha$. Hence $(a, c) \in \theta$. Therefore $(a, b) \in \theta$ and $(b, c) \in \theta$ implies that $(a, c) \in \theta$.

Hence $\theta \in \mathrm{Equiv}(X)$. □

Definition 2.3.0.8. Let X be a set. Suppose that $X \neq \varnothing$. Let $\nu \in \mathcal{R}^2(X)$.

- We define the **equivalence relations containing** ν, denoted $\mathcal{E}(\nu)$, by $\mathcal{E}(\nu) := \{\theta \in \mathrm{Equiv}(X) : \nu \subset \theta\}$.
- We define the **equivalence relation on** X **generated by** ν, denoted $\epsilon(\nu) \in \mathcal{R}^2(X)$ by $\epsilon(\nu) := \bigcap_{\theta \in \mathcal{E}(\nu)} \theta$.

Note 2.3.0.9.

- A previous ex implies that $\nabla_X \in \mathcal{E}(\nu)$.
- A previous ex (about intersections of equiv relats) implies that $\epsilon(\nu) \in \mathcal{E}(\nu)$.

Exercise 2.3.0.10. Let X be a set. Suppose that $X \neq \varnothing$. Let $\nu \in \mathcal{R}^2(X)$. Define $(\nu_n)_{n\in\mathbb{N}_0} \subset \mathcal{R}^2(X)$ by

$$\nu_n := \begin{cases} \Delta_X \cup \nu \cup \nu^*, & n = 0 \\ \nu_{n-1} \circ \nu_{n-1}, & n \geq 1. \end{cases}$$

Then

1. (a) for each $n \in \mathbb{N}$, ν_n is reflexive
   (b) for each $n \in \mathbb{N}$, ν_n is symmetric
   (c) for each $n \in \mathbb{N}_0$, $\nu_n \subset \nu_{n+1}$
2. $\epsilon(\nu) = \bigcup_{n\in\mathbb{N}_0} \nu_n$.

Proof.

1. (a)
   - **Base Case:**
     Since
     $$\begin{aligned} \Delta_X &\subset \Delta_X \cup \nu \cup \nu^* \\ &= \nu_0, \end{aligned}$$
     a previous ex implies that ν_0 is reflexive.
   - **Induction Step:**
     Let $n \in \mathbb{N}_0$. Suppose that ν_n is reflexive. Then that same prev ex implies that $\Delta_X \subset \nu_n$. two prev ex in section on relations about composition with identity and subsets then imply that
     $$\begin{aligned} \Delta_X &= \Delta_X \circ \Delta_X \\ &\subset \nu_n \circ \nu_n \\ &= \nu_{n+1}. \end{aligned}$$
     Another application of that prev ex implies that ν_{n+1} is reflexive.

   By induction, for each $n \in \mathbb{N}_0$, ν_n is reflexive.

   (b)
   - **Base Case:**
     a previous ex about converse relations and unions, double converse and since $\Delta_X \in \mathrm{Equiv}(X)$ and an ex about symmetric relations implies that
     $$\begin{aligned} \nu_0^* &= (\Delta_X \cup \nu \cup \nu^*)^* \\ &= \Delta_X^* \cup \nu^* \cup (\nu^*)^* \\ &= \Delta_X^* \cup \nu^* \cup \nu \\ &\subset \Delta_X \cup \nu^* \cup \nu \\ &= \Delta_X \cup \nu \cup \nu^* \\ &= \nu_0. \end{aligned}$$
     that same previous ex about symmetric relations implies that ν_0 is symmetric.
   - **Induction Step:**
     Let $n \in \mathbb{N}_0$. Suppose that ν_n is symmetric. Then that same prev ex about symmetric and converse relations implies that $\nu_n^* \subset \nu_n$. Therefore prev exs imply
     $$\begin{aligned} \nu_{n+1}^* &= (\nu_n \circ \nu_n)^* \\ &= \nu_n^* \circ \nu_n^* \\ &\subset \nu_n \circ \nu_n. \end{aligned} \qquad = \nu_{n+1}.$$

Another application of that prev ex implies that ν_{n+1} is symmetric.

By induction, for each $n \in \mathbb{N}_0$, ν_n is symmetric.

(c) Let $n \in \mathbb{N}_0$. Since ν_n is reflexive, prev ex implies that $\Delta_X \subset \nu_n$. Therefore prev ex implies

$$\begin{aligned}\nu_n &= \Delta_X \circ \nu_n \\ &\subset \nu_n \circ \nu_n \\ &= \nu_{n+1}.\end{aligned}$$

Since $n \in \mathbb{N}_0$ is arbitrary, we have that for each $n \in \mathbb{N}_0$, $\nu_n \subset \nu_{n+1}$.

2. Set $\theta_0 := \bigcup_{n \in \mathbb{N}_0} \nu_n$.

- – **Base Case:**
  Since $\epsilon(\nu) \in \mathcal{E}(\nu)$, we have that $\epsilon(\nu) \in \mathrm{Equiv}(X)$ and $\nu \subset \epsilon(\nu)$. Since $\epsilon(\nu) \in \mathrm{Equiv}(X)$, we have that $\epsilon(\nu)$ is reflexive and symmetric. A ref ex then implies that $\Delta_X \subset \epsilon(\nu)$ and $\epsilon(\nu)^* \subset \epsilon(\nu)$. Therefore a prev ex about converse relations and subsets

$$\begin{aligned}\nu_0 &= \Delta_X \cup \nu \cup \nu^* \\ &\subset \epsilon(\nu) \cup \epsilon(\nu) \cup \epsilon(\nu)^* \\ &\subset \epsilon(\nu) \cup \epsilon(\nu) \cup \epsilon(\nu) \\ &= \epsilon(\nu).\end{aligned}$$

  – **Induction Step:**
  Let $n \in \mathbb{N}_0$. Suppose that $\nu_n \subset \epsilon(\nu)$. Then ref prev ex

$$\begin{aligned}\nu_{n+1} &= \nu_n \circ \nu_n \\ &\subset \epsilon(\nu) \circ \epsilon(\nu) \\ &= \epsilon(\nu).\end{aligned}$$

  By induction, for each $n \in \mathbb{N}_0$, $\nu_n \subset \epsilon(\nu)$. Therefore

$$\begin{aligned}\theta_0 &= \bigcup_{n \in \mathbb{N}_0} \nu_n \\ &\subset \epsilon(\nu).\end{aligned}$$

- – Since ν_0 is reflexive, we have that

$$\begin{aligned}\Delta_X &\subset \nu_0 \\ &\subset \bigcup_{n \in \mathbb{N}_0} \nu_n \\ &= \theta_0.\end{aligned}$$

  Hence θ_0 is reflexive.

  – Since for each $n \in \mathbb{N}_0$, ν_n is symmetric, we have that that ref exs for each $n \in \mathbb{N}_0$, $\nu_n^* \subset \nu_n$. Therefore

$$\begin{aligned}\theta_0^* &= (\bigcup_{n \in \mathbb{N}_0} \nu_n)^* \\ &= \bigcup_{n \in \mathbb{N}_0} \nu_n^* \\ &\subset \bigcup_{n \in \mathbb{N}_0} \nu_n \\ &= \theta_0.\end{aligned}$$

  Hence θ_0 is symmetric.

- Let $a, b, c \in X$. Suppose that $(a, b) \in \theta_0$ and $(b, c) \in \theta_0$. Then there exist $n_1, n_2 \in \mathbb{N}_0$ such that $(a, b) \in \nu_{n_1}$ and $(b, c) \in \nu_{n_2}$. Set $n_0 := \max\{n_1, n_2\}$. Since for each $n \in \mathbb{N}$, $\nu_n \subset \nu_{n+1}$, we have that

$$\begin{aligned}(a, b) &\in \nu_{n_1} \\ &\subset \nu_{n_0}.\end{aligned}$$

  and

$$\begin{aligned}(b, c) &\in \nu_{n_2} \\ &\subset \nu_{n_0}.\end{aligned}$$

  Since $(a, b), (b, c) \in \nu_{n_0}$, we have that

$$\begin{aligned}(a, c) &\in \nu_{n_0} \circ \nu_{n_0} \\ &= \nu_{n_0+1} \\ &\subset \bigcup_{n \in \mathbb{N}_0} \nu_n \\ &= \theta_0.\end{aligned}$$

  Since $a, b, c \in X$ with $(a, b) \in \theta_0$ and $(b, c) \in \theta_0$ are arbitrary, we have that for each $a, b, c \in X$, $(a, b) \in \theta_0$ and $(b, c) \in \theta_0$ implies that $(a, c) \in \theta_0$. Hence θ_0 is transitive.

Since θ_0 is reflexive, symmetric and transitive, $\theta_0 \in \mathrm{Equiv}(X)$. Since

$$\begin{aligned}\nu &\subset \Delta_X \cup \nu \cup \nu^* \\ &= \nu_0 \\ &\subset \bigcup_{n \in \mathbb{N}_0} \nu_n \\ &= \theta_0,\end{aligned}$$

we have that $\theta_0 \in \mathcal{E}(\nu)$. Therefore

$$\begin{aligned}\epsilon(\nu) &= \bigcap_{\theta \in \mathcal{E}(\nu)} \theta \\ &\subset \theta_0.\end{aligned}$$

Since $\theta_0 \subset \epsilon(\nu)$ and $\epsilon(\nu) \subset \theta_0$, we have that $\theta_0 = \epsilon(\nu)$.

$\square$

2.4 Products

Chapter 3

Ordered Sets

3.1 To Do

3.1.1 To Do

at this point, there should be a simple structure capturing prosets and $\downarrow S$, measurable spaces and $\sigma(\mathcal{A})$, topological spaces and $\tau(\mathcal{E})$, groups/rings/$\cdots$/algebras and $\langle S\rangle/(E)/\cdots/(F)$. It seems like we need a category $\mathcal{C}$ with some structure, another category $\mathcal{C}_F$ which is like $\mathcal{C}$, but less structured and some forgetful-like functor $F : \mathcal{C} \to \mathcal{C}_F$. eg, $F : \mathbf{Pro} \to \mathbf{Set}$ or $F : \mathbf{Top} \to \mathbf{Set}$ and we need an associated poset $\mathcal{P}$ containing the structure forgotten by F, e.g. the lower sets of X or the set of topologies on X and we need a "generating" or "contained" object $\mathcal{E}$ in $\mathcal{C}_F$ in the sense that there is at least one object A in $\mathcal{C}$ and monomirphism $\iota_A : \mathcal{E} \to F(A)$ in $\mathcal{C}_F$. We then need a minimal element in the structure poset $\mathcal{P}$ in some universal sense relating to these monomorphisms. Ask people who know category theory

3.2 Prosets

3.2.1 Introduction

Definition 3.2.1.1. Preordered Set:
Let X be a set and $\leq \subset X \times X$ a binary relation on X. Then

- $\leq$ is said to be a **preorder on** X if
  1. for each $a \in X$, $a \leq a$
  2. for each $a, b, c \in X$, $a \leq b$ and $b \leq c$ implies that $a \leq c$
- $(X, \leq)$ is said to be a **preordered set** or **proset** if $\leq$ is a preorder on X.

Definition 3.2.1.2. Let $(X, \leq)$ be a proset. We define the **dual order of** $\leq$ **on** X, denoted $\leq^*$, by $a \leq^* b$ iff $b \leq a$.

Exercise 3.2.1.3. Let $(X, \leq)$ be a proset. Then $\leq^*$ is a preorder on X.

Proof.

1. Let $a \in X$. Since $a \leq a$, we have that $a \leq^* a$.
2. Let $a, b, c \in X$. Suppose that $a \leq^* b$ and $b \leq^* c$. Then $b \leq a$ and $c \leq b$. Hence $c \leq a$. Thus $a \leq^* c$.

Therefore $\leq^*$ is a preorder on X. $\square$

3.2.2 Products

Definition 3.2.2.1. Let $(A, \leq_A)$ and $(B, \leq_B)$ be prosets. We define the

- **product preorder of $\leq_A$ and $\leq_B$ on $A \times B$**, denoted $\leq_A \otimes \leq_B$ by $(a_1, b_1) \leq_A \otimes \leq_B (a_2, b_2)$ iff $a_1 \leq_A a_2$ and $b_1 \leq_B b_2$.
- **product proset of $(A, \leq_A)$ and $(B, \leq_B)$**, denoted $(A, \leq_A) \otimes (B, \leq_B)$ by $(A, \leq_A) \otimes (B, \leq_B) := (A \times B, \leq_A \otimes \leq_B)$

Exercise 3.2.2.2. probably need to change notation since $\otimes$ might be reserved for something else. Let $(A, \leq_A)$ and $(B, \leq_B)$ be prosets. Then

1. $\leq_A \otimes \leq_B$ is a preorder on $A \times B$,
2. $(A, \leq_A) \otimes (B, \leq_B)$ is a proset.

Proof.

1. (a) Let $(a, b) \in A \times B$. Then $a \leq_A a$ and $b \leq_B b$. Therefore $(a, b) \leq_A \otimes \leq_B (a, b)$.
   (b) Let $(a_1, b_1), (a_2, b_2), (a_3, b_3) \in A \times B$. Suppose that $(a_1, b_1) \leq_A \otimes \leq_B (a_2, b_2)$ and $(a_2, b_2) \leq_A \otimes \leq_B (a_3, b_3)$. Then $a_1 \leq_A a_2$, $a_2 \leq_A a_3$, $b_1 \leq_B b_2$ and $b_2 \leq_B b_3$. Therefore $a_1 \leq_A a_3$ and $b_1 \leq_B b_3$. Hence $(a_1, b_1) \leq_A \otimes \leq_B (a_3, b_3)$.

   Hence $\leq_A \otimes \leq_B$ is a preorder on $A \times B$.
2. Since $\leq_A \otimes \leq_B$ is a preorder on $A \times B$, $(X, \leq)$ is a proset.

□

Definition 3.2.2.3. Let $(X, \leq)$ be a proset. We define $\sim_{\leq} \subset X \times X$ by $a \sim_{\leq} b$ iff $a \leq b$ and $b \leq a$.

Exercise 3.2.2.4. Let $(X, \leq)$ be a proset. Then $\sim_{\leq}$ is an equivalence relation on X.

Proof. Let $x, y, z \in X$.

1. Since $x \leq x$, $x \sim_{\leq} x$.
2. Suppose that $x \sim_{\leq} y$. Then $x \leq y$ and $y \leq x$. Thus $y \sim_{\leq} x$.
3. Suppose that $x \sim_{\leq} y$ and $y \sim_{\leq} z$. Then $x \leq y$, $y \leq x$, $y \leq z$ and $z \leq y$. Therefore $x \leq z$ and $z \leq x$. Hence $x \sim_{\leq} z$.

□

3.2.3 Quotients

3.2.4 Functions

Let X be a set and (Y, leq) a proset. Define $\leq_{Y^X}$ by $f \leq_{Y^X} g$ iff for each $x \in X$, $f(x) \leq g(x)$., this notion is later used in galois connections subsection

3.2.5 Upper and Lower Sets

Definition 3.2.5.1. Let $(X, \leq)$ be a proset and $A \subset X$. Then

- A is said to be a **$\leq$-upper set** if for each $a \in A$ and $x \in X$, $a \leq x$ implies that $x \in A$.
- A is said to be an **$\leq$-lower set** if for each $a \in A$ and $x \in X$, $x \leq a$ implies that $x \in A$.

Note 3.2.5.2. When the context is clear, we say A is a

- "upper set" instead of "$\leq$-upper set"
- "lower set" instead of "$\leq$-lower set"

Exercise 3.2.5.3. Let $(X, \leq)$ be a proset. Then

1. X is a $\leq$-upper set
2. X is a $\leq$-lower set

Proof.

1. Let $a, x \in X$. Suppose that $a \leq x$. By assumption, $x \in X$. Since $a, x \in X$ with $a \leq x$ are arbitrary, we have that for each $a, x \in X$, $a \leq x$ implies that $x \in A$. Hence X is a $\leq$-upper set.
2. Similar to (1).

□

Exercise 3.2.5.4. Let $(X, \leq)$ be a proset and $A \subset X$. Then

1. A is a $\leq$-upper set iff A is a $\leq^*$-lower set
2. A is a $\leq$-lower set iff A is a $\leq^*$-upper set

Proof.

1.
   - $(\implies)$:
     Suppose that A is a $\leq$-upper set. Let $a \in A$ and $x \in X$. Suppose that $x \leq^* a$. Then $a \leq x$. Since A is a $\leq$-upper set, $x \in A$. Since $a \in A$ and $x \in X$ with $x \leq^* a$ is arbitrary, we have that for each $a \in A$ and $x \in X$, $x \leq^* a$ implies that $x \in A$. Hence A is a $\leq^*$-lower set.
   - $(\impliedby)$:
     Suppose that A is a $\leq^*$-lower set. Let $a \in A$ and $x \in X$. Suppose that $a \leq x$. Then $x \leq^* a$. Since A is a $\leq^*$-lower set, $x \in A$. Since $a \in A$ and $x \in X$ with $a \leq x$ is arbitrary, we have that for each $a \in A$ and $x \in X$, $a \leq x$ implies that $x \in A$. Hence A is a $\leq$-upper set.
2. Similar to (1).

□

Exercise 3.2.5.5. Let $(X, \leq)$ be a proset and $A \subset X$. Then

1. A is an upper set iff A^c is a lower set
2. A is a lower set iff A^c is an upper set

Proof.

1.
   - $(\implies)$:
     Suppose that A is an upper set. Let $b \in A^c$ and $x \in X$. Suppose that $x \leq b$. For the sake of contradiction, suppose that $x \in A$. Since $x \leq b$ and A is an upper set, $b \in A$. This is a contradiction since $b \in A^c$. Hence $x \in A^c$. Since $b \in A^c$ and $x \in X$ with $x \leq b$ are arbitrary, we have that for each $b \in A^c$ and $x \in X$, $x \leq b$ implies that $x \in A^c$. Thus A^c is a lower set.
   - $(\impliedby)$:
     Suppose that A^c is a $\leq$-lower set. Exercise 3.2.5.4 implies that A^c is a $\leq^*$-upper set. The previous part implies that A is a $\leq^*$-lower set. Another application of Exercise 3.2.5.4 implies that A is a $\leq$-upper set.
2. Similar to (1).

□

Exercise 3.2.5.6. Let $(X, \leq)$ be a proset, $(E_\alpha)_{\alpha \in A} \subset \mathcal{P}(X)$.

1. If for each $\alpha \in A$, E_α is an upper set, then
   (a) $\bigcup\limits_{\alpha \in A} E_\alpha$ is an upper set

(b) $\bigcap_{\alpha \in A} E_\alpha$ is an upper set

2. If for each $\alpha \in A$, E_α is a lower set, then

   (a) $\bigcup_{\alpha \in A} E_\alpha$ is a lower set

   (b) $\bigcap_{\alpha \in A} E_\alpha$ is an lower set

Proof.

1. Suppose that for each $\alpha \in A$, E_α is an upper set. Set $E := \bigcup_{\alpha \in A} E_\alpha$.

   (a) Let $e \in E$ and $x \in X$. Suppose that $e \leq x$. Since $e \in E$, there exists $\alpha \in A$ such that $e \in E_\alpha$. Since E_α is an upper set and $e \leq x$, we have that

$$\begin{aligned} x &\in E_\alpha \\ &\subset \bigcup_{\alpha \in A} E_\alpha \\ &= E. \end{aligned}$$

   Since $e \in E$ and $x \in X$ with $e \leq x$ are arbitrary, we have that for each $e \in E$ and $x \in X$, $e \leq x$ implies that $x \in E$. Hence E is an upper set.

   (b) Let $\alpha \in A$. Since E_α is a $\leq$-upper set, Exercise 3.2.5.5 implies that E_α^c is a $\leq$-lower set. Exercise 3.2.5.4 then implies that E_α^c is a $\leq^*$-upper set. Since $\alpha \in A$ is arbitrary, we have that for each $\alpha \in A$, E_α^c is a $\leq^*$-upper set. The previous part implies that $\bigcup_{\alpha \in A} E_\alpha^c$ is a $\leq^*$-upper set. Since $\bigcap_{\alpha \in A} E_\alpha = \left(\bigcup_{\alpha \in A} E_\alpha^c \right)^c$, Exercise 3.2.5.5 implies that $\bigcap_{\alpha \in A} E_\alpha$ is a $\leq^*$-lower set. Exercise 3.2.5.4 then implies that $\bigcap_{\alpha \in A} E_\alpha$ is a $\leq$-upper set.

2. Similar to (1).

□

Definition 3.2.5.7. Let $(X, \leq)$ be a proset and $S \subset X$. We define the

- **$\leq$-upper sets containing** S, denoted $\mathcal{U}(S, \leq) \subset \mathcal{P}(X)$, by $\mathcal{U}(S, \leq) := \{U \subset X : U \text{ is a } \leq\text{-upper set and } S \subset U\}$
- **$\leq$-lower sets containing** S, denoted $\mathcal{L}(S, \leq) \subset \mathcal{P}(X)$, by $\mathcal{L}(S, \leq) := \{L \subset X : L \text{ is a } \leq\text{-lower set and } S \subset L\}$.
- **$\leq$-upper set generated by** S, denoted $\uparrow(S, \leq) \subset X$, by $\uparrow(S, \leq) := \bigcap_{U \in \mathcal{U}(S, \leq)} U$
- **$\leq$-lower set generated by** S, denoted $\downarrow(S, \leq) \subset X$, by $\downarrow(S, \leq) := \bigcap_{L \in \mathcal{L}(S, \leq)} L$

Note 3.2.5.8.

- When the context is clear, we write $\mathcal{U}(S)$, $\mathcal{L}(S)$, $\uparrow S$ and $\downarrow S$ in place of $\mathcal{U}(S, \leq)$, $\mathcal{L}(S, \leq)$, $\uparrow(S, \leq)$ and $\downarrow(S, \leq)$ respectively.
- If $S = \{s\}$, we write $\uparrow s$ and $\downarrow s$ in place of $\uparrow S$ and $\downarrow S$ respectively.
- Exercise 3.2.5.3 implies that $X \in \mathcal{U}(S)$ and $X \in \mathcal{L}(S)$.
- Exercise 3.2.5.6 implies that $\uparrow S \in \mathcal{U}(S)$ and $\downarrow S \in \mathcal{L}(S)$.

Exercise 3.2.5.9. Let $(X, \leq)$ be a proset and $S \subset X$. Then

1. $\mathcal{U}(S, \leq^*) = \mathcal{L}(S, \leq)$,
2. $\mathcal{L}(S, \leq^*) = \mathcal{U}(S, \leq)$,

3. $\uparrow(S, \leq^*) = \downarrow(S, \leq)$,
4. $\downarrow(S, \leq^*) = \uparrow(S, \leq)$.

Proof.

1. - Let $L \in \mathcal{U}(S, \leq^*)$. Then $S \subset L$ and L is a $\leq^*$-upper set. Exercise 3.2.5.4 then implies that L is a $\leq$-lower set. Hence $L \in \mathcal{L}(S, \leq)$. Since $L \in \mathcal{U}(S, \leq^*)$ is arbitrary, we have that for each $L \in \mathcal{U}(S, \leq^*)$, $L \in \mathcal{L}(S, \leq)$. Thus $\mathcal{U}(S, \leq^*) \subset \mathcal{L}(S, \leq)$.
   - Let $L \in \mathcal{L}(S, \leq)$. Then $S \subset L$ and L is a $\leq$-lower set. Another application of Exercise 3.2.5.4 implies that L is a $\leq^*$-upper set. Hence $L \in \mathcal{U}(S, \leq^*)$. Since $L \in \mathcal{L}(S, \leq)$ is arbitrary, we have that for each $L \in \mathcal{L}(S, \leq)$, $L \in \mathcal{U}(S, \leq^*)$. Thus $\mathcal{L}(S, \leq) \subset \mathcal{U}(S, \leq^*)$.

   Since $\mathcal{U}(S, \leq^*) \subset \mathcal{L}(S, \leq)$ and $\mathcal{L}(S, \leq) \subset \mathcal{U}(S, \leq^*)$, we have that $\mathcal{U}(S, \leq^*) = \mathcal{L}(S, \leq)$.
2. By (1), we have that
$$\begin{aligned}\mathcal{L}(S, \leq^*) &= \mathcal{U}(S, (\leq^*)^{\mathrm{op}}) \\ &= \mathcal{U}(S, \leq).\end{aligned}$$
3. Part (1) implies that
$$\begin{aligned}\uparrow(S, \leq^*) &= \bigcap_{U \in \mathcal{U}(S, \leq^*)} U \\ &= \bigcap_{U \in \mathcal{L}(S, \leq)} U \\ &= \downarrow(S, \leq^*).\end{aligned}$$
4. Similar to (3).

□

Exercise 3.2.5.10. Let $(X, \leq)$ be a proset and $S \subset X$. Then

1. $S \in \mathcal{U}(S, \leq)$ iff $\uparrow S = S$
2. $S \in \mathcal{L}(S, \leq)$ iff $\downarrow S = S$

Proof.

1. - $(\Longrightarrow)$:
     Suppose that $S \in \mathcal{U}(S, \leq)$. Then
$$\begin{aligned}\uparrow S &= \bigcap_{U \in \mathcal{U}(S, \leq)} U \\ &\subset S \\ &\subset \uparrow S.\end{aligned}$$
     Hence $\uparrow S = S$.
   - $(\Longleftarrow)$:
     Suppose that $\uparrow S = S$. Then
$$\begin{aligned}S &= \uparrow S \\ &\in \mathcal{U}(S, \leq).\end{aligned}$$
2. Similar to (1).

□

Exercise 3.2.5.11. Let $(X, \leq)$ be a proset and $a \in X$. Then

1. $\uparrow a = \{x \in X : a \leq x\}$
2. $\downarrow a = \{x \in X : x \leq a\}$

Proof.

1. Define $U \subset X$ by $U := \{x \in X : a \leq x\}$.
   - Let $u \in U$ and $x \in X$. Suppose that $u \leq x$. Then
   $$\begin{aligned} a &\leq u \\ &\leq x. \end{aligned}$$
   Hence $x \in U$. Since $u \in U$ and $x \in X$ with $u \leq x$ are arbitrary, we have that for each $u \in U$ and $x \in X$, $u \leq x$ implies that $x \in U$. Hence U is an upper set. Since $a \leq a$, $a \in U$. By definition, $U \in \mathcal{U}(a)$. Therefore
   $$\begin{aligned} \uparrow a &= \bigcap_{U' \in \mathcal{U}(a)} U' \\ &\subset U. \end{aligned}$$
   - Let $x \in U$ and $U' \in \mathcal{U}(a)$. Since $x \in U$, $a \leq x$. Since $U' \in \mathcal{U}(a)$, U' is an upper set and $a \in U'$. Thus $x \in U'$. Since $U' \in \mathcal{U}(a)$ is arbitrary, we have that for each $U' \in \mathcal{U}(a)$, $x \in U'$. Thus
   $$\begin{aligned} x &\in \bigcap_{U' \in \mathcal{U}(a)} U' \\ &= \uparrow a. \end{aligned}$$
   Since $x \in U$ is arbitrary, we have that for each $x \in U$, $x \in \uparrow a$. Hence $U \subset \uparrow a$.

   Since $\uparrow a \subset U$ and $U \subset \uparrow a$, we have that $\uparrow a = U$.
2. Exercise 3.2.5.4 and (1) imply that
$$\begin{aligned} \downarrow(a, \leq) &= \uparrow(a, \leq^*) \\ &= \{x \in X : a \leq^* x\} \\ &= \{x \in X : x \leq a\}. \end{aligned}$$

□

Exercise 3.2.5.12. Let $(X, \leq)$ be a poset and $a, b \in X$.

1. If $\uparrow a = \uparrow b$, then $a \sim_{\leq} b$
2. If $\downarrow a = \downarrow b$, then $a \sim_{\leq} b$

Proof. 1. Suppose that $\uparrow a = \uparrow b$. Then
$$\begin{aligned} a &\in \uparrow a \\ &= \uparrow b. \end{aligned}$$
Hence $a \leq b$. Similarly, $b \leq a$. Hence $a \sim_{\leq} b$.

2.

□

Exercise 3.2.5.13. Let $(X, \leq)$ be a proset and $S_1, S_2 \subset X$. If $S_1 \subset S_2$, then

1. $\uparrow S_1 \subset \uparrow S_2$
2. $\downarrow S_1 \subset \downarrow S_2$

Proof. Suppose that $S_1 \subset S_2$.

1. Since $\uparrow S_2 \in \mathcal{U}(S_2)$, $S_2 \subset \uparrow S_2$ and $\uparrow S_2$ is an upper set. Since
$$\begin{aligned} S_1 &\subset S_2 \\ &\subset \uparrow S_2, \end{aligned}$$
we have that $\uparrow S_2 \in \mathcal{U}(S_1)$. Therefore
$$\begin{aligned} \uparrow S_1 &= \bigcap_{U \in \mathcal{U}(S_1)} U \\ &\subset \uparrow S_2. \end{aligned}$$
2. Part (1) and Exercise 3.2.5.9 implies that
$$\begin{aligned} \downarrow(S_1, \leq) &= \uparrow(S_1, \leq^*) \\ &\subset \uparrow(S_2, \leq^*) \\ &= \downarrow(S_2, \leq). \end{aligned}$$

□

Exercise 3.2.5.14. Let $(X, \leq)$ be a proset and $a, b \in X$. Then the following are equivalent:

1. $a \leq b$,
2. $\uparrow b \subset \uparrow a$,
3. $\downarrow a \subset \downarrow b$.

Proof.

1. (1) $\implies$ (2):
Suppose that $a \leq b$. Since $\uparrow a \in \mathcal{U}(a)$, we have that $a \in \uparrow a$ and $\uparrow a$ is an upper set. Since $a \leq b$, we have that $b \in \uparrow a$. Hence $\uparrow a \in \mathcal{U}(b)$. Therefore
$$\begin{aligned} \uparrow b &= \bigcap_{U \in \mathcal{U}(b)} U \\ &\subset \uparrow a. \end{aligned}$$
2. (2) $\implies$ (3):
Suppose that $\uparrow(b, \leq) \subset \uparrow(a, \leq)$. Exercise 3.2.5.11 then implies that
$$\begin{aligned} b &\in \uparrow(b, \leq) \\ &\subset \uparrow(a, \leq) \\ &= \{x \in X : a \leq x\}. \end{aligned}$$
Hence $a \leq b$. Thus $b \leq^* a$. Exercise 3.2.5.9 and part (1) $\implies$ (2) then imply that
$$\begin{aligned} \downarrow(a, \leq) &= \uparrow(a, \leq^*) \\ &\subset \uparrow(b, \leq^*) \\ &= \downarrow(b, \leq). \end{aligned}$$

3. (3) $\implies$ (1):
   Suppose that $\downarrow a \subset \downarrow b$. Exercise 3.2.5.11 then implies that

$$\begin{aligned} a &\in \downarrow a \\ &\subset \downarrow b \\ &= \{x \in X : x \leq b\}. \end{aligned}$$

   Hence $a \leq b$.

□

Exercise 3.2.5.15. Let $(X, \leq)$ be a proset and $S \subset X$. Then

1. $\uparrow S = \bigcup_{s \in S} \uparrow s$
2. $\downarrow S = \bigcup_{s \in S} \downarrow s$

Proof.

1. Define $U \subset X$ by $U := \bigcup_{s \in S} \uparrow s$.

   - Since for each $s \in S$, $\uparrow s$ is an upper set, Exercise 3.2.5.6 implies that U is an upper set. Let $s \in S$. Then

$$\begin{aligned} s &\in \uparrow s \\ &\subset \bigcup_{s \in S} \uparrow s \\ &= U. \end{aligned}$$

     Since $s \in S$ is arbitrary, we have that for each $s \in S$, $s \in U$. Hence $S \subset U$. Therefore $U \in \mathcal{U}(S)$ and

$$\begin{aligned} \uparrow S &= \bigcap_{U' \in \mathcal{U}(S)} U' \\ &\subset U. \end{aligned}$$

   - Let $x \in U$ and $U' \in \mathcal{U}(S)$. Since $x \in U$, there exists $s \in S$ such that $x \in \uparrow s$. Exercise 3.2.5.11 then implies that $s \leq x$. Since $U' \in \mathcal{U}(S)$, U' is an upper set and $S \subset U'$. Then

$$\begin{aligned} s &\in S \\ &\subset U'. \end{aligned}$$

     Since U' is an upper set and $s \leq x$, $x \in U'$. Since $U' \in \mathcal{U}(S)$ is arbitrary, we have that for each $U' \in \mathcal{U}(S)$, $x \in U'$. Thus

$$\begin{aligned} x &\in \bigcap_{U' \in \mathcal{U}(S)} U' \\ &= \uparrow S. \end{aligned}$$

     Since $x \in U$ is arbitrary, we have that for each $x \in U$, $x \in \uparrow S$. Hence $U \subset \uparrow S$.

   Since $\uparrow S \subset U$ and $U \subset \uparrow S$, we have that $\uparrow S = U$.

2. Similar to (1).

□

Exercise 3.2.5.16. Let $(X, \leq)$ be a proset and $(E_\alpha)_{\alpha \in A} \subset \mathcal{P}(X)$. Then

1. $\uparrow \bigcup_{\alpha \in A} E_\alpha = \bigcup_{\alpha \in A} \uparrow E_\alpha$
2. $\downarrow \bigcup_{\alpha \in A} E_\alpha = \bigcup_{\alpha \in A} \downarrow E_\alpha$

Proof.

1.
   - Let $x \in \uparrow \bigcup_{\alpha \in A} E_\alpha$. Exercise 3.2.5.15 implies that there exists $y \in \bigcup_{\alpha \in A} E_\alpha$ such that $x \in \uparrow y$. Then there exists $\alpha_0 \in A$ such that $y \in E_{\alpha_0}$. Since $\{y\} \subset E_{\alpha_0}$, Exercise 3.2.5.13 implies that $\uparrow y \subset \uparrow E_{\alpha_0}$. Therefore
   $$\begin{aligned} x &\in \uparrow y \\ &\subset \uparrow E_{\alpha_0} \\ &\subset \bigcup_{\alpha \in A} \uparrow E_\alpha. \end{aligned}$$
   Since $x \in \uparrow \bigcup_{\alpha \in A} E_\alpha$ is arbitrary, we have that for each $x \in \uparrow \bigcup_{\alpha \in A} E_\alpha$, $x \in \bigcup_{\alpha \in A} \uparrow E_\alpha$. Hence $\uparrow \bigcup_{\alpha \in A} E_\alpha \subset \bigcup_{\alpha \in A} \uparrow E_\alpha$.
   - Let $x \in \bigcup_{\alpha \in A} \uparrow E_\alpha$. Then there exists $\alpha_0 \in A$ such that $x \in \uparrow E_{\alpha_0}$. Exercise 3.2.5.15 implies that there exists $y \in E_{\alpha_0}$ such that $x \in \uparrow y$. Since
   $$\begin{aligned} \{y\} &\subset E_{\alpha_0} \\ &\subset \bigcup_{\alpha \in A} E_\alpha, \end{aligned}$$
   Exercise 3.2.5.13 implies that
   $$\begin{aligned} x &\in \uparrow y \\ &\subset \uparrow \bigcup_{\alpha \in A} E_\alpha. \end{aligned}$$
   Since $x \in \bigcup_{\alpha \in A} \uparrow E_\alpha$ is arbitrary, we have that for each $x \in \bigcup_{\alpha \in A} \uparrow E_\alpha$, $x \in \uparrow \bigcup_{\alpha \in A} E_\alpha$. Hence $\bigcup_{\alpha \in A} \uparrow E_\alpha \subset \uparrow \bigcup_{\alpha \in A} E_\alpha$.

   Since $\uparrow \bigcup_{\alpha \in A} E_\alpha \subset \bigcup_{\alpha \in A} \uparrow E_\alpha$ and $\bigcup_{\alpha \in A} \uparrow E_\alpha \subset \uparrow \bigcup_{\alpha \in A} E_\alpha$, we have that $\uparrow \bigcup_{\alpha \in A} E_\alpha = \bigcup_{\alpha \in A} \uparrow E_\alpha$.
2. Similar to (1).

□

Definition 3.2.5.17. Let $(X, \leq)$ be a proset and $a \in X$. Then a is said to be

- **$\leq$-maximal** if for each $x \in X$, $a \leq x$ implies that $x \sim_\leq a$.
- **$\leq$-minimal** if for each $x \in X$, $x \leq a$ implies that $x \sim_\leq a$.

Note 3.2.5.18. When the context is clear, we write "maximal" and "minimal" instead of "$\leq$-maximal" and "$\leq$-minimal" respectively.

Exercise 3.2.5.19. Let $(X, \leq)$ be a proset and $a \in X$. Then

1. a is maximal iff $\uparrow a = \pi_{X/\sim_\leq}(a)$
2. a is minimal iff $\downarrow a = \pi_{X/\sim_\leq}(a)$

Proof.

1.
   - $(\Longrightarrow)$:
     Suppose that a is maximal.

- Let $x \in \uparrow a$. Exercise 3.2.5.11 implies $a \leq x$. Since a is maximal, $x \sim_{\leq} a$. Thus $x \in \pi_{X/\sim_{\leq}}(a)$. Since $x \in \uparrow a$ is arbitrary, we have that for each $x \in \uparrow a$, $x \in \pi_{X/\sim_{\leq}}(a)$. Hence $\uparrow a \subset \pi_{X/\sim_{\leq}}(a)$.
- Let $x \in \pi_{X/\sim_{\leq}}(a)$. Then $a \sim_{\leq} x$. Hence $a \leq x$ and $x \leq a$. Since $a \leq x$, we have that $x \in \uparrow a$. Since $x \in \pi_{X/\sim_{\leq}}(a)$ is arbitrary, we have that for each $x \in \pi_{X/\sim_{\leq}}(a)$, $x \in \uparrow a$. Hence $\pi_{X/\sim_{\leq}}(a) \subset \uparrow a$.

Since $\uparrow a \subset \pi_{X/\sim_{\leq}}(a)$ and $\pi_{X/\sim_{\leq}}(a) \subset \uparrow a$, we have that $\uparrow a = \pi_{X/\sim_{\leq}}(a)$.

- ($\Longleftarrow$):
Suppose that $\uparrow a = \pi_{X/\sim_{\leq}}(a)$. Let $x \in X$. Suppose that $a \leq x$. Then

$$\begin{aligned} x &\in \uparrow a \\ &= \pi_{X/\sim_{\leq}}(a). \end{aligned}$$

Thus $x \sim_{\leq} a$. Since $x \in X$ with $a \leq x$ is arbitrary, we have that for each $x \in X$, $a \leq x$ implies that $x \sim_{\leq} a$. Hence a is maximal.

2. Similar to (1).

□

Definition 3.2.5.20. Let $(X, \leq)$ be a proset and $A \subset X$.

- Let $x \in X$. Then x is said to be a
  - **$\leq$-upper bound of** A if for each $a \in A$, $a \leq x$
  - **$\leq$-lower bound of** A if for each $a \in A$, $x \leq a$
- We define
  - $\mathrm{ub}(A, \leq) := \{x \in X : x \text{ is a } \leq\text{-upper bound of } A\}$
  - $\mathrm{lb}(A, \leq) := \{x \in X : x \text{ is a } \leq\text{-lower bound of } A\}$
- Then A is said to be
  - **$\leq$-bounded above** if $\mathrm{ub}(A, \leq) \neq \varnothing$
  - **$\leq$-bounded below** if $\mathrm{lb}(A, \leq) \neq \varnothing$

Note 3.2.5.21. When the context is clear, we write

- "upper bound" and "lower bound" instead of "$\leq$-upper bound" and "$\leq$-lower bound" respectively
- $\mathrm{ub}\, A$ and $\mathrm{lb}\, A$ in place of $\mathrm{ub}(A, \leq)$ and $\mathrm{lb}(A, \leq)$ respectively
- "bounded above" and "bounded below" instead of "$\leq$-bounded above" and "$\leq$-bounded below" respectively

Exercise 3.2.5.22. Let $(X, \leq)$ be a proset and $A \subset X$. Then

1. $\mathrm{ub}(A, \leq) = \mathrm{lb}(A, \leq^*)$
2. $\mathrm{ub}(A, \leq^*) = \mathrm{lb}(A, \leq)$

Proof.

1. We note that for each $x \in X$,

$$\begin{aligned} x \in \mathrm{ub}(A, \leq) &\iff \text{for each } a \in A,\ a \leq x \\ &\iff \text{for each } a \in A,\ x \leq^* a \\ &\iff x \in \mathrm{lb}(A, \leq^*). \end{aligned}$$

Hence $\mathrm{ub}(A, \leq) = \mathrm{lb}(A, \leq^*)$.

2. Similar to (1) ref.

□

Exercise 3.2.5.23. Let $(X, \leq)$ be a proset and $A \subset X$. Then

1. $\operatorname{ub} A = \bigcap_{x \in A} \uparrow x$
2. $\operatorname{lb} A = \bigcap_{x \in A} \downarrow x$

Proof.

1.
   - Let $a \in \operatorname{ub} A$ and $x \in A$. Then $x \leq a$. Hence $a \in \uparrow x$. Since $x \in A$ is arbitrary, we have that for each $x \in A$, $a \in \uparrow x$. Thus $a \in \bigcap_{x \in A} \uparrow x$. Since $a \in \operatorname{ub} A$ is arbitrary, we have that for each $a \in \operatorname{ub} A$, $a \in \bigcap_{x \in A} \uparrow x$. Hence $\operatorname{ub} A \subset \bigcap_{x \in A} \uparrow x$.
   - Let $a \in \bigcap_{x \in A} \uparrow x$ and $x_0 \in A$. Then $a \in \uparrow x_0$. Hence $x_0 \leq a$. Since $x_0 \in A$ is arbitrary, we have that for each $x_0 \in A$, $x_0 \leq a$. Hence $a \in \operatorname{ub} A$. Since $a \in \bigcap_{x \in A} \uparrow x$ is arbitrary, we have that for each $a \in \bigcap_{x \in A} \uparrow x$, $a \in \operatorname{ub} A$. Thus $\bigcap_{x \in A} \uparrow x \subset \operatorname{ub} A$.

   Since $\operatorname{ub} A \subset \bigcap_{x \in A} \uparrow x$ and $\bigcap_{x \in A} \uparrow x \subset \operatorname{ub} A$, we have that $\operatorname{ub} A = \bigcap_{x \in A} \uparrow x$.
2. Similar to (1).

□

Exercise 3.2.5.24. Let $(X, \leq)$ be a proset and $A, B \subset X$. If $A \subset B$, then

1. $\operatorname{ub} B \subset \operatorname{ub} A$
2. $\operatorname{lb} B \subset \operatorname{lb} A$

Proof. Suppose that $A \subset B$.

1. Exercise 3.2.5.23 implies that

$$\begin{aligned} \operatorname{ub} B &= \bigcap_{x \in B} \uparrow x \\ &\subset \bigcap_{x \in A} \uparrow x \\ &= \operatorname{ub} A. \end{aligned}$$

2. Similar to (1). ref

□

Exercise 3.2.5.25. Let $(X, \leq)$ be a proset and $(E_\alpha)_{\alpha \in A} \subset \mathcal{P}(X)$. Then $\operatorname{ub} \bigcup_{\alpha \in A} E_\alpha = \bigcap_{\alpha \in A} \operatorname{ub} E_\alpha$.

Proof. Define $E_\vee \subset X$ by $E := \bigcup_{\alpha \in A} E_\alpha$. Exercise 3.2.5.23 then implies that

$$\begin{aligned} \operatorname{ub} \bigcup_{\alpha \in A} E_\alpha &= \bigcap_{x \in E} \uparrow x \\ &= \bigcap_{\alpha \in A} \left[\bigcap_{x \in E_\alpha} \uparrow x \right] \\ &= \bigcap_{\alpha \in A} \operatorname{ub} E_\alpha. \end{aligned}$$

□

3.2.6 Monotone Maps

Definition 3.2.6.1. Let $(X, \leq_X)$ $(Y, \leq_Y)$ be prosets and $f : X \to Y$. Then f is said to be

- $(\leq_X, \leq_Y)$**-order preserving** or $(\leq_X, \leq_Y)$**-monotone** if for each $a, b \in X$, $a \leq_X b$ implies that $f(a) \leq_Y f(b)$.
- $(\leq_X, \leq_Y)$**-order reversing** or $(\leq_X, \leq_Y)$**-antitone** if for each $a, b \in X$, $a \leq_X b$ implies that $f(b) \leq_Y f(a)$.

Note 3.2.6.2. When the context is clear we say that f is "monotone" and "antitone" instead of "$(\leq_X, \leq_Y)$-monotone" and "$(\leq_X, \leq_Y)$-antitone" respectively.

Exercise 3.2.6.3. Let $(X, \leq_X)$ $(Y, \leq_Y)$ be prosets and $f : X \to Y$. Then

1. the folloing are equivalent:
   (a) f is $(\leq_X, \leq_Y)$-antitone
   (b) f is $(\leq_X, \leq_Y^*)$-monotone
   (c) f is $(\leq_X^*, \leq_Y)$-monotone
   (d) f is $(\leq_X^*, \leq_Y^*)$-antitone
2. the folloing are equivalent:
   (a) f is $(\leq_X, \leq_Y)$-monotone
   (b) f is $(\leq_X, \leq_Y^*)$-antitone
   (c) f is $(\leq_X^*, \leq_Y)$-antitone
   (d) f is $(\leq_X^*, \leq_Y^*)$-monotone

Proof.

1.
   - $(a) \implies (b)$:
     Suppose that f is $(\leq_X, \leq_Y)$-antitone. Let $a, b \in X$. Then
     $$\begin{aligned} a \leq_X b &\implies f(b) \leq_Y f(a) \\ &\iff f(a) \leq_Y^* f(b). \end{aligned}$$
     Since $a, b \in X$ are arbitrary, we have that for each $a, b \in X$, $a \leq_X b$ implies that $f(a) \leq_Y^* f(b)$. Hence f is $(\leq_X, \leq_Y^*)$-monotone.
   - $(b) \implies (c)$:
     Suppose that f is $(\leq_X, \leq_Y^*)$-monotone. Let $a, b \in X$. Then
     $$\begin{aligned} a \leq_X^* b &\iff b \leq_X^* a \\ &\implies f(b) \leq_Y^* f(a) \\ &\iff f(a) \leq_Y f(b). \end{aligned}$$
     Since $a, b \in X$ are arbitrary, we have that for each $a, b \in X$, $a \leq_X^* b$ implies that $f(a) \leq_Y f(b)$. Hence f is $(\leq_X^*, \leq_Y)$-monotone.
   - $(c) \implies (d)$:
     Suppose that f is $(\leq_X^*, \leq_Y)$-monotone. Let $a, b \in X$. Then
     $$\begin{aligned} a \leq_X^* b &\implies f(a) \leq_Y f(b) \\ &\iff f(b) \leq_Y^* f(a). \end{aligned}$$
     Since $a, b \in X$ are arbitrary, we have that for each $a, b \in X$, $a \leq_X^* b$ implies that $f(b) \leq_Y^* f(a)$. Hence f is $(\leq_X^*, \leq_Y^*)$-antitone.

- $(d) \implies (a)$:
  Suppose that f is $(\leq_X^*, \leq_Y^*)$-antitone. Let $a, b \in X$. Then

$$\begin{aligned} a \leq_X b &\iff b \leq_X^* a \\ &\implies f(a) \leq_Y^* f(b) \\ &\iff f(b) \leq_Y f(a). \end{aligned}$$

  Since $a, b \in X$ are arbitrary, we have that for each $a, b \in X$, $a \leq_X b$ implies that $f(b) \leq_Y f(a)$. Hence f is $(\leq_X, \leq_Y)$-antitone.

2. Set $\leq_Y' := \leq_Y^*$. Part (1) implies that

$$\begin{aligned} f \text{ is } (\leq_X, (\leq_Y')^{\mathrm{op}})\text{-monotone} &\iff f \text{ is } (\leq_X, \leq_Y')\text{-antitone} \quad (\text{by } 1(b) \iff 1(a)) \\ &\iff f \text{ is } (\leq_X^*, (\leq_Y')^{\mathrm{op}})\text{-antitone} \quad (\text{by } 1(a) \iff 1(d)) \\ &\iff f \text{ is } (\leq_X^*, \leq_Y')\text{-monotone} \quad (\text{by } 1(d) \iff 1(c)). \end{aligned}$$

   Since $(\leq_Y')^{\mathrm{op}}) = \leq_Y$, we have that

$$\begin{aligned} f \text{ is } (\leq_X, \leq_Y)\text{-monotone} &\iff f \text{ is } (\leq_X, \leq_Y^*)\text{-antitone} \\ &\iff f \text{ is } (\leq_X^*, \leq_Y)\text{-antitone} \\ &\iff f \text{ is } (\leq_X^*, \leq_Y^*)\text{-monotone.} \end{aligned}$$

□

Exercise 3.2.6.4. Let $(X, \leq_X)$ $(Y, \leq_Y)$ be prosets and $f : X \to Y$. Then f is $(\leq_X, \leq_Y)$-monotone iff f is $(\leq_X^*, \leq_Y^*)$-monotone.

Proof.

- $(\implies)$:
  Suppose that f is $(\leq_X, \leq_Y)$-monotone. Let $a, b \in X$. Suppose that $a \leq_X^* b$. Then $b \leq_X a$. Hence $f(b) \leq_Y f(a)$. Thus $f(a) \leq_Y^* f(b)$. Since $a, b \in X$ with $a \leq_X^* b$ are arbitrary, we have that for each $a, b \in X$, $a \leq_X^* b$ implies that $f(a) \leq_Y^* f(b)$. Therefore f is $(\leq_X^*, \leq_Y^*)$-monotone.

- $(\impliedby)$:
  Suppose that f is $(\leq_X^*, \leq_Y^*)$-monotone. Since $(\leq_X^*)^{\mathrm{op}} = \leq_X$, the previous part implies that f is $(\leq_X, \leq_Y)$-monotone.

□

Exercise 3.2.6.5. Let $(X, \leq_X)$ $(Y, \leq_Y)$ be prosets and $f : X \to Y$. Then for each $U \subset Y$, U is a $\leq_Y$-upper set implies that $f^{-1}(U)$ is a $\leq_X$-upper set iff for each $L \subset Y$, L is a $\leq_Y$-lower set implies that $f^{-1}(L)$ is a $\leq_X$-lower set.

Proof.

- $(\implies)$:
  Suppose that for each $U \subset Y$, U is a $\leq_Y$-upper set implies that $f^{-1}(U)$ is a $\leq_X$-upper set. Let $L \subset Y$. Suppose that L is a $\leq_Y$-lower set. Exercise 3.2.5.5 then implies that L^c is a $\leq_Y$-upper set. By assumption, $f^{-1}(L^c)$ is a $\leq_X$-upper set. Another application of Exercise 3.2.5.5 implies that $f^{-1}(L^c)^c$ is a $\leq_X$-lower set. Since

$$\begin{aligned} f^{-1}(L) &= [f^{-1}(L)^c]^c \\ &= f^{-1}(L^c)^c, \end{aligned}$$

  we have that $f^{-1}(L)$ is a $\leq_X$-lower set. Since $L \subset Y$ such that L is a $\leq_Y$-lower set is arbitrary, we have that for each $L \subset Y$, L is a $\leq_Y$-lower set implies that $f^{-1}(L)$ is a $\leq_X$-lower set.

- $(\impliedby)$:
  Similar to $(\implies)$.

□

Exercise 3.2.6.6. Let $(X, \leq_X)$ $(Y, \leq_Y)$ be posets and $f : X \to Y$. If f is $(\leq_X, \leq_Y)$-monotone, then for each $U \subset Y$, U is a $\leq_Y$-upper set implies that $f^{-1}(U)$ is a $\leq_X$-upper set.

Proof. Suppose that f is $(\leq_X, \leq_Y)$-monotone. Let $U \subset Y$. Suppose that U is a $\leq_Y$-upper set. Let $a \in f^{-1}(U)$ and $x \in X$. Suppose that $a \leq x$. Since $a \in f^{-1}(U)$, $f(a) \in U$. Since f is monotone, $f(a) \leq f(x)$. Since U is a $\leq_Y$-upper set and $f(a) \in U$, we have that $f(x) \in U$. Hence $x \in f^{-1}(U)$. Since $a \in f^{-1}(U)$ and $x \in X$ with $a \leq x$ are arbitrary, we have that for each $a \in f^{-1}(U)$ and $x \in X$, $a \leq x$ implies that $x \in f^{-1}(U)$. Thus $f^{-1}(U)$ is a $\leq_X$-upper set. Since $U \subset Y$ such that U is a $\leq_Y$-upper set is arbitrary, we have that for each $U \subset Y$, U is a $\leq_Y$-upper set implies that $f^{-1}(U)$ is a $\leq_X$-upper set. □

Exercise 3.2.6.7. Let $(X, \leq_X)$ $(Y, \leq_Y)$ be posets, $f : X \to Y$ and $A \subset X$. If f is $(\leq_X, \leq_Y)$-monotone, then

1. $f(\mathrm{ub}(A, \leq_X)) \subset \mathrm{ub}(f(A), \leq_Y)$,
2. $f(\mathrm{lb}(A, \leq_X)) \subset \mathrm{lb}(f(A), \leq_Y)$

Proof. Suppose that f is $(\leq_X, \leq_Y)$-monotone.

1. Let $x \in \mathrm{ub}(A, \leq_X)$ and $y_0 \in f(A)$. Then there exists $x_0 \in A$ such that $f(x_0) = y$. Since $x \in \mathrm{ub}(A, \leq_X)$, $x_0 \leq x$. Since f is $(\leq_X, \leq_Y)$-monotone,

$$\begin{aligned} y_0 &= f(x_0) \\ &\leq f(x). \end{aligned}$$

   Since $y_0 \in f(A)$ is arbitrary, we have that for each $y_0 \in f(A)$, $y_0 \leq f(x)$. Therefore $f(x) \in \mathrm{ub}(F(A), \leq_Y)$. Since $x \in \mathrm{ub}(A, \leq_X)$ is arbitrary, ew have that for each $x \in \mathrm{ub}(A, \leq_X)$, $f(x) \in \mathrm{ub}(F(A), \leq_Y)$. Hence $f(\mathrm{ub}(A, \leq_X)) \subset \mathrm{ub}(f(A), \leq_Y)$.

2. Exercise 3.2.5.22 and the prev part imply that

$$\begin{aligned} f(\mathrm{lb}(A, \leq_X)) &= f(\mathrm{ub}(A, \leq_X^*)) \\ &\subset \mathrm{ub}(f(A), \leq_Y^*) \\ &= \mathrm{lb}(f(A), \leq_Y). \end{aligned}$$

□

3.2.7 Galois Connections

Definition 3.2.7.1. Let $(X, \leq_X)$ and $(Y, \leq_Y)$ be posets, $f : X \to Y$ and $g : Y \to X$. Then

- (f, g) is said to be a $(\leq_X, \leq_Y)$**-monotone Galois connection** if for each $x \in X$ and $y \in Y$,

$$f(x) \leq_Y y \text{ iff } x \leq_X g(y),$$

- (f, g) is said to be a $(\leq_X, \leq_Y)$**-antitone Galois connection** if for each $x \in X$ and $y \in Y$,

$$f(x) \leq_Y y \text{ iff } g(y) \leq_X x,$$

- f is said to be a $(\leq_X, \leq_Y)$**-left adjoint of** g if (f, g) is a $(\leq_X, \leq_Y)$-monotone Galois connection,
- g is said to be a $(\leq_X, \leq_Y)$**-right adjoint of** f if (f, g) is a $(\leq_X, \leq_Y)$-monotone Galois connection.

Definition 3.2.7.2. Let $(X, \leq_X)$ and $(Y, \leq_Y)$ be posets, $f : X \to Y$ and $g : Y \to X$. Then

- f is said to be a **left adjoint of** g if (f, g) is a monotone Galois connection,
- g is said to be a **right adjoint of** f if (f, g) is a monotone Galois connection,

Note 3.2.7.3. When the context is clear, we say

- "monotone (resp. antitone) Galois connection" instead of "$(\leq_X, \leq_Y)$-monotone (resp. $(\leq_X, \leq_Y)$-antitone) Galois connection"
- "left (resp. right) adjoint" instead of "$(\leq_X, \leq_Y)$-left (resp. $(\leq_X, \leq_Y)$-right) adjoint"

Exercise 3.2.7.4. Let $(X, \leq_X)$ and $(Y, \leq_Y)$ be prosets, $f : X \to Y$ and $g : Y \to X$. Then

1. (f, g) is a $(\leq_X, \leq_Y)$-monotone Galois connection iff (g, f) is a $(\leq_Y^*, \leq_X^*)$-monotone Galois connection
2. (f, g) is a $(\leq_X, \leq_Y)$-antitone Galois connection iff (f, g) is a $(\leq_X^*, \leq_Y)$-monotone Galois connection
3. (f, g) is a $(\leq_X, \leq_Y)$-antitone Galois connection iff (g, f) is a $(\leq_Y, \leq_X)$-antitone Galois connection

Proof.

1. - $(\implies)$:
   Suppose that (f, g) is a $(\leq_X, \leq_Y)$-monotone Galois connection. Let $y \in Y$ and $x \in X$. Then by assumption,
   $$\begin{aligned} g(y) \leq_X^* x &\iff x \leq_X g(y) \\ &\iff f(x) \leq_Y y \\ &\iff y \leq_Y^* f(x). \end{aligned}$$
   Since $y \in Y$ and $x \in X$ are arbitrary, we have that for each $y \in Y$ and $x \in X$, $g(y) \leq_X^* x$ iff $y \leq_Y^* f(x)$. Therefore (g, f) is a $(\leq_Y^*, \leq_X^*)$-monotone Galois connection.
   - $(\impliedby)$:
   Suppose that (g, f) is a $(\leq_Y^*, \leq_X^*)$-monotone Galois connection. Let $x \in X$ and $y \in Y$. Then by assumption,
   $$\begin{aligned} f(x) \leq_Y y &\iff y \leq_Y^* f(x) \\ &\iff g(y) \leq_X^* x \\ &\iff x \leq_X g(y). \end{aligned}$$
   Since $x \in X$ and $y \in Y$ are arbitrary, we have that for each $x \in X$ and $y \in Y$, $f(x) \leq_Y y$ iff $x \leq_X g(y)$. Therefore (f, g) is a $(\leq_X, \leq_Y)$-monotone Galois connection.
2. - $(\implies)$:
   Suppose that (f, g) is a $(\leq_X, \leq_Y)$-antitone Galois connection. Let $x \in X$ and $y \in Y$. Then by assumption,
   $$\begin{aligned} f(x) \leq_Y y &\iff g(y) \leq_X x \\ &\iff x \leq_X^* g(y). \end{aligned}$$
   Since $x \in X$ and $y \in Y$ are arbitrary, we have that for each $x \in X$ and $y \in Y$, $f(x) \leq_Y y$ iff $x \leq_X^* g(y)$. Therefore (f, g) is a $(\leq_X^*, \leq_Y)$-monotone Galois connection.
   - $(\impliedby)$:
   Suppose that (f, g) is a $(\leq_X^*, \leq_Y)$-monotone Galois connection. Let $x \in X$ and $y \in Y$. Then by assumption,
   $$\begin{aligned} f(x) \leq_Y y &\iff x \leq_X^* g(y) \\ &\iff g(y) \leq_X x. \end{aligned}$$
   Since $x \in X$ and $y \in Y$ are arbitrary, we have that for each $x \in X$ and $y \in Y$, $f(x) \leq_Y y$ iff $g(y) \leq_X x$. Therefore (f, g) is a $(\leq_X, \leq_Y)$-antitone Galois connection.
3. By (1) and (2),
   $$\begin{aligned} (f, g) \text{ is a } (\leq_X, \leq_Y)\text{-antitone Galois connection} &\iff (f, g) \text{ is a } (\leq_X^*, \leq_Y)\text{-monotone Galois connection} \\ &\iff (g, f) \text{ is a } (\leq_Y^*, (\leq_X^*)^{\mathrm{op}})\text{-monotone Galois connection} \\ &\iff (g, f) \text{ is a } (\leq_Y^*, \leq_X)\text{-monotone Galois connection} \\ &\iff (g, f) \text{ is a } (\leq_Y, \leq_X)\text{-antitone Galois connection.} \end{aligned}$$

□

Exercise 3.2.7.5. Let $(X, \leq_X)$ and $(Y, \leq_Y)$ be prosets, $f : X \to Y$ and $g : Y \to X$. Then

1. (f, g) is a $(\leq_X, \leq_Y)$-monotone Galois connection iff
   (a) i. $\mathrm{id}_X \leq_X g \circ f$
   ii. $f \circ g \leq_Y \mathrm{id}_Y$
   (b) f is $(\leq_X, \leq_Y)$-monotone and g is $(\leq_Y, \leq_X)$-monotone
2. (f, g) is a $(\leq_X, \leq_Y)$-antitone Galois connection iff
   (a) i. $g \circ f \leq_X \mathrm{id}_X$
   ii. $f \circ g \leq_Y \mathrm{id}_Y$
   (b) f is $(\leq_X, \leq_Y)$-antitone and g is $(\leq_Y, \leq_X)$-antitone

Proof.

1. • $(\Longrightarrow)$:
   Suppose that (f, g) us a $(\leq_X, \leq_Y)$-monotone Galois connection.
   (a) i. Let $x \in X$. Define $y \in Y$ by $y := f(x)$. Since $f(x) \leq_Y f(x)$, we have that $f(x) \leq_Y y$. Therefore
   $$\begin{aligned} x &\leq_X g(y) \\ &= g \circ f(x). \end{aligned}$$
   Since $x \in X$ is arbitrary, we have that for each $x \in X$, $x \leq_X g \circ f(x)$. Hence $\mathrm{id}_X \leq_X g \circ f$.
   ii. Let $y \in Y$. the previous ex implies that (g, f) is a $(\leq_Y^*, \leq_X^*)$-monotone Galois connection. Part $1(a)$ then implies that for each $y \in Y$, $y \leq_Y^* f \circ g(y)$. Therefore for each $y \in Y$, $f \circ g(y) \leq_Y y$. Hence $f \circ g \leq_Y \mathrm{id}_Y$.
   (b) – Let $a, b \in X$. Define $x \in X$ and $y \in Y$ by $x := a$ and $y := f(b)$. Suppose that $a \leq_X b$. Part $1(a)$ implies that
   $$\begin{aligned} x &= a \\ &\leq_X b \\ &\leq_X g(f(x)) \\ &= g(y). \end{aligned}$$
   Since (f, g) is a $(\leq_X, \leq_Y)$-monotone Galois connection, we have that
   $$\begin{aligned} f(a) &= f(x) \\ &\leq_Y y \\ &= f(b). \end{aligned}$$
   Since $a, b \in X$ with $a \leq_X b$ are arbitrary, we have that for each $a, b \in X$, $a \leq_X b$ implies that $f(a) \leq_Y f(b)$. Thus f is $(\leq_X, \leq_Y)$-monotone.
   – Since (f, g) is a $(\leq_X, \leq_Y)$-monotone Galois connection, a previous ex implies that (g, f) is a $(\leq_Y^*, \leq_X^*)$-monotone Galois connection. The previous part implies that g is $(\leq_Y^*, \leq_X^*)$-monotone. Exercise 3.2.6.3 then implies that g is $(\leq_Y, \leq_X)$-monotone.

   • $(\Longleftarrow)$:
   Suppose that
   (a) i. for each $x \in X$, $x \leq_X g \circ f(x)$
   ii. for each $y \in Y$, $f \circ g(y) \leq_Y y$
   (b) f is $(\leq_X, \leq_Y)$-monotone and g is $(\leq_Y, \leq_X)$-monotone.
   Let $x \in X$ and $y \in Y$.

- ($\Longrightarrow$):
  Suppose that $f(x) \leq_Y y$. Then by (a)i and (a)ii,

$$\begin{aligned} x &\leq_X g \circ f(x) \\ &\leq_X g(y). \end{aligned}$$

- ($\Longleftarrow$):
  Suppose that $x \leq_X g(y)$. Then by (a)i and (a)ii,

$$\begin{aligned} f(x) &\leq_Y f \circ g(y) \\ &\leq_Y y. \end{aligned}$$

Since $x \in X$ and $y \in Y$ are arbitrary, we have that for each $x \in X$ and $y \in Y$,

$$f(x) \leq_Y y \text{ iff } f(x) \leq_Y y.$$

Therefore (f, g) is a $(\leq_X, \leq_Y)$-Galois connection.

2. the previous ex implies that (f, g) is a $(\leq_X, \leq_Y)$-antitone Galois connection iff (f, g) is a $(\leq_X^*, \leq_Y)$-monotone Galois connection. Part (1) implies that (f, g) is a $(\leq_X^*, \leq_Y)$-monotone Galois connection iff

   (a) i. for each $x \in X$, $x \leq_X^* g \circ f(x)$
   ii. for each $y \in Y$, $f \circ g(y) \leq_Y y$

   (b) f is $(\leq_X^*, \leq_Y)$-monotone and g is $(\leq_Y, \leq_X^*)$-monotone.

   two previous exercise about monotonic functions and the dual order then imply that (f, g) is a $(\leq_X, \leq_Y)$-antitone Galois connection iff

   (a) i. for each $x \in X$, $g \circ f(x) \leq_X x$
   ii. for each $y \in Y$, $f \circ g(y) \leq_Y y$

   (b) f is $(\leq_X, \leq_Y)$-antitone and g is $(\leq_Y, \leq_X)$-antitone.

□

Exercise 3.2.7.6. Let $(X, \leq_X)$ and $(Y, \leq_Y)$ be prosets, $f : X \to Y$ and $g : Y \to X$. If (f, g) is a $(\leq_X, \leq_Y)$-monotone Galois connection or (f, g) is a $(\leq_X, \leq_Y)$-antitone Galois connection, then $f \sim_{\leq_Y} f \circ g \circ f$ and $g \sim_{\leq_X} g \circ f \circ g$.

Proof.

- Suppose that (f, g) a $(\leq_X, \leq_Y)$-monotone Galois connection.
  - Let $x \in X$.
    * Define $a \in X$ and $b \in Y$ by $a := g \circ f(x)$ and $b := f(x)$. Since

$$\begin{aligned} g \circ f(x) \leq g \circ f(x) &\iff a \leq_X g(b) \\ &\iff f(a) \leq_Y b \\ &\iff f \circ g \circ f(x) \leq_Y f(x), \end{aligned}$$

      we have that $f \circ g \circ f(x) \leq_Y f(x)$.
    * The prev ex implies that $x \leq_X g \circ f(x)$ and f is $(\leq_X, \leq_Y)$-monotone. Therefore $f(x) \leq_Y f \circ g \circ f(x)$.

    Since $f \circ g \circ f(x) \leq_Y f(x)$ and $f(x) \leq_Y f \circ g \circ f(x)$, we have that $f \circ g \circ f(x) \sim_{\leq_Y} f(x)$, Since $x \in X$ is arbitrary, we have that for each $x \in X$, $f \circ g \circ f(x) \sim_{\leq_Y} f(x)$. Therefore $f \sim_{\leq_Y} f \circ g \circ f$.
  - Let $y \in Y$.

* Define $a \in X$ and $b \in Y$ by $a := g(y)$ and $b := f \circ g(y)$. Since

$$\begin{aligned} f \circ g(x) \leq_Y f \circ g(x) &\iff f(a) \leq_Y b \\ &\iff a \leq_X g(b) \\ &\iff g(y) \leq_X g \circ f \circ g(y), \end{aligned}$$

we have that $g(y) \leq_X g \circ f \circ g(y)$.

* The prev ex implies that $f \circ g(y) \leq_Y y$ and g is $(\leq_Y, \leq_X)$-monotone. Therefore $g \circ f \circ g(y) \leq_X g(y)$.

Since $g(y) \leq_X g \circ f \circ g(y)$ and $g \circ f \circ g(y) \leq_X g(y)$, we have that $g(y) \sim_{\leq_X} g \circ f \circ g(y)$, Since $y \in Y$ is arbitrary, we have that for each $y \in Y$, $g(y) \sim_{\leq_X} g \circ f \circ g(y)$. Therefore $g \sim_{\leq_X} g \circ f \circ g$.

- Suppose that (f, g) a $(\leq_X, \leq_Y)$-antitone Galois connection. The prev ex then implies that (f, g) is a $(\leq_X^*, \leq_Y)$-monotone Galois connection. Part (1) then implies that $f \sim_{\leq_Y} f \circ g \circ f$ and $g \sim_{\leq_X^*} g \circ f \circ g$. Make previous ex that $x \sim_{\leq} y$ iff $x \sim_{\leq^*} y$ then implies that $g \sim_{\leq_X} g \circ f \circ g$.

FINSIH!! □

Exercise 3.2.7.7. Let $(X, \leq_X)$ and $(Y, \leq_Y)$ be prosets, $f : X \to Y$ and $g : Y \to X$.

1. If (f, g) is a $(\leq_X, \leq_Y)$-monotone Galois connection, then
   (a) for each $A \subset X$, $\mathrm{ub}(f(A), \leq_Y) = g^{-1}(\mathrm{ub}(A, \leq_X))$,
   (b) for each $B \subset Y$, $\mathrm{lb}(g(B), \leq_X) = f^{-1}(\mathrm{lb}(B, \leq_Y))$.
2. If (f, g) is a $(\leq_X, \leq_Y)$-antitone Galois connection, then
   (a) for each $A \subset X$, $\mathrm{ub}(f(A), \leq_Y) = g^{-1}(\mathrm{lb}(A, \leq_X))$,
   (b) for each $B \subset Y$, $\mathrm{ub}(g(B), \leq_X) = f^{-1}(\mathrm{lb}(B, \leq_Y))$.

Proof. Since (f, g) is a $(\leq_X, \leq_Y)$-monotone Galois connection, a prev ex implies that f is $(\leq_X, \leq_Y)$-monotonoe, g is $(\leq_Y, \leq_X)$-monotone, $\mathrm{id}_X \leq_X g \circ f$ and $f \circ g \leq_Y \mathrm{id}_Y$.

1. Suppose that (f, g) is a $(\leq_X, \leq_Y)$-monotone Galois connection.
   (a) Let $A \subset X$.
   - Let $y \in \mathrm{ub}(f(A), \leq_Y)$ and $a \in A$. Then $f(a) \leq_Y y$. Hence

$$\begin{aligned} a &\leq_X g \circ f(a) \\ &= g(f(a)) \\ &\leq_X g(y). \end{aligned}$$

   Since $a \in A$ is arbitrary, we have that for each $a \in A$, $a \leq_X g(y)$. Thus $g(y) \in \mathrm{ub}(A, \leq_X)$. Hence $y \in g^{-1}(\mathrm{ub}(A, \leq_X))$. Since $y \in \mathrm{ub}(f(A), \leq_Y)$ is arbitrary, we have that for each $y \in \mathrm{ub}(f(A), \leq_Y)$, $y \in g^{-1}(\mathrm{ub}(A, \leq_X))$. Thus $\mathrm{ub}(f(A), \leq_Y) \subset g^{-1}(\mathrm{ub}(A, \leq_X))$.
   - Let $y \in g^{-1}(\mathrm{ub}(A, \leq_X))$. Then $g(y) \in \mathrm{ub}(A, \leq_X)$. Let $b \in f(A)$. Then there exists $a \in A$ such that $b = f(a)$. Then $a \leq_X g(y)$. Therefore

$$\begin{aligned} b &= f(a) \\ &\leq_Y f(g(y)) \\ &= f \circ g(y) \\ &= \leq_Y y. \end{aligned}$$

   Since $b \in f(A)$ is arbitrary, we have that for each $b \in f(A)$, $b \leq_Y y$. Thus $y \in \mathrm{ub}(f(A), \leq_Y)$. Since $y \in g^{-1}(\mathrm{ub}(A, \leq_X))$ is arbitrary, we have that for each $y \in g^{-1}(\mathrm{ub}(A, \leq_X))$, $y \in \mathrm{ub}(f(A), \leq_Y)$. Hence $g^{-1}(\mathrm{ub}(A, \leq_X)) \subset \mathrm{ub}(f(A), \leq_Y)$.

Since $\mathrm{ub}(f(A), \leq_Y) \subset g^{-1}(\mathrm{ub}(A, \leq_X))$ and $g^{-1}(\mathrm{ub}(A, \leq_X)) \subset \mathrm{ub}(f(A), \leq_Y)$, we have that

$$\mathrm{ub}(f(A), \leq_Y) = g^{-1}(\mathrm{ub}(A, \leq_X)).$$

(b) Since (f, g) is a $(\leq_X, \leq_Y)$-monotone Galois connection, a prev rex implies that (g, f) is a $(\leq_Y^*, \leq_X^*)$-monotone Galois connection. The previous part (ref) and Exercise 3.2.5.22 then imply that for each $B \subset Y$,

$$\begin{aligned}\mathrm{lb}(g(B), \leq_X) &= \mathrm{ub}(g(B), \leq_X^*)\\ &= f^{-1}(\mathrm{ub}(B, \leq_Y^*))\\ &= f^{-1}(\mathrm{lb}(B, \leq_Y)).\end{aligned}$$

2. Suppose that (f, g) is a $(\leq_X, \leq_Y)$-antitone Galois connection. A prev ex then implies that (f, g) is a $(\leq_X^*, \leq_Y)$-monotone Galois connection and (g, f) is a $(\leq_Y^*, \leq_X)$-monotone Galois connection. The prev part implies that

   (a) for each $A \subset X$,

$$\begin{aligned}\mathrm{ub}(f(A), \leq_Y) &= g^{-1}(\mathrm{ub}(A, \leq_X^*))\\ &= g^{-1}(\mathrm{lb}(A, \leq_X)).\end{aligned}$$

   (b) for each $B \subset Y$,

$$\begin{aligned}\mathrm{ub}(g(B), \leq_X) &= f^{-1}(\mathrm{ub}(B, \leq_Y^*))\\ &= f^{-1}(\mathrm{lb}(B, \leq_Y)).\end{aligned}$$

□

3.2.8 Suprema and Infima

relabel ex/def references

Definition 3.2.8.1. Let $(X, \leq)$ be a proset and $A \subset X$.

- Let $x \in X$. Then x is said to be a
  - **$\leq$-supremum of** A or **$\leq$-least upper bound of** A, if
    1. $x \in \mathrm{ub}(A, \leq)$
    2. for each $y \in \mathrm{ub}(A, \leq)$, $x \leq y$
  - **$\leq$-infimum of** A or **$\leq$-greatest lower bound of** A, if
    1. $x \in \mathrm{lb}(A, \leq)$
    2. for each $y \in \mathrm{lb}(A, \leq)$, $y \leq x$.
-
  - We define $\mathrm{supSet}(A, \leq) \subset X$ by $\mathrm{supSet}(A, \leq) := \{x \in X : x \text{ is a } \leq\text{-supremum of } A\}$
  - We define $\mathrm{infSet}(A, \leq) \subset X$ by $\mathrm{infSet}(A, \leq) := \{x \in X : x \text{ is an } \leq\text{-infimum of } A\}$

Note 3.2.8.2. When the context is clear,

- we say "supremum" and "infimum" in place of "$\leq$-supremum" and "$\leq$-infimum" respectively,
- we write supSet A and infSet A in place of $\mathrm{supSet}(A, \leq)$ and $\mathrm{infSet}(A, \leq)$ respectively.

Exercise 3.2.8.3. Let $(X, \leq)$ be a poset and $A \subset X$. Then

1. for each $x, y \in \sup A$, $x \sim_\leq y$.
2. for each $x, y \in \inf A$, $x \sim_\leq y$.

Proof.

1. Let $x, y \in \sup A$. Then $x, y \in \operatorname{ub} A$. Since $x \in \sup A$ and $y \in \operatorname{ub} A$, $x \leq y$. Since $y \in \sup A$ and $x \in \operatorname{ub} A$, $y \leq x$. Hence $x \sim_{\leq} y$.
2. Similar to (1).

□

Exercise 3.2.8.4. Let $(X, \leq)$ be a poset and $A \subset X$. Then

1. $\operatorname{supSet} A = \operatorname{ub} A \cap \operatorname{lb}(\operatorname{ub} A)$
2. $\operatorname{infSet} A = \operatorname{lb} A \cap \operatorname{ub}(\operatorname{lb} A)$

Proof.

1. We note that for each $x \in X$,

$$\begin{aligned} x \in \operatorname{supSet} A &\iff x \in \operatorname{ub} A \text{ and for each } y \in \operatorname{ub} A,\ x \leq y \\ &\iff x \in \operatorname{ub} A \text{ and } x \in \operatorname{lb}(\operatorname{ub} A) \\ &\iff x \in \operatorname{ub} A \cap \operatorname{lb}(\operatorname{ub} A). \end{aligned}$$

   Therefore $\operatorname{supSet} A = \operatorname{ub} A \cap \operatorname{lb}(\operatorname{ub} A)$.

2. Similar to (1).

□

3.3 Posets

3.3.1 Introduction

Definition 3.3.1.1. Poset:
Let X be a set and $\leq \subset X \times X$ a binary relation on X. Then

- $\leq$ is said to be a **partial order on** X if
  1. $\leq$ is a preorder on X
  2. for each $a, b \in X$, $a \leq b$ and $b \leq a$ implies that $a = b$,
- $(X, \leq)$ is $(X, \leq)$ is said to be a **partially ordered set** or **poset** if $\leq$ is a partial ordering on X.

Exercise 3.3.1.2. Let $(X, \leq)$ be a poset and $a, b \in X$. Then $a \sim_{\leq} b$ iff $a = b$.

Proof.

- $(\implies)$:
  Suppose that $a \sim_{\leq} b$. Then $a \leq b$ and $b \leq a$. Since $\leq$ is a partial order, $a = b$.
- $(\impliedby)$:
  Since $\sim_{\leq}$ is reflexive, $a = b$ implies that $a \sim_{\leq} b$.

□

Definition 3.3.1.3. Let $(X, \leq_X)$ be a proset. Set $Y := X/\sim_{\leq_X}$. We define $\leq_Y \subset Y \times Y$ by $y_1 \leq_Y y_2$ iff there exists $x_1 \in y_1$ and $x_2 \in y_2$ such that $x_1 \leq_X x_2$.

Exercise 3.3.1.4. Let $(X, \leq_X)$ be a proset. Set $Y := X/\sim_{\leq_X}$. Then $(Y, \leq_Y)$ is a poset.

Proof.

1. (a) Let $y \in Y$. Then there exists $x \in y$. Since $x \leq_X x$, we have that $y \leq_Y y$.
   (b) Let $y_1, y_2, y_3 \in Y$. Suppose that $y_1 \leq_Y y_2$ and $y_2 \leq_Y y_3$. Then there exist $x_1 \in y_1$, $x_2 \in y_2$ and $x_3 \in y_3$ such that $x_1 \leq_X x_2$ and $x_2 \leq_X x_3$. Therefore $x_1 \leq x_3$. Hence $y_1 \leq_Y y_3$.

   Thus $\leq_Y$ is a preorder on Y.
2. Let $y_1, y_2 \in Y$. Suppose that $y_1 \leq_Y y_2$ and $y_2 \leq_Y y_1$. Then there exist $a_1, b_1 \in y_1$, $a_2, b_2 \in y_2$ such that $a_1 \leq_X a_2$ and $b_2 \leq_X b_1$. Since $a_1, b_1 \in y_1$ and $a_2, b_2 \in y_2$, $a_1 \sim_{\leq_X} b_1$ and $a_2 \sim_{\leq_X} b_2$. Thus
$$\begin{aligned} a_1 &\leq_X a_2 \\ &\leq_X b_2. \end{aligned}$$
   and
$$\begin{aligned} b_2 &\leq_X b_1 \\ &\leq_X a_1. \end{aligned}$$
   Hence $a_1 \sim_{\leq_X} b_2$ and
$$\begin{aligned} y_1 &= \pi(a_1) \\ &= \pi(b_2) \\ &= y_2. \end{aligned}$$

Therefore $\leq_Y$ is a partial order on Y. □

Exercise 3.3.1.5. Let $(X, \leq)$ be a poset and $a, b \in X$.

1. If $\uparrow a = \uparrow b$, then $a = b$
2. If $\downarrow a = \downarrow b$, then $a = b$

Proof.
1. Suppose that $\uparrow a = \uparrow b$. Exercise 3.2.5.12 then implies that $a \sim_{\leq} b$. Since $(X, \leq)$ is a poset, we have that $a = b$.
2. Similar to (1).

□

3.3.2 Galois Connections

Exercise 3.3.2.1. Let $(X, \leq_X)$ and $(Y, \leq_Y)$ be posets. If (f, g) is a $(\leq_X, \leq_Y)$-monotone Galois connection or (f, g) is a $(\leq_X, \leq_Y)$-antitone Galois connection, then $f = f \circ g \circ f$ and $g = g \circ f \circ g$.

Proof. Sppose that (f, g) is a $(\leq_X, \leq_Y)$-monotone Galois connection or (f, g) is a $(\leq_X, \leq_Y)$-antitone Galois connection. a prev ex implies that $f \sim_{\leq_Y} f \circ g \circ f$ and $g \sim_{\leq_X} g \circ f \circ g$. Since $(X, \leq_X)$ and $(Y, \leq_Y)$ are posets, we have tthat $f = f \circ g \circ f$ and $g = g \circ f \circ g$. □

Exercise 3.3.2.2. Let $(X, \leq)$ be a proset. Define $f, g : \mathcal{P}(X) \to \mathcal{P}(X)$ by $f := \mathrm{ub}(\cdot, \leq)$ and $g := \mathrm{lb}(\cdot, \leq)$. Then (f, g) is a $(\subset^*, \subset^*)$-antitone Galois connection.

Proof.

1. make previous exercise implies that $A \subset g \circ f(A)$ and $A \subset f \circ g(A)$. Then for each $A \in \mathcal{P}(X)$, $g \circ f(A) \subset^* A$ and $f \circ g(A) \subset^* A$.

2. make previous exercises imply that f and g are $(\subset^*, \subset^*)$-antitone

the previous exercise implies that (f, g) FINISH!!! □

Exercise 3.3.2.3. Let $(X, \leq_X)$ and $(Y, \leq_Y)$ be posets, $f : X \to Y$ and $g : Y \to X$.

1. For each $g' : Y \to X$, if g and g' are right adjoints of f, then $g = g'$.

2. For each $f' : X \to Y$, if f and f' are left adjoints of g, then $f = f'$.

Proof.

1. Let $g' : Y \to X$. Suppose that g and g' are right adjoints of f. hen (f, g) and (f, g') are Galois connections. a previous ex implies that $f \circ g' \circ f = f$ and for each $x \in X$, $g' \circ f(x) \leq_Y g \circ f(g' \circ f(x))$. Therefore for each $x \in X$,

$$\begin{aligned} g' \circ f(x) &\leq g \circ f(g' \circ f(x)) \\ &= g(f \circ g' \circ f(x)) \\ &= g(f(x)) \\ &= g \circ f(x). \end{aligned}$$

   Thus $g' \circ f \leq g \circ f$. Similarly, $g \circ f \leq g' \circ f$. Hence $g' \circ f = g \circ f$. Therefore

$$\begin{aligned} g &= g \circ f \circ g \\ &= g' \circ f \circ g. \end{aligned}$$

   and

$$\begin{aligned} g' &= g' \circ f \circ g' \\ &= g \circ f \circ g'. \end{aligned}$$

   Let $y \in Y$. Since $f \circ g'(y) \leq y$, we have that

$$\begin{aligned} g'(y) &= g \circ f \circ g'(y) \\ &= g(f \circ g'(y)) \\ &\leq g(y). \end{aligned}$$

   Similarly, $g(y) \leq g'(y)$. Hence $g(y) = g'(y)$. Since $y \in Y$ is arbitrary, we have that for each $y \in Y$, $g(y) = g'(y)$. Thus $g = g'$.

2. Similar to (1).

TIDY UP!!!! □

Exercise 3.3.2.4. Let $(X, \leq_X)$ and $(Y, \leq_Y)$ be posets, $f : X \to Y$ and $g : Y \to X$. Suppose that (f, g) is a monotone Galois connection. Then

1. for each $A \subset X$, if $\sup(A, \leq_X)$ exists, then $f(\sup(A, \leq_X)) = \sup(f(A), \leq_Y)$.
2. for each $B \subset Y$, if $\sup(B, \leq_Y)$ exists, then $g(\sup(B, \leq_Y)) = \sup(g(B), \leq_X)$.

Proof.

1. Let $A \subset X$. Suppose that $\sup(A, \leq_X)$ exists.
2.

□

Exercise 3.3.2.5. Let $(X, \leq_X)$ and $(Y, \leq_Y)$ be posets and $f : X \to Y$. Then there exists $g : Y \to X$ such that (f, g) is a monotone Galois connection iff for each $y \in Y$, there exists $x \in X$ such that $f^{-1}(\downarrow y) = \downarrow x$.

Proof.

- $(\implies)$:
  Suppose that there exists $g : Y \to X$ such that (f, g) is a monotone Galois connection. a prev ex in galois connections on prosets section Exercise B.0.0.3 then implies that f and g are monotone, $\mathrm{id}_X \leq_X g \circ f$ and $f \circ g \leq_Y \mathrm{id}_Y$. Let $y \in Y$. Define $x \in X$ by $x := g(y)$.
  - We note that for each $a \in X$,
    $$\begin{aligned} a \in f^{-1}(\downarrow y) &\iff f(a) \in \downarrow y \\ &\iff f(a) \leq_Y y. \end{aligned}$$
    Let $a \in f^{-1}(\downarrow y)$. Then
    $$\begin{aligned} a &\leq_X g \circ f(a) \\ &= g(f(a)) \\ &\leq_X g(y) \\ &= x. \end{aligned}$$
    Therefore $a \in \downarrow x$. Since $a \in f^{-1}(\downarrow y)$ is arbitrary, we have that for each $a \in f^{-1}(\downarrow y)$, $a \in \downarrow x$. Hence $a \in f^{-1}(\downarrow y) \subset \downarrow x$. Then $f(a) \in \downarrow$.
  - Let $a \in \downarrow x$. Then $a \leq_X x$. Hence
    $$\begin{aligned} f(a) &\leq_Y f(x) \\ &= f(g(y)) \\ &= f \circ g(y) \\ &\leq_Y y. \end{aligned}$$
    Thus $f(a) \in \downarrow y$ and $a \in f^{-1}(\downarrow y)$. Since $a \in \downarrow x$ is arbitrary, we have that for each $a \in \downarrow x$, $a \in f^{-1}(\downarrow y)$. Hence $\downarrow x \subset f^{-1}(\downarrow y)$.

  Since $f^{-1}(\downarrow y) \subset \downarrow x$ and $\downarrow x \subset f^{-1}(\downarrow y)$, we have that $f^{-1}(\downarrow y) = \downarrow x$. Since $y \in Y$ is arbitrary, we have that for each $y \in Y$, there exists $x \in X$ such that $f^{-1}(\downarrow y) = \downarrow x$.

- $(\impliedby)$:
  Suppose that for each $y \in Y$, there exists $x \in X$ such that $f^{-1}(\downarrow y) = \downarrow x$. The axiom of choice implies that there exists $g : Y \to X$ such that for each $y \in Y$, $f^{-1}(\downarrow y) = \downarrow g(y)$. Then for each $x \in X$ and $y \in Y$,
  $$\begin{aligned} f(x) \leq_X y &\iff f(x) \in \downarrow y \\ &\iff x \in f^{-1}(\downarrow y) \\ &\iff x \in \downarrow g(y) \\ &\iff x \leq_Y g(y). \end{aligned}$$
  Thus (f, g) is a monotone Galois connection.

□

3.3.3 Suprema and Infima

Exercise 3.3.3.1. Let $(X, \leq)$ be a poset and $A \subset X$. Then

1. for each $x, y \in \operatorname{supSet} A$, $x = y$.
2. for each $x, y \in \operatorname{infSet} A$, $x = y$.

Proof.

1. Let $x, y \in \operatorname{supSet} A$. Exercise 3.2.8.3 implies that $x \sim_{\leq} y$. Exercise 3.3.1.2 then implies that $x = y$.
2. Similar to (1).

□

Definition 3.3.3.2. Let $(X, \leq)$ be a poset and $A \subset X$.

- If $\operatorname{supSet} A \neq \varnothing$, we define the **supremum of** A, denoted $\sup A$, to be the unique $x \in X$ such that $x \in \operatorname{supSet} A$.
- If $\operatorname{infSet} A \neq \varnothing$, we define the **infimum of** A, denoted $\inf A$, to be the unique $x \in X$ such that $x \in \operatorname{infSet} A$.
- We say that $\sup A$ **(resp.** $\inf A$**) exists** if $\operatorname{supSet} A \neq \varnothing$ (resp. $\operatorname{infSet} A \neq \varnothing$).

Exercise 3.3.3.3. Let $(X, \leq)$ be a poset and $A \subset X$. Then

1. (a) $\operatorname{supSet} A = \operatorname{infSet}(\operatorname{ub} A)$
   (b) $\sup A$ exists iff $\operatorname{infSet}(\operatorname{ub} A)$ exists
   (c) $\sup A$ exists implies that $\sup A = \inf(\operatorname{ub} A)$
2. (a) $\operatorname{infSet} A = \operatorname{supSet}(\operatorname{lb} A)$
   (b) $\inf A$ exists iff $\operatorname{supSet}(\operatorname{lb} A)$ exists
   (c) $\inf A$ exists implies that $\inf A = \sup(\operatorname{lb} A)$

Proof.

1. (a) A previous ex in subsection on suprema and infima implies that
   - $\operatorname{supSet} A = \operatorname{ub} A \cap \operatorname{lb}(\operatorname{ub} A)$
   - $\operatorname{infSet}(\operatorname{ub} A) = \operatorname{lb}(\operatorname{ub} A) \cap \operatorname{ub}(\operatorname{lb}(\operatorname{ub} A))$.

   Define $f, g : \mathcal{P}(X) \to \mathcal{P}(X)$ by $f := \operatorname{ub}(\cdot, \leq)$ and $g := \operatorname{lb}(\cdot, \leq)$. A previous exercis in poset galois connection subsection implies that (f, g) is a $(\subset^*, \subset^*)$-antitone Galois connection. Another prev ex in poset galois subsection implies that for each $A \in \mathcal{P}(X)$, $\operatorname{ub}(\operatorname{lb}(\operatorname{ub} A)) = \operatorname{ub} A$. Therefore

$$\begin{aligned}
\operatorname{supSet} A &= \operatorname{ub} A \cap \operatorname{lb}(\operatorname{ub} A) \\
&= \operatorname{ub}(\operatorname{lb}(\operatorname{ub} A)) \cap \operatorname{lb}(\operatorname{ub} A) \\
&= \operatorname{lb}(\operatorname{ub} A) \cap \operatorname{ub}(\operatorname{lb}(\operatorname{ub} A)) \\
&= \operatorname{infSet}(\operatorname{ub} A).
\end{aligned}$$

   (b) The previous part implies that

$$\begin{aligned}
\sup A \text{ exists} &\iff \operatorname{supSet} A \neq \varnothing \\
&\iff \operatorname{infSet}(\operatorname{ub} A) \neq \varnothing \\
&\iff \inf(\operatorname{ub} A) \text{ exists.}
\end{aligned}$$

(c) Suppose that $\sup A$ exists. the previous part (a) then implies that $\inf(\operatorname{ub} A)$ exists. a previous ex implies that $\operatorname{supSet} A = \{\sup A\}$ and $\operatorname{infSet}(\operatorname{ub} A) = \{\inf(\operatorname{ub} A)\}$. The previous parts (a) then implies that

$$\begin{aligned}\{\sup A\} &= \operatorname{supSet} A \\ &= \operatorname{infSet}(\operatorname{ub} A) \\ &= \{\inf(\operatorname{ub} A)\}.\end{aligned}$$

Hence $\sup A = \inf(\operatorname{ub} A)$.

2. Similar to (1).

□

Exercise 3.3.3.4. Associativity of Supremum:
Let $(X, \leq)$ be a poset and $(E_\alpha)_{\alpha \in A} \subset \mathcal{P}(X)$.

1. Suppose that
   - for each $\alpha \in A$, $\sup E_\alpha$ exists
   - $\sup\limits_{\alpha \in A} \left[\sup E_\alpha\right]$ exists.

   Then $\sup \bigcup\limits_{\alpha \in A} E_\alpha$ exists and
$$\sup \bigcup_{\alpha \in A} E_\alpha = \sup_{\alpha \in A} \left[\sup E_\alpha\right]$$
2. Suppose that
   - for each $\alpha \in A$, $\inf E_\alpha$ exists
   - $\inf\limits_{\alpha \in A} \left[\inf E_\alpha\right]$ exists.

   Then $\inf \bigcup\limits_{\alpha \in A} E_\alpha$ exists
$$\inf \bigcup_{\alpha \in A} E_\alpha = \inf_{\alpha \in A} \left[\inf E_\alpha\right]$$

Proof.

1. Define $s \in X$ by $s := \sup\limits_{\alpha \in A} \left[\sup E_\alpha\right]$.
   - Let $x \in \bigcup\limits_{\alpha \in A} E_\alpha$. Then there exists $\alpha_0 \in A$ such that $x \in E_{\alpha_0}$. We note by definition, $s \in \operatorname{ub}\{\sup E_\alpha : \alpha \in A\}$. Hence for each $\alpha \in A$, $s \geq \sup E_\alpha$. Since $\sup E_{\alpha_0} \in \operatorname{ub} E_{\alpha_0}$, we have that
$$\begin{aligned} x &\leq \sup E_{\alpha_0} \\ &\leq s. \end{aligned}$$
     Since $x \in \bigcup\limits_{\alpha \in A} E_\alpha$ is arbitrary, we have that for each $x \in \bigcup\limits_{\alpha \in A} E_\alpha$, $x \leq s$. Hence $s \in \operatorname{ub} \bigcup\limits_{\alpha \in A} E_\alpha$.
   - Let $x \in \operatorname{ub} \bigcup\limits_{\alpha \in A} E_\alpha$ and $\alpha_0 \in A$. Since $E_{\alpha_0} \subset \bigcup\limits_{\alpha \in A} E_\alpha$, Exercise 3.2.5.24 implies that
$$\begin{aligned} x &\in \operatorname{ub} \bigcup_{\alpha \in A} E_\alpha \\ &\subset \operatorname{ub} E_{\alpha_0}. \end{aligned}$$
     Therefore $\sup E_{\alpha_0} \leq x$. Since $\alpha_0 \in A$ is arbitrary, we have that for each $\alpha_0 \in A$, $\sup E_{\alpha_0} \leq x$. Hence $x \in \operatorname{ub}\{\sup E_\alpha : \alpha \in A\}$. Therefore $s \leq x$. Since $x \in \operatorname{ub} \bigcup\limits_{\alpha \in A} E_\alpha$ is arbitrary, we have that for each $x \in \operatorname{ub} \bigcup\limits_{\alpha \in A} E_\alpha$, $s \leq x$.

Therefore $s \in \text{supSet} \bigcup_{\alpha \in A} E_\alpha$ and $\sup \bigcup_{\alpha \in A} E_\alpha = s$.

2. Similar to (1). Maybe fill out, use duality

□

Exercise 3.3.3.5. Let $(X, \leq)$ be a poset and $a, b \in X$. Then

1. $\sup\{a\}$ exists and $\sup\{a\} = a$
2. $a \leq b$ iff $\sup\{a, b\}$ exists and $b = \sup\{a, b\}$.

Proof.

1. Since $a \leq a$, $a \in \text{ub}\{a\}$. By definition, for each $x \in \text{ub}\{a\}$, $a \leq x$. Hence $a \in \text{supSet}\{a\}$. Thus $\sup\{a\}$ exists and $a = \sup\{a\}$.
2.
   - $(\implies)$:
     Suppose that $a \leq b$. Since $b \leq b$, we have that for each $c \in \{a, b\}$, $c \leq b$. Hence $b \in \text{lb}\{a, b\}$. Let $c \in \text{ub}\{a, b\}$. Then $b \leq c$. Since $c \in \text{ub}\{a, b\}$ is arbitrary, we have that for each $c \in \text{ub}\{a, b\}$, $b \leq c$. Hence $b \in \text{supSet}\{a, b\}$ and $b = \sup\{a, b\}$.
   - $(\impliedby)$:
     Suppose that $\sup\{a, b\}$ exists and $b = \sup\{a, b\}$. Then $b \in \text{ub}\{a, b\}$. Therefore $a \leq b$.

□

Exercise 3.3.3.6. Let $(X, \leq)$ be a poset. Then for each $a, b \in X$, $\sup\{a, b\}$ exists iff for each $n \in \mathbb{N}$ and $(a_k)_{k=1}^n \in X^n$, $\sup\{a_1, \ldots, a_n\}$ exists and

$$\sup\{a_1, \ldots, a_n\} = \begin{cases} a_1, & n = 1 \\ \sup\left\{\sup\{a_1, \ldots a_{n-1}\}, a_n\right\}, & n \geq 2. \end{cases}$$

Proof.

- $(\implies)$:
  Suppose that for each $a, b \in X$, $\sup\{a, b\}$ exists. Let $n \in \mathbb{N}$ and $(a_k)_{k=1}^n \in X^n$.
  - **Base Case:**
    Suppose that $n = 1$. Exercise 3.3.3.5 implies that $\sup\{a_1\}$ exists and $\sup\{a_1\} = a_1$.
  - **Induction Step:**
    Suppose that $n \geq 2$ and $\sup\{a_1, \ldots, a_{n-1}\}$ exists. Then by assumption, $\sup\left\{\sup\{a_1, \ldots a_{n-1}\}, a_n\right\}$ exists. Exercise 3.3.3.4 then implies that $\sup\{a_1, \ldots, a_n\}$ exists and

$$\begin{aligned} \sup\{a_1, \ldots, a_n\} &= \sup\left\{\{a_1, \ldots a_{n-1}\} \cup \{a_n\}\right\} \\ &= \sup\left\{\sup\{a_1, \ldots a_{n-1}\}, \sup\{a_n\}\right\} \\ &= \sup\left\{\sup\{a_1, \ldots a_{n-1}\}, a_n\right\}. \end{aligned}$$

  By induction, for each $n \in \mathbb{N}$ and $(a_k)_{k=1}^n \in X^n$, $\sup\{a_1, \ldots, a_n\}$ exists and

$$\sup\{a_1, \ldots, a_n\} = \begin{cases} a_1, & n = 1 \\ \sup\left\{\sup\{a_1, \ldots a_{n-1}\}, a_n\right\}, & n \geq 2. \end{cases}$$

- ($\Longleftarrow$):
  Clear.

□

Exercise 3.3.3.7. Let $(X, \leq)$ be a poset. Then

1. for each $a, b \in \operatorname{ub} X$, $a = b$,
2. for each $a \in \operatorname{ub} X$, $\sup X = a$,
3. for each $a, b \in \operatorname{lb} X$, $a = b$,
4. for each $a \in \operatorname{lb} X$, $\inf X = a$

Proof.

1. Let $a, b \in \operatorname{ub} X$. Since $a \in X$ and $b \in \operatorname{ub} X$, $a \leq b$. Similarly, $b \leq a$. Hence $a = b$.
2. Let $a, b \in \operatorname{ub} X$. the previous part implies that $a = b$. Since $b \in \operatorname{ub} X$ is arbitrary, we have that for each $b \in \operatorname{ub} X$, $a \leq b$. Thus $a = \sup X$.
3. Similar to (1) ref.
4. Similar to (2) ref.

□

3.3.4 Monotone Maps

3.4 Directed Sets

Definition 3.4.0.1. Directed Set:
Let A be a set and $\leq \subset A \times A$ a binary relation on A. Then

- $\leq$ is said to be a **direction on** A if
  1. $\leq$ is a preorder on A
  2. for each $\alpha, \beta \in A$, $\{\alpha, \beta\}$ is bounded above.
- $(A, \leq)$ is said to be a **directed set** if
  1. $A \neq \varnothing$
  2. $\leq$ is a direction on A

Exercise 3.4.0.2. Let $(A, \leq_A)$ and $(B, \leq_B)$ be directed sets. Then

1. $\leq_A \otimes \leq_B$ is a direction on $A \times B$,
2. $(A, \leq_A) \otimes (B, \leq_B)$ is a directed set.

Proof.

1. (a) Exercise **??** implies that $\leq_A \otimes \leq_B$ is a preorder of $A \times B$.
   (b) Let $(a_1, b_1), (a_2, b_2) \in A \times B$. Then there exist $a \in A$ and $b \in B$ such that $a_1, a_2 \leq_A a$ and $b_1, b_2 \leq_B b$. Hence $(a_1, b_1), (a_2, b_2) \leq_A \otimes \leq_B (a, b)$.

   Hence $\leq_A \otimes \leq_B$ is a direction on $A \times B$.
2. (a) Since $A \neq \varnothing$ and $B \neq \varnothing$, we have that $A \times B \neq \varnothing$.
   (b) From above, $\leq_A \otimes \leq_B$ is a direction on $A \times B$.

   Hence $(A \times B, \leq_A \otimes \leq_B)$ is a directed set.

$\square$

Part II

Algebraic Structures

Chapter 4

Lattices

4.0.1 To Do

- maybe cover join/meet presemilattices, i.e. join/meet semilattices but with preorder instead of partial order, this would prbably need to move exercises around and rework stuff
- explain that we assume lattices are nonempty due to the convention in universal algebra (i guess for some slick way to state important results without having to make extra cases/exceptions?)

4.1 Introduction

4.1.1 Skew Semilattices

cite Leech, skew lattices in rings

Definition 4.1.1.1. Let X be a set and $\diamond : X \times X \to X$ a binary operator on X. Then

- $\diamond$ is said to be a **skew semilattice operator on** X if $\diamond$ is associative and idempotent.
- $(X, \diamond)$ is said to be a **skew semilattice** if
  1. $X \neq \varnothing$
  2. $\diamond$ is a skew semilattice operator on X

Definition 4.1.1.2. Let $(X, \diamond)$ be an skew semilattice. We define the

- **join preorder on** X **induced by** $\diamond$, denoted $\leq_\diamond^\vee \subset X \times X$, by $a \leq_\diamond^\vee b$ iff $(b \diamond a) \diamond b = b$
- **meet preorder on** X **induced by** $\diamond$, denoted $\leq_\diamond^\wedge \subset X \times X$, by $a \leq_\diamond^\wedge b$ iff $(a \diamond b) \diamond a = a$.

Exercise 4.1.1.3. Let $(X, \diamond)$ be an skew semilattice. Then

1. $\leq_\diamond^\vee$ is a preorder on X.
   Hint: If $cbc = c$ and $bab = b$, then
   (a) $c = (cb)(abc)(abc)$ and $c = cabc$
   (b) $cac = (ca)(ca)(bc)$
2. $\leq_\diamond^\wedge$ is a preorder on X.

Proof.

1. Let $a, b, c \in X$.

(a) Since $\diamond$ is idempotent, we have that

$$\begin{aligned}(a \diamond a) \diamond a &= a \diamond a \\ &= a\end{aligned}$$

and therefore $a \leq_\diamond^\vee a$.

(b) Suppose that $a \leq_\diamond^\vee b$ and $b \leq_\diamond^\vee c$. Then $(b \diamond a) \diamond b = b$ and $(c \diamond b) \diamond c = c$. Since $\diamond$ is associative and idempotent,

$$\begin{aligned}c &= cbc \\ &= cbabc \\ &= (cb)(abc) \\ &= (cb)(abc)(abc) \\ &= c(bab)cabc \\ &= cbcabc \\ &= (cbc)abc \\ &= cabc.\end{aligned}$$

Therefore

$$\begin{aligned}cac &= (ca)(c) \\ &= (ca)(cabc) \\ &= (ca)(ca)(bc) \\ &= (ca)(bc) \\ &= cabc \\ &= c.\end{aligned}$$

Hence $a \leq_\diamond^\vee c$.

So $\leq_\diamond^\vee$ is a preorder on X.

2. Similar to (1).

□

Exercise 4.1.1.4. Let $(X, \diamond)$ be an skew semilattice. Then

1. $(\leq_\diamond^\vee)^{\mathrm{op}} = \leq_\diamond^\wedge$
2. $(\leq_\diamond^\wedge)^{\mathrm{op}} = \leq_\diamond^\vee$

Proof.

1. Let $a, b \in X$. Then

$$\begin{aligned}a(\leq_\diamond^\vee)^{\mathrm{op}}b &\iff b \leq_\diamond^\vee a \\ &\iff a \diamond b \diamond a = a \\ &\iff a \leq_\diamond^\wedge b.\end{aligned}$$

2. Similar to (1).

□

Exercise 4.1.1.5. Let $(X, \diamond)$ be an skew semilattice. Then for each $a, b \in X$,

1. $a \diamond b \in \mathrm{ub}(\{a, b\}, \leq_\diamond^\vee)$

2. $a \diamond b \in \mathrm{lb}(\{a, b\}, \leq_{\diamond}^{\wedge})$

Proof. Let $a, b \in X$.

1. We note that

$$\begin{aligned}(a \diamond b) \diamond [a \diamond (a \diamond b)] &= (a \diamond b) \diamond [(a \diamond a) \diamond b] \\ &= (a \diamond b) \diamond (a \diamond b) \\ &= a \diamond b.\end{aligned}$$

   Similarly, $[(a \diamond b) \diamond b] \diamond (a \diamond b) = a \diamond b$. Therefore $a \leq_{\diamond}^{\vee} a \diamond b$ and $b \leq_{\diamond}^{\vee} a \diamond b$. Hence $a \diamond b \in \mathrm{ub}(\{a, b\}, \leq_{\diamond}^{\vee})$.

2. FINISH!!!

□

Definition 4.1.1.6. Let X be a set and $\vee, \wedge$ skew semilattice operators on X. Then $\vee$ and $\wedge$ are said to **satisfy the skew lattice absorption identities** if for each $a, b \in X$,

1. $a \wedge (a \vee b) = a$
2. $a \vee (a \wedge b) = a$
3. $(b \vee a) \wedge a = a$
4. $(b \wedge a) \vee a = a$

4.1.2 Skew Semilattice Homomorphisms

Definition 4.1.2.1. Let $(X, \diamond_X)$ and $(Y, \diamond_Y)$ be skew semilattices and $f : X \to Y$. Then f is said to be a $(\diamond_X, \diamond_Y)$-**homormorphism** if for each $a, b \in X$, $f(a \diamond_X b) = f(a) \diamond_Y f(b)$.

Note 4.1.2.2. When the context is clear, we say homomorphism instead of $(\diamond_X, \diamond_Y)$-homomorphism.

Exercise 4.1.2.3. Let $(X, \diamond_X)$ and $(Y, \diamond_Y)$ be skew semilattices and $f : X \to Y$. If f is a $(\diamond_X, \diamond_Y)$-homomorphism, then

1. f is $(\leq_{\diamond_X}^{\vee}, \leq_{\diamond_Y}^{\vee})$-monotone
2. f is $(\leq_{\diamond_X}^{\wedge}, \leq_{\diamond_Y}^{\wedge})$-monotone

Proof. Suppose that f is a $(\diamond_X, \diamond_Y)$-homomorphism.

1. Let $a, b \in X$. Suppose that $a \leq_{\diamond_X}^{\vee} b$. Then $(b \diamond_X a) \diamond_X b = b$. Therefore

$$\begin{aligned}f(b) &= (b \diamond_X a) \diamond_X b \\ &= f(b \diamond_X a) \diamond_Y f(b) \\ &= (f(b) \diamond_Y f(a)) \diamond_Y f(b)\end{aligned}$$

   Thus $f(a) \leq_{\diamond_Y}^{\vee} f(b)$. Since $a, b \in X$ with $a \leq_{\diamond_X}^{\vee} b$ are arbitrary, we have that for each $a, b \in X$, $a \leq_{\diamond_X}^{\vee} b$ implies that $f(a) \leq_{\diamond_Y}^{\vee} f(b)$. Hence f is $(\leq_{\diamond_X}^{\vee}, \leq_{\diamond_Y}^{\vee})$-monotone.

2. use duality?, FINISH!!!

□

4.1.3 Skew Lattices

Exercise 4.1.3.1. Let X be a set and $\vee, \wedge$ skew semilattice operators on X. Then $\vee$ and $\wedge$ satisfy the skew lattice absorption identities iff for each $a, b \in X$,

1. $a \vee b = b$ iff $a \wedge b = a$
2. $a \vee b = a$ iff $a \wedge b = b$

Proof.

- ($\Longrightarrow$):
  Suppose that $\vee$ and $\wedge$ satisfy the skew lattice absorption identities. Let $a, b \in X$.

  1. – ($\Longrightarrow$):
     If $a \vee b = b$, then

     $$\begin{aligned} a \wedge b &= a \wedge (a \vee b) \\ &= a. \end{aligned}$$

     – ($\Longleftarrow$):
     If $a \wedge b = a$, then

     $$\begin{aligned} a \vee b &= (a \wedge b) \vee b \\ &= b. \end{aligned}$$

     Thus $a \vee b = b$ iff $a \wedge b = a$.
  2. Similar to (1)

- ($\Longleftarrow$):
  Suppose that for each $a, b \in X$,

  1. $a \vee b = b$ iff $a \wedge b = a$,
  2. $a \vee b = a$ iff $a \wedge b = b$.

  Let $a, b \in X$.

  1. Set $c := a \vee b$. By assumption,

     $$\begin{aligned} a \wedge (a \vee b) = a &\iff a \wedge c = a \\ &\iff a \vee c = c \\ &\iff a \vee (a \vee b) = c \\ &\iff (a \vee a) \vee b = c \\ &\iff a \vee b = c \\ &\iff c = c. \end{aligned}$$

     Since $c = c$, we have that $a \wedge (a \vee b) = a$.
  2. Similar to (1)
  3. Similar to (1)
  4. Similar to (1)

□

Exercise 4.1.3.2. Let X be a set and $\vee, \wedge$ skew semilattice operators on X. If $\vee$ and $\wedge$ satisfy the skew lattice absorption identities, then for each $a, b \in X$, $b \vee a \vee b = b$ iff $a \wedge b \wedge a = a$.
Hint: Exercise 4.1.3.1 implies that $b \vee (a \vee b) = b$ iff $b \wedge (a \vee b) = a \vee b$. Use, skew lattice absorption identities and associativity.

Proof. Suppose that $\vee$ and $\wedge$ satisfy the skew lattice absorption identities. Let $a, b \in X$.

- ($\Longrightarrow$):
  Suppose that $b \vee a \vee b = b$. Then Exercise 4.1.3.1 implies that $b \wedge (a \vee b) = (a \vee b)$. Therefore

$$\begin{aligned} a \wedge b &= (a \wedge b) \vee [(a \wedge b) \wedge (a \vee b)] \quad \text{(absorption)} \\ &= (a \wedge b) \vee [a \wedge (b \wedge [a \vee b])] \quad \text{(associativity)} \\ &= (a \wedge b) \vee [a \wedge (a \vee b)] \quad \text{(from earlier)} \\ &= (a \wedge b) \vee a \quad \text{(absorption).} \end{aligned}$$

  Another application of Exercise 4.1.3.1 implies that $(a \wedge b) \wedge a = a$.

- ($\Longleftarrow$):
  Similar to ($\Longrightarrow$).

□

Definition 4.1.3.3. Let X be a set and $\vee, \wedge$ skew semilattice operators on X. Then $(X, \vee, \wedge)$ is said to be a **skew lattice** if

1. $X \neq \varnothing$
2. $\vee$ and $\wedge$ satisfy the skew lattice absorption identities

Exercise 4.1.3.4. Let $(X, \vee, \wedge)$ be an skew lattice. Then

1. $\leq_{\vee}^{\vee} = \leq_{\wedge}^{\wedge}$.
2. $\leq_{\vee}^{\wedge} = \leq_{\wedge}^{\vee} = (\leq_{\vee}^{\vee})^{\mathrm{op}}$.

Proof.

1. Let $a, b \in X$. Since $(X, \vee, \wedge)$ is an skew lattice, $\vee$ and $\wedge$ satisfy the skew lattice absorption identities. Exercise 4.1.3.2 then implies that

$$\begin{aligned} a \leq_{\vee}^{\vee} b &\iff b \vee a \vee b = b \\ &\iff a \wedge b \wedge a = a \\ &\iff a \leq_{\wedge}^{\wedge} b. \end{aligned}$$

   Since $a, b \in X$ are arbitrary, we have that for each $a, b \in X$, $a \leq_{\vee}^{\vee} b$ iff $a \leq_{\wedge}^{\wedge} b$. Hence $\leq_{\vee}^{\vee} = \leq_{\wedge}^{\wedge}$.

2. Exercise 4.1.1.4 and part (1) imply that

$$\begin{aligned} \leq_{\vee}^{\wedge} &= (\leq_{\vee}^{\vee})^{\mathrm{op}} \\ &= (\leq_{\wedge}^{\wedge})^{\mathrm{op}} \\ &= \leq_{\wedge}^{\vee}. \end{aligned}$$

□

Note 4.1.3.5. Let $(X, \vee, \wedge)$ be an skew lattice. When the context is clear, we write $\leq$ in place of $\leq_{\vee}^{\vee}$.

4.1.4 Semilattices

Definition 4.1.4.1. Let X be a set and $\diamond : X \times X \to X$. Then

- $\diamond$ is said to be a **semilattice operator on** X if $\diamond$ associative, idempotent and commutative.
- $(X, \diamond)$ is said to be a **semilattice** if
  1. $X \neq \varnothing$
  2. $\diamond$ is a semilattice operator on X.

Exercise 4.1.4.2. Let X be a set and $\diamond : X \times X \to X$. Then $\diamond$ is a semilattice operator on X iff

1. $\diamond$ is a skew semilattice operator on X,
2. $\diamond$ is commutative.

Proof. Clear by definition. □

Exercise 4.1.4.3. Let $(X, \diamond)$ be a semilattice. Then

1. for each $a, b \in X$,
   (a) $a \leq_\diamond^\vee b$ iff $a \diamond b = b$
   (b) $a \leq_\diamond^\wedge b$ iff $a \diamond b = a$
2. (a) $\leq_\diamond^\vee$ is a partial order on X
   (b) $\leq_\diamond^\wedge$ is a partial order on X

Proof.

1. Let $a, b \in X$.
   (a) Since $\diamond$ is commutative, associative and idempotent,
   $$\begin{aligned} a \leq_\diamond^\vee b &\iff (b \diamond a) \diamond b = b \\ &\iff (a \diamond b) \diamond b = b \\ &\iff a \diamond (b \diamond b) = b \\ &\iff a \diamond b = b. \end{aligned}$$
   (b) Similar to 1(a).
2. (a) Exercise 4.1.1.3 implies that $\leq_\diamond^\vee$ is a preorder on X. Let $a, b \in X$. Suppose that $a \leq_\diamond^\vee b$ and $b \leq_\diamond^\vee a$. Part (1) then implies that $a \diamond b = b$ and $b \diamond a = a$. Since $\diamond$ is commutative,
   $$\begin{aligned} a &= b \diamond a \\ &= a \diamond b \\ &= b. \end{aligned}$$
   Since $a, b \in X$ are arbitrary, we have that for each $a, b \in X$, if $a \leq_\diamond^\vee b$ and $b \leq_\diamond^\vee a$, then $a = b$. Thus $\leq_\diamond^\vee$ is a a partial order on X.
   (b) Similar to 2(a).

□

Exercise 4.1.4.4. Let $(X, \diamond)$ be a semilattice. Then for each $a, b \in X$,

1. $a \diamond b = \sup(\{a, b\}, \leq_\diamond^\vee)$
2. $a \diamond b = \inf(\{a, b\}, \leq_\diamond^\wedge)$

Proof. Let $a, b \in X$.

1. Exercise 4.1.1.5 implies that $a \diamond b \in \mathrm{ub}(\{a, b\}, \leq_\diamond^\vee)$. Let $c \in \mathrm{ub}(\{a, b\}, \leq_\diamond^\vee)$. Since $a \leq_\diamond^\vee c$ and $b \leq_\diamond^\vee c$, Exercise 4.1.4.3 implies that $a \diamond c = c$ and $b \diamond c = c$. Therefore

$$\begin{aligned}(a \diamond b) \diamond c &= a \diamond (b \diamond c) \\ &= a \diamond c \\ &= c.\end{aligned}$$

   Another application of Exercise 4.1.4.3 implies that $a \diamond b \leq_\diamond^\vee c$. Since $c \in \mathrm{ub}(\{a, b\}, \leq_\diamond^\vee)$ is arbitrary, we have that for each $c \in \mathrm{ub}(\{a, b\}, \leq_\diamond^\vee)$, $a \diamond b \leq_\diamond^\vee c$. Therefore $a \diamond b = \sup(\{a, b\}, \leq_\diamond^\vee)$.

2. FINISH!!!

□

Definition 4.1.4.5. Let $(X, \leq)$ be a poset. Then $(X, \leq)$ is said to be a

- **join semilattice** if
  1. $X \neq \varnothing$
  2. for each $a, b \in X$, $\sup\{a, b\}$ exists.
- **meet semilattice** if
  1. $X \neq \varnothing$
  2. for each $a, b \in X$, $\inf\{a, b\}$ exists.

Exercise 4.1.4.6. Let $(X, \diamond)$ be a semilattice. Then

1. $(X, \leq_\diamond^\vee)$ is a join semilattice
2. $(X, \leq_\diamond^\wedge)$ is a meet semilattice

Proof. Exercise 4.1.4.4 implies that

1. For each $a, b \in X$, $\sup(\{a, b\}, \leq_\diamond^\vee) = a \diamond b$. Thus $(X, \leq_\diamond^\vee)$ is a join semilattice.
2. for each $a, b \in X$, $\inf(\{a, b\}, \leq_\diamond^\wedge) = a \diamond b$. Thus $(X, \leq_\diamond^\wedge)$ is a meet semilattice.

□

Definition 4.1.4.7. Let $(X, \leq)$ be a poset.

- If $(X, \leq)$ is a join semilattice, we define the **join operator on X induced by $\leq$**, denoted $\vee_\leq : X \times X \to X$, by $a \vee_\leq b := \sup\{a, b\}$.
- If $(X, \leq)$ is a meet semilattice, we define the **meet operator on X induced by $\leq$**, denoted $\wedge_\leq : X \times X \to X$, by $a \wedge_\leq b := \inf\{a, b\}$.

Exercise 4.1.4.8. Let $(X, \leq)$ be a poset.

1. If $(X, \leq)$ an join semilattice, then $\vee_\leq$ is a semilattice operator on X.
2. If $(X, \leq)$ an meet semilattice, then $\wedge_\leq$ is a semilattice operator on X.

Proof.

1. Suppose that $(X, \leq)$ an join semilattice.

(a)
- Let $a, b, c \in X$. Exercise 3.3.3.4 implies that

$$\begin{aligned}(a \vee_{\leq} b) \vee_{\leq} c &= \sup\{\sup\{a, b\}, c\} \\ &= \sup\{\sup\{a, b\}, \sup\{c\}\} \\ &= \sup\{a, b, c\} \\ &= \sup\{\sup\{a\}, \sup\{b, c\}\} \\ &= \sup\{a, \sup\{b, c\}\} \\ &= a \vee_{\leq} (b \vee_{\leq} c).\end{aligned}$$

  Hence $\vee_{\leq}$ is associative.
- Let $a \in X$. Then

$$\begin{aligned}a \vee_{\leq} a &= \sup\{a, a\} \\ &= \sup\{a\} \\ &= a.\end{aligned}$$

  Hence $\vee_{\leq}$ is idempotent.
- Let $a, b \in X$. Then

$$\begin{aligned}a \vee_{\leq} b &= \sup\{a, b\} \\ &= \sup\{b, a\} \\ &= b \vee_{\leq} a.\end{aligned}$$

  Hence $\vee_{\leq}$ is commutative.

Since $\vee_{\leq}$ is associative, idempotent and commutative, $\vee_{\leq}$ is a semilattice operator on X.

2. Similar to (1).

□

Exercise 4.1.4.9. Let $(X, \leq)$ be a poset.

1. If $(X, \leq)$ is a join semilattice, then
   (a) $\leq^{\vee}_{\vee_{\leq}} = \leq$,
   (b) $\leq^{\wedge}_{\vee_{\leq}} = \leq^{*}$.
2. If $(X, \leq)$ is a meet semilattice, then
   (a) $\leq^{\wedge}_{\wedge_{\leq}} = \leq$,
   (b) $\leq^{\vee}_{\wedge_{\leq}} = \leq^{*}$.

Proof.

1. Suppose that $(X, \leq)$ is a join semilattice.
   (a) Let $a, b \in X$. Then

$$\begin{aligned}a \leq^{\vee}_{\vee_{\leq}} b &\iff a \vee_{\leq} b = b \quad \text{(Exercise 4.1.4.3)} \\ &\iff \sup(\{a, b\}, \leq) = b \quad \text{(Definition 4.1.4.7)} \\ &\iff a \leq b \quad \text{(Exercise 3.3.3.5)}.\end{aligned}$$

   Since $a, b \in X$ are arbitrary, we have that $\leq^{\vee}_{\vee_{\leq}} = \leq$.

(b) Exercise 4.1.1.4 implies that

$$\begin{aligned} \leq^{\wedge}_{\vee_{\leq}} &= (\leq^{\vee}_{\vee_{\leq}})^{\mathrm{op}} \\ &= \leq^{*}. \end{aligned}$$

2. Similar to (1).

□

Exercise 4.1.4.10. Let $(X, \diamond)$ be a semilattice. Then

1. $\leq^{\vee}_{\diamond}$ is the unique partial order $\leq$ on X such that $(X, \leq)$ is a join semilattice and for each $a, b \in X$, $a \diamond b = \sup(\{a, b\}, \leq)$.

2. $\leq^{\wedge}_{\diamond}$ is the unique partial order $\leq$ on X such that $(X, \leq)$ is an meet semilattice and for each $a, b \in X$, $a \diamond b = \inf(\{a, b\}, \leq)$.

Proof.

1. Let $\leq$ be partial order on X. Suppose that $(X, \leq)$ is a join semilattice and for each $a, b \in X$, $a \diamond b = \sup(\{a, b\}, \leq)$. Let $a, b \in X$. Then

$$\begin{aligned} a \leq b &\iff b = \sup(\{a, b\}, \leq) \quad \text{(Exercise 3.3.3.5)} \\ &\iff a \diamond b = b \quad \text{(by assumption)} \\ &\iff a \leq^{\vee}_{\diamond} b \quad \text{(Exercise 4.1.4.3).} \end{aligned}$$

Since $a, b \in X$ are arbitrary, we have that for each $a, b \in X$, $a \leq b$ iff $a \leq^{\vee}_{\diamond} b$. Hence $\leq = \leq^{\vee}_{\diamond}$.

2. Similar to (1).

□

Exercise 4.1.4.11. Let $(X, \leq)$ be a poset.

1. If $(X, \leq)$ is a join semilattice, then $\vee_{\leq}$ is the unique semilattice operator $\diamond$ on X such that for each $a, b \in X$, $a \diamond b = \sup(\{a, b\}, \leq)$.

2. If $(X, \leq)$ is a meet semilattice, then $\wedge_{\leq}$ is the unique semilattice operator $\diamond$ on X such that for each $a, b \in X$, $a \diamond b = \inf(\{a, b\}, \leq)$.

Proof.

1. Let $\diamond : X \times X \to X$ be a semilattice operator on X. Suppose that $a, b \in X$, $a \diamond b = \sup(\{a, b\}, \leq)$. Then for each $a, b \in X$,

$$\begin{aligned} a \diamond b &= \sup(\{a, b\}, \leq) \quad \text{(by assumption)} \\ &= a \vee_{\leq} b \quad \text{(Definition 4.1.4.7).} \end{aligned}$$

Hence $\diamond = \vee_{\leq}$.

2. Similar to (1).

□

4.1.5 Semilattice Homomorphisms

Definition 4.1.5.1. Let $(X, \leq_X), (Y, \leq_Y)$ be posets and $f : X \to Y$.

1. Suppose that $(X, \leq_X), (Y, \leq_Y)$ are join semilattices. Then f is said to be $(\leq_X, \leq_Y)$**-binary join preserving** if for each $a, b \in X$, $f(\sup(\{a, b\}, \leq_X)) = \sup(\{f(a), f(b)\}, \leq_Y)$.

2. Suppose that $(X, \leq_X), (Y, \leq_Y)$ are meet semilattices. Then f is said to be $(\leq_X, \leq_Y)$**-binary meet preserving** if for each $a, b \in X$, $f(\inf(\{a, b\}, \leq_X)) = \inf(\{f(a), f(b)\}, \leq_Y)$.

Exercise 4.1.5.2. Let $(X, \leq_X), (Y, \leq_Y)$ be posets and $f : X \to Y$.

1. Suppose that $(X, \leq_X), (Y, \leq_Y)$ are join semilattices. Then f is finite $(\leq_X, \leq_Y)$-join preserving iff f is a $(\vee_{\leq_X}, \vee_{\leq_Y})$-homomorphism.

2. Suppose that $(X, \leq_X), (Y, \leq_Y)$ are meet semilattices. Then f is finite $(\leq_X, \leq_Y)$-meet preserving iff f is a $(\wedge_{\leq_X}, \wedge_{\leq_Y})$-homomorphism.

Proof. Clear. □

Exercise 4.1.5.3. Let $(X, \diamond_X)$ and $(Y, \diamond_Y)$ be semilattices and $f : X \to Y$. Then the following are equivalent:

1. f is a $(\diamond_X, \diamond_Y)$-homomorphism
2. f is finite $(\leq^{\vee}_{\diamond_X}, \leq^{\vee}_{\diamond_Y})$-join preserving
3. f is finite $(\leq^{\wedge}_{\diamond_X}, \leq^{\wedge}_{\diamond_Y})$-meet preserving

Proof.

- (1) $\implies$ (2):
  Suppose that f is a $(\diamond_X, \diamond_Y)$-homomorphism. make/ref previous ex here implies that $\vee_{\leq^{\vee}_{\diamond_X}} = \diamond_X$ and $\vee_{\leq^{\vee}_{\diamond_Y}} = \diamond_Y$. Therefore f is a $(\vee_{\leq^{\vee}_{\diamond_X}}, \vee_{\leq^{\vee}_{\diamond_Y}})$-homomorphism. the previous ex implues that f is finite $(\leq^{\vee}_{\diamond_X}, \leq^{\vee}_{\diamond_Y})$-join preserving.

- (2) $\implies$ (3):
  Suppose that FINISH!!!

- (3) $\implies$ (1):

□

Exercise 4.1.5.4. Let $(X, \leq_X), (Y, \leq_Y)$ be posets and $f : X \to Y$.

1. Suppose that $(X, \leq_X)$ and $(Y, \leq_Y)$ are join semilattices. Then f is $(\leq_X, \leq_Y)$-monotone iff for each $a, b \in X$, $f(a) \vee_{\leq_Y} f(b) \leq_Y f(a \vee_{\leq_X} b)$.

2. Suppose that $(X, \leq_X)$ and $(Y, \leq_Y)$ are meet semilattices. Then f is $(\leq_X, \leq_Y)$-monotone iff for each $a, b \in X$, $f(a) \vee_{\leq_Y} f(b) \leq_Y f(a \vee_{\leq_X} b)$.

Proof.

1. - ($\implies$):
     Suppose that f is $(\leq_X, \leq_Y)$-monotone. Let $a, b \in X$. Since $a \vee_{\leq_X} b \in \mathrm{ub}(\{a, b\}, \leq_X)$, Exercise 3.2.6.7 implies that

     $$\begin{aligned} f(a \vee_{\leq_X} b) &\in f(\mathrm{ub}(\{a, b\}, \leq_X)) \\ &\subset \mathrm{ub}(f(\{a, b\}), \leq_Y) \\ &= \mathrm{ub}(\{f(a), f(b)\}, \leq_Y) \end{aligned}$$

     Therefore

     $$\begin{aligned} f(a) \vee_{\leq_Y} f(b) &= \sup(\{f(a), f(b)\}, \leq_Y) \\ &\leq f(a \vee_{\leq_X} b). \end{aligned}$$

     Since $a, b \in X$ are arbitrary, we have that for each $a, b \in X$, $f(a) \vee_{\leq_Y} f(b) \leq_Y f(a \vee_{\leq_X} b)$.

- ($\Longleftarrow$):
  Suppose that for each $a, b \in X$, $f(a) \vee_{\leq_Y} f(b) \leq_Y f(a \vee_{\leq_X} b)$. Let $a, b \in X$. Suppose that $a \leq_X b$. Exercise 3.3.3.5 then implies that $a \vee_{\leq_X} b = b$. By assumption,

$$\begin{aligned} f(a) \vee_{\leq_Y} f(b) &\leq_Y f(a \vee_{\leq_X} b) \\ &= f(b) \\ &\leq f(a) \vee_{\leq_Y} f(b). \end{aligned}$$

  Therefore $f(a) \vee_{\leq_Y} f(b) = f(b)$. Another application of Exercise 3.3.3.5 implies that $f(a) \leq_Y f(b)$. Since $a, b \in X$ with $a \leq b$ are arbitrary, we have that for each $a, b \in X$, $a \leq_X b$ implies that $f(a) \leq_Y f(b)$. Hence f is $(\leq_X, \leq_Y)$-monotone.

2. use duality FINISH!!!

□

Exercise 4.1.5.5. Let $(X, \leq_X), (Y, \leq_Y)$ be posets and $f : X \to Y$.

1. Suppose that $(X, \leq_X)$ and $(Y, \leq_Y)$ are join semilattices. If f preserves finite joins, then f is $(\leq_X, \leq_Y)$-monotone.
2. Suppose that $(X, \leq_X)$ and $(Y, \leq_Y)$ are meet semilattices. If f preserves finite meets, then f is $(\leq_X, \leq_Y)$-monotone.

Proof.

1. Suppose that f preserves finite joins. Then the previous exercise implies that

$$\begin{aligned} \text{for each } a, b \in X,\ f(a) \vee_{\leq_Y} f(b) = f(a \vee_{\leq_X} b) &\implies \text{for each } a, b \in X,\ f(a) \vee_{\leq_Y} f(b) \leq f(a \vee_{\leq_X} b) \\ &\implies f \text{ is } (\leq_X, \leq_Y)\text{-monotone} \end{aligned}$$

2. Suppose that $(X, \leq_X)$ and $(Y, \leq_Y)$ are meet semilattices. If f preserves finite meets, then f is $(\leq_X, \leq_Y)$-monotone.

□

maybe make exercise about galois connections and right adjoints preserving joins, left preserving meets

4.1.6 Lattices

Definition 4.1.6.1. Let X be a set and $\vee, \wedge$ semilattice operators on X. Then $\vee$ and $\wedge$ are said to **satisfy the lattice absorption identities** if for each $a, b \in X$,

1. $a \wedge (a \vee b) = a$
2. $a \vee (a \wedge b) = a$

Exercise 4.1.6.2. Let X be a set and $\vee, \wedge$ semilattice operators on X. Then $\vee$ and $\wedge$ satisfy the lattice absorption identities iff $\vee$ and $\wedge$ satisfy the skew lattice absorption identities.

Proof.

- ($\Longrightarrow$):
  Clear by commutativity.
- ($\Longleftarrow$):
  Immediate.

□

Definition 4.1.6.3. Let X be a set and $\vee, \wedge$ semilattice operators on X. Then $(X, \vee, \wedge)$ is said to be an **lattice** if

1. $X \neq \varnothing$

2. $\vee$ and $\wedge$ satisfy the skew lattice absorption identities.

Definition 4.1.6.4. Let $(X, \leq)$ be a poset. Then $(X, \leq)$ is said to be an **ordered lattice** if $(X, \leq)$ is a join semilattice and $(X, \leq)$ is a meet semilattice.

Exercise 4.1.6.5. Let $(X, \vee, \wedge)$ be a lattice. Then $(X, \leq_{\vee}^{\vee})$ is an ordered lattice.

Proof. Exercise 4.1.4.6 implies that $(X, \leq_{\vee}^{\vee})$ is a join semilattice and $(X, \leq_{\wedge}^{\wedge})$ is a meet semilattice. Exercise 4.1.3.4 implies that $\leq_{\vee}^{\vee} = \leq_{\wedge}^{\wedge}$. Therefore $(X, \leq_{\vee}^{\vee})$ is a join semilattice and $(X, \leq_{\vee}^{\vee})$ is a meet semilattice. Hence $(X, \leq_{\vee}^{\vee})$ is an ordered lattice. □

Exercise 4.1.6.6. Let $(X, \leq)$ be an ordered lattice. Then $(X, \vee_{\leq}, \wedge_{\leq})$ is a lattice.

Proof. Let $a, b \in X$.

1.
   - Since $\sup\{a, b\} \in \mathrm{ub}\{a, b\}$ is an upper bound of $\{a, b\}$, we have that $a \leq \sup\{a, b\}$. Since $a \leq a$, we have that $a \in \mathrm{lb}\{a, \sup\{a, b\}\}$. Hence $a \leq \inf\{a, \sup\{a, b\}\}$.
   - Since $\inf\{a, \sup\{a, b\}\} \in \mathrm{lb}\{a, \sup\{a, b\}\}$, we have that $\inf\{a, \sup\{a, b\}\} \leq a$.

   Since $a \leq \inf\{a, \sup\{a, b\}\}$ and $\inf\{a, \sup\{a, b\}\} \leq a$, we have that

$$\begin{aligned} a \wedge_{\leq} (a \vee_{\leq} b) &= \inf\{a, \sup\{a, b\}\} \\ &= a. \end{aligned}$$

2. Similarly, $a \vee_{\leq} (a \wedge_{\leq} b) = a$.

Since $a, b \in X$ are arbitrary, we have that for each $a, b \in X$,

1. $a \wedge_{\leq} (a \vee_{\leq} b) = a$
2. $a \vee_{\leq} (a \wedge_{\leq} b) = a$

Hence $\vee_{\leq}$ and $\wedge_{\leq}$ satisfy the lattice absorption identities. Therefore $(X, \vee_{\leq}, \wedge_{\leq})$ is an lattice. □

Exercise 4.1.6.7. Let $(X, \leq)$ be an ordered lattice. Then

1. $\leq_{\vee_{\leq}}^{\vee} = \leq_{\wedge_{\leq}}^{\wedge} = \leq$
2. $\leq_{\vee_{\leq}}^{\wedge} = \leq_{\wedge_{\leq}}^{\vee} = \leq^{*}$

Proof. Since $(X, \leq)$ is an ordered lattice, $(X, \leq)$ is a join semilattice $(X, \leq)$ is a meet semilattice. Exercise 4.1.4.9 then implies that

1.

$$\begin{aligned} \leq_{\vee_{\leq}}^{\vee} &= \leq \\ &= \leq_{\wedge_{\leq}}^{\wedge} \end{aligned}$$

2.

$$\begin{aligned} \leq_{\vee_{\leq}}^{\wedge} &= \leq^{*} \\ &= \leq_{\wedge_{\leq}}^{\vee} \end{aligned}$$

□

Exercise 4.1.6.8. Let $(X, \vee, \wedge)$ be an lattice. Then $\leq_{\vee}^{\vee}$ is the unique partial order $\leq$ on X such that $(X, \leq)$ is an ordered lattice and for each $a, b \in X$, $a \vee b = \sup(\{a, b\}, \leq)$.

Proof. Let $\leq$ be a partial order on X. Suppose that $(X, \leq)$ is an ordered lattice and for each $a, b \in X$, $a \vee b = \sup(\{a, b\}, \leq)$. Since $(X, \vee)$ is a semilattice, $(X, \leq_{\vee}^{\vee})$ and $(X, \leq)$ are join semilattices and for each $a, b \in X$, $a \vee b = \sup(\{a, b\}, \leq_{\vee}^{\vee})$ and $a \vee b = \sup(\{a, b\}, \leq)$, Exercise 4.1.4.10 implies that $\leq = \leq_{\vee}^{\vee}$. □

Exercise 4.1.6.9. Let $(X, \leq)$ be an ordered lattice. Then $\vee_{\leq}, \wedge_{\leq}$ are the unique semilattice operators $\vee, \wedge$ on X such that $(X, \vee, \wedge)$ is an lattice and for each $a, b \in X$, $a \vee b = \sup(\{a, b\}, \leq)$

Proof. Let $\vee, \wedge$ be semilattice operators on X. Suppose that $(X, \vee, \wedge)$ is a lattice and for each $a, b \in X$, $a \vee b = \sup(\{a, b\}, \leq)$. Then for each $a, b \in X$,

$$\begin{aligned} a \vee_{\leq} b &= \sup(\{a, b\}, \leq) \quad \text{(Definition 4.1.4.7)} \\ &= a \vee b \quad \text{(by assumption)}. \end{aligned}$$

Thus $\vee = \vee_{\leq}$. Since $(X, \vee, \wedge)$ is a lattice and for each $a, b \in X$, $a \vee b = \sup(\{a, b\}, \leq)$, Exercise 4.1.6.8 implies that $\leq_{\vee}^{\vee} = \leq$. Exercise 4.1.3.4 then implies that

$$\begin{aligned} \leq_{\wedge}^{\wedge} &= \leq_{\vee}^{\vee} \\ &= \leq . \end{aligned}$$

Therefore for each $a, b \in X$,

$$\begin{aligned} a \wedge b &= \inf(\{a, b\}, \leq_{\wedge}^{\wedge}) \quad \text{(Exercise 4.1.4.4)} \\ &= \inf(\{a, b\}, \leq) \\ &= a \wedge_{\leq} b \quad \text{(Definition 4.1.4.7)}. \end{aligned}$$

Hence $\wedge = \wedge_{\leq}$. □

4.1.7 Lattice Homomorphisms

???

4.2 Basic Properties

4.2.1 Bounded Lattices

mayb define bounded prosets,

Definition 4.2.1.1. Let $(L, \vee, \wedge)$ be a lattice.

- Let $a \in L$. Then
  - a is said to be a $(\vee, \wedge)$**-zero** if for each $x \in L$, $x \vee a = x$.
  - a is said to be a $(\vee, \wedge)$**-one** if for each $x \in L$, $x \wedge a = x$
- Then
  - $(L, \vee, \wedge)$ is said to **have a zero** if there exists $a \in L$ such that a is a $(\vee, \wedge)$-zero
  - $(L, \vee, \wedge)$ is said to **have a one** if there exists $a \in L$ such that a is a $(\vee, \wedge)$-one

Note 4.2.1.2. We the context is clear, we say "zero" and "one" inplace of "$(\vee, \wedge)$-zer" and "$(\vee, \wedge)$-one" respectively.

Exercise 4.2.1.3. Let L be a lattice and $a, b \in L$.

1. If a, b are zeros, then $a = b$.
2. If a, b are ones, then $a = b$.

Proof.

1. Suppose that a, b are zeros. Then

$$\begin{aligned} a &= a \wedge b \\ &= b \wedge a \\ &= b. \end{aligned}$$

2. Similar to (1).

□

Note 4.2.1.4.

- If L has a zero, we denote the unique zero of L by 0.
- If L has a one, we denote the unique one of L by 1.

Definition 4.2.1.5. Let $(L, \vee, \wedge)$ be a lattice. Then

- $(L, \vee, \wedge)$ is said to be a **lattice with** 0 if L has a $(\vee, \wedge)$-zero
- $(L, \vee, \wedge)$ is said to be a **lattice with** 1 if $(L, \vee, \wedge)$ has a $(\vee, \wedge)$-one
- $(L, \vee, \wedge, 0, 1)$ is said to be a **bounded lattice** if 0 is a $(\vee, \wedge)$-zero and 1 is a $(\vee, \wedge)$-one

Exercise 4.2.1.6. Let L be a lattice and $a \in L$. Then

1. a is a one of L iff $\sup L = a$.
2. a is a zero of L iff $\inf L = a$.

Proof.

1. - $(\implies)$:
     Suppose that a is a one of L. Let $x \in L$. Since $x \wedge a = x$, we have that $x \leq a$. Since $x \in L$ is arbitrary, we have that for each $x \in L$, $x \leq a$. Hence $a \in \operatorname{ub} L$. Exercise 3.3.3.7 then implies that $\sup L = a$.

- ($\Longleftarrow$):
  Suppose that $\sup L = a$. Let $x \in L$. Then Since $a \in \operatorname{ub} L$, $x \leq a$. Therefore

$$\begin{aligned} x \wedge a &= \inf\{x, a\} \\ &= x. \end{aligned}$$

  Since $x \in L$ is arbitrary, we have that for each $x \in L$, $x \wedge a = x$. Hence a is a one of L.

2. Similar to (1).

□

Exercise 4.2.1.7. Let $(L, \leq)$ be an ordered lattice. (maybe change to join/meet lattice) Then

1. $(L, \vee_{\leq}, \wedge_{\leq})$ is a lattice with 0 iff for each $A \subset L$, A is finite implies that $\sup A$ exists.
2. $(L, \vee_{\leq}, \wedge_{\leq})$ is a lattice with 1 iff for each $A \subset L$, A is finite implies that $\inf A$ exists.

Proof.

- ($\Longrightarrow$):
  Suppose that $(L, \vee_{\leq}, \wedge_{\leq})$ is a lattice with 0. Let $A \subset L$. Suppose that A is finite.
  - Suppose that $\#A = 0$. Then $A = \varnothing$. a previous ex implies that

$$\begin{aligned} \sup A \text{ exists} &\iff \inf(\operatorname{ub} A) \text{ exists} \\ &\iff \inf(\operatorname{ub} \varnothing) \text{ exists} \\ &\iff \inf L \text{ exists.} \end{aligned}$$

    Since L has a zero, the previous ex implies that $\inf L$ exists. Therefore $\sup A$ exists.
  - Suppose that $\#A \geq 1$. Since $(L, \leq)$ is an join semilattice, we have that for each $a, b \in L$, $\sup\{a, b\}$ exists. Exercise 3.3.3.6 then implies that $\sup A$ exists.
- ($\Longleftarrow$):
  Suppose that for each $A \subset L$, A is finite implies that $\sup A$ exists. Set $z := \sup \varnothing$. a prev ex implies that $z = \inf L$. Then

FINISH!!!! □

4.2.2 Bounded Lattice Homomorphisms

Definition 4.2.2.1. move to section on semilattice homomorphisms or make section on semilattice homomorphisms with 0 or 1 and move it there Let $(X, \leq_X), (Y, \leq_Y)$ be posets and $f : X \to Y$.

1. Suppose that $(X, \leq_X), (Y, \leq_Y)$ are join semilattices with 0. Then f is said to be $(\leq_X, \leq_Y)$**-finite join preserving** if for each $A \subset X$, A is finite implies that $f(\sup(A, \leq_X)) = \sup(f(A), \leq_Y)$.
2. Suppose that $(X, \leq_X), (Y, \leq_Y)$ are meet semilattices with 1. Then f is said to be $(\leq_X, \leq_Y)$**-finite meet preserving** if for each $A \subset X$, A is finite implies that $f(\inf(A, \leq_X)) = \inf(f(A), \leq_Y)$.

Exercise 4.2.2.2. Let $(L, \leq_L)$ and $(M, \leq_M)$ be posets and $f : L \to M$.

1. Suppose that $(L, \leq_L)$ and $(M, \leq_M)$ are join semilattices with 0. Then f is $(\leq_L, \leq_M)$-finite join preserving iff f is $(\leq_L, \leq_M)$-binary join preserving and $f(0_L) = 0_M$.
2. Suppose that $(L, \leq_L)$ and $(M, \leq_M)$ are meet semilattices with 1. Then f is $(\leq_L, \leq_M)$-finite meet preserving iff f is $(\leq_L, \leq_M)$-binary meet preserving and $f(1_L) = 1_M$.

Proof.

1.
   - ($\Longrightarrow$):
     Suppose that f is $(\leq_L, \leq_M)$-finite join preserving. Then clearly f is $(\leq_L, \leq_M)$-binary join preserving. Since $\varnothing$ is finite, a prev ex about $\sup \varnothing = 0$ implies that

$$\begin{aligned} f(0_L) &= f(\sup(\varnothing, \leq_L)) \\ &= \sup(f(\varnothing), \leq_M) \\ &= \sup(\varnothing, \leq_M) \\ &= 0_M. \end{aligned}$$

   - ($\Longleftarrow$):
     Suppose that f is $(\leq_L, \leq_M)$-binary join preserving and $f(0_L) = 0_M$. Let $A \subset L$. Suppose that A is finite.
     - Suppose that $\#A = 0$. Then $A = \varnothing$ and

$$\begin{aligned} f(\sup(A, \leq_L)) &= f(\sup(\varnothing, \leq_L)) \\ &= f(0_L) \\ &= 0_M \\ &= \sup(\varnothing, \leq_M) \\ &= \sup(f(\varnothing), \leq_M) \\ &= \sup(f(A), \leq_M). \end{aligned}$$

     - Suppose that $\#A = 1$. Then there exists $a \in L$ such that $A = \{a\}$. Hence

$$\begin{aligned} f(\sup(A, \leq_L)) &= f(\sup(\{a\}, \leq_L)) \\ &= f(a) \\ &= \sup(\{f(a)\}, \leq_M) \\ &= \sup(f(A), \leq_M). \end{aligned}$$

     - Suppose that $\#A \geq 2$. Then there exists $n \in \mathbb{N}$ and $a_1, \ldots, a_n \in L$ such that $n \geq 2$ and $A = \{a_1, \ldots, a_n\}$. ref prev ex implies that $\sup(A, \leq_L) = (\cdots((a_1 \vee_{\leq_L} a_2) \vee_{\leq_L} a_3) \cdots a_n)$. Then

$$\begin{aligned} f(\sup(A, \leq_L)) &= f(\cdots((a_1 \vee_{\leq_L} a_2) \vee_{\leq_L} a_3) \vee_{\leq_L} \cdots \vee_{\leq_L} a_n) \\ &= \end{aligned}$$

       do by induction on $\#A$ so it is less messy.

2.

FINISH!!! □

4.2.3 Irreducibility and Primality

Definition 4.2.3.1. Let $(L, \leq)$ be a poset.

- Suppose that $(L, \leq)$ is a join semilattice.
  - Let $x \in L$. Then x is said to be
    - $\leq$**-join irreducible** if
      1. x is not a zero of L,
      2. for each $a, b \in L$, $x = a \vee_{\leq} b$ implies that $x = a$ or $x = b$
    - $\leq$**-join prime**
      1. x is not a zero of L,
      2. if for each $a, b \in L$, $x \leq a \vee_{\leq} b$ implies that $x \leq a$ or $x \leq b$
  - We define

    * $\text{irred}_\vee(L, \leq) := \{x \in L : x \text{ is } \leq\text{-join irreducible.}\}$
    * $\text{prime}_\vee(L, \leq) := \{x \in L : x \text{ is } \leq\text{-join prime.}\}$

- Suppose that $(L, \leq)$ is a meet semilattice.
  - Let $x \in L$. Then x is said to be
    * **$\leq$-meet irreducible**
      1. x is not a one of L,
      2. if for each $a, b \in L$, $x = a \wedge_\leq b$ implies that $x = a$ or $x = b$
    * **$\leq$-meet prime** if
      1. x is not a one of L,
      2. for each $a, b \in L$, $a \wedge_\leq b \leq x$ implies that $a \leq x$ or $b \leq x$
  - We define
    * $\text{irred}_\wedge(L, \leq) := \{x \in L : x \text{ is } \leq\text{-meet irreducible.}\}$
    * $\text{prime}_\wedge(L, \leq) := \{x \in L : x \text{ is } \leq\text{-meet prime.}\}$

Note 4.2.3.2. When the context is clear,

- we say "join irreducible", "join prime", "meet irreducible" and "meet prime" in place of "$\leq$-join irreducible", "$\leq$-join prime", "$\leq$-meet irreducible" and "$\leq$-meet prime" respectively,
- we write $\text{irred}_\vee(L)$, $\text{prime}_\vee(L)$, $\text{irred}_\wedge(L)$ and $\text{prime}_\wedge(L)$ in place of $\text{irred}_\vee(L, \leq)$, $\text{prime}_\vee(L, \leq)$, $\text{irred}_\wedge(L, \leq)$ and $\text{prime}_\wedge(L, \leq)$ respectively

Exercise 4.2.3.3. Let $(L, \leq)$ be a poset.

1. Suppose that $(L, \leq)$ is a join semilattice. Then $\text{prime}_\vee(L) \subset \text{irred}_\vee(L)$.
2. Suppose that $(L, \leq)$ is a meet semilattice. Then $\text{prime}_\wedge(L) \subset \text{irred}_\wedge(L)$.

Proof.

1. Let $x \in \text{prime}_\vee(L)$.
   (a) Since x is join prime, x is not at zero of L.
   (b) Let $a, b \in L$. Suppose that $x = a \vee b$. Then $x \leq a \vee b$. Since x is join prime, $x \leq a$ or $x \leq b$.
   - Suppose that $x \leq a$. Then
   $$\begin{aligned} a \vee b &= x \\ &\leq a \\ &\leq a \vee b \end{aligned}$$
     and therefore $x = a$. Thus $x \leq a$ implies that $x = a$.
   - Similarly, $x \leq b$ implies that $x = b$.

   Since $x \leq a$ or $x \leq b$, we have that $x = a$ or $x = b$. Since $a, b \in L$ with $x = a \vee b$ are arbitrary, we have that for each $a, b \in L$, $x = a \vee b$ implies that $x = a$ or $x = b$.

   Thus $x \in \text{irred}_\vee(L)$. Since $x \in \text{prime}_\vee(L)$ is arbitrary, we have that for each $x \in \text{prime}_\vee(L)$, $x \in \text{irred}_\vee(L)$. Hence $\text{prime}_\vee(L) \subset \text{irred}_\vee(L)$.
2. Similar to (1) use duality.

□

4.3 Lattice Ideals and Filters

4.3.1 Introduction

Definition 4.3.1.1. Let L be a lattice.

- Let $J \subset L$. Then
  - J is said to be an **ideal of** L if
    1. $J \neq \varnothing$,
    2. for each $a, b \in J$, $a \vee b \in J$,
    3. for each $x \in L$ and $x \in J$, $x \wedge a \in J$,
  - J is said to be an **filter of** L if
    1. $J \neq \varnothing$,
    2. for each $a, b \in J$, $a \wedge b \in J$,
    3. for each $x \in L$ and $a \in J$, $x \vee a \in J$.
- We define
  - $\mathcal{I}(L) := \{I \subset L : I \text{ is an ideal of } L\}$
  - $\mathcal{F}(L) := \{F \subset L : F \text{ is a filter of } L\}$

Exercise 4.3.1.2. Let L be a lattice. If $L \neq \varnothing$, then

1. $L \in \mathcal{I}(L)$,
2. $L \in \mathcal{F}(L)$.

Proof.

1. (a) Since L is a lattice, $L \neq \varnothing$.
   (b) Clearly for each $a, b \in L$, $a \vee b \in L$
   (c) Clearly for each $x \in L$ and $b \in L$, $a \wedge b \in L$
2. Similar to (1) or use dualtity?.

□

Exercise 4.3.1.3. Let L be a lattice. Then for each $x \in L$,

1. $\downarrow x \in \mathcal{I}(L)$
2. $\uparrow x \in \mathcal{F}(L)$

Proof. Let $x \in L$.

1. (a) Since $x \in \downarrow x$, $\downarrow x \neq \varnothing$.
   (b) Let $a, b \in \downarrow x$. Then $a \leq x$ and $b \leq x$. Hence $x \in \mathrm{ub}(\{a, b\})$. Therefore
   $$\begin{aligned} a \vee b &= \sup\{a, b\} \\ &\leq x. \end{aligned}$$
   Thus $a \wedge b \in \downarrow x$. Since $a, b \in \downarrow x$ are arbitrary, we have that for each $a, b \in \downarrow x$, $a \vee b \in \downarrow x$.
   (c) Let $y \in L$ and $a \in \downarrow x$. Then $a \leq x$. Therefore
   $$\begin{aligned} y \wedge a &= \inf(\{y, a\}) \\ &\leq a \\ &\leq x. \end{aligned}$$
   Hence $y \wedge a \in \downarrow x$. Since $y \in L$ and $a \in \downarrow x$ are arbitrary, we have that for each $y \in L$ and $a \in \downarrow x$, $y \wedge a \in \downarrow x$.

2. use duality

□

Exercise 4.3.1.4. Let L be a lattice and $J \subset L$. Set $\leq_J := \leq|_{J^2}$. Then

1. $J \in \mathcal{I}(L)$ iff J is a $\leq$-lower set and $(J, \leq_J)$ is upward directed. rework after making some exercises about subprosets, subdirected sets and subposets and showing facts about sup inf of subprosets and upper/lower sets in sub-prosets

2. $J \in \mathcal{F}(L)$ iff J is an $\leq$-upper set and $(J, \leq_J)$ is downward directed.

Proof.

1. - ($\Longrightarrow$):
   Suppose that $J \in \mathcal{I}(L)$.
   - Let $x \in L$ and $a \in J$. Suppose that $x \leq a$. Then $x = \inf(\{a, b\}, \leq)$ and therefore
   $$\begin{aligned} x &= \inf(\{x, a\}, \leq) \\ &= x \wedge_{\leq} a \\ &\in J. \end{aligned}$$
   Since $a \in J$ and $x \in L$ with $x \leq a$ are arbitrary, we have that J is a down set.
   - Since $J \in \mathcal{I}(L)$, we have that $J \neq \varnothing$. Let $a, b \in J$. Since $J \in \mathcal{I}(L)$,
   $$\begin{aligned} &\sup(\{a, b\}, \leq) \\ &\qquad = a \vee_{\leq} b \\ &\qquad \in J. \end{aligned}$$
   Hence (ref ex here)
   $$\begin{aligned} \sup(\{a, b\}, \leq) &= \sup(\{a, b\}, \leq_J) \\ &\in \mathrm{ub}(\{a, b\}, \leq_J) \end{aligned}$$
   and $\mathrm{ub}(\{a, b\}, \leq_J) \neq \varnothing$. Since $a, b \in J$ are arbitrary, we have that for each $a, b \in J$, $\mathrm{ub}(\{a, b\}, \leq_J) \neq \varnothing$. Since $J \neq \varnothing$ and for each $a, b \in J$, $\mathrm{ub}(J, \leq_J) \neq \varnothing$, we have that $(J, \leq_J)$ is upward directed.
   - ($\Longleftarrow$):
   Suppose that J is a $\leq$-lower set and $(J, \leq_J)$ is upward directed.
   (a) Since $(J, \leq_J)$ is upward directed, $J \neq \varnothing$.
   (b) Let $a, b \in J$. Since $(J, \leq_J)$ is upward directed, $\mathrm{ub}(\{a, b\}, \leq_J) \neq \varnothing$. Thus there exists (ref ex here)
   $$\begin{aligned} c &\in \mathrm{ub}(\{a, b\}, \leq_J) && (4.1) \\ &\subset \mathrm{ub}(\{a, b\}, \leq). && (4.2) \end{aligned}$$
   Therefore $\sup(\{a, b\}, \leq) \leq c$. Since J is a $\leq$-lower set, $c \in J$ and $\sup(\{a, b\}, \leq) \leq c$, we have that
   $$\begin{aligned} a \vee b &= \sup(\{a, b\}, \leq) \\ &\in J. \end{aligned}$$
   Since $a, b \in J$ are arbitrary, we have that for each $a, b \in J$, $a \vee b \in J$. rework after making some exercises about subprosets, subdirected sets and subposets and showing facts about sup inf of subprosets and upper/lower sets in sub-prosets
   (c) Let $x \in L$ and $a \in J$. Then $x \wedge a \leq a$. Since J is a $\leq$-lower set, $x \wedge a \in J$. Since $x \in L$ and $a \in J$ are arbitrary, we have that for each $x \in L$ and $a \in J$, $x \wedge a \in J$.

   Therefore $J \in \mathcal{I}(L)$.

2.

□

Exercise 4.3.1.5. Let L be a lattice and $J \subset L$.

1. If L has a zero and $J \in \mathcal{I}(L)$, then $0 \in J$.
2. If L has a one and $J \in \mathcal{F}(L)$, then $1 \in J$.

Proof.

1. Suppose that L has a zero and $J \in \mathcal{I}(L)$. Since J is an ideal of L, $J \neq \varnothing$ and a previous ex implies J is a lower set. Hence there exists $a \in J$. a prev ex implies that $0 \in \operatorname{lb} L$. Thus $0 \leq a$. Since J is a lower set, $0 \in J$.
2. Similar to (1).

□

Exercise 4.3.1.6. Let L be a lattice.

1. Let $(I_\alpha)_{\alpha \in A} \subset \mathcal{I}(L)$.
   (a) If $\bigcap_{\alpha \in A} I_\alpha \neq \varnothing$, then $\bigcap_{\alpha \in A} I_\alpha \in \mathcal{I}(L)$.
   (b) If $A = \{1, 2\}$, then $\bigcap_{\alpha \in A} I_\alpha \in \mathcal{I}(L)$.
   (c) If L has a zero, then $\bigcap_{\alpha \in A} I_\alpha \in \mathcal{I}(L)$.
2. Let $(F_\alpha)_{\alpha \in A} \subset \mathcal{F}(L)$.
   (a) If $\bigcap_{\alpha \in A} F_\alpha \neq \varnothing$, then $\bigcap_{\alpha \in A} F_\alpha \in \mathcal{F}(L)$.
   (b) If $A = \{1, 2\}$, then $\bigcap_{\alpha \in A} F_\alpha \in \mathcal{F}(L)$.
   (c) If L has a one, then $\bigcap_{\alpha \in A} F_\alpha \in \mathcal{F}(L)$.

Proof.

1. Set $I := \bigcap_{\alpha \in A} I_\alpha$.
   (a) Suppose that $I \neq \varnothing$.
      i. By assumption $I \neq \varnothing$.
      ii. Let $a, b \in I$ and $\alpha \in A$. Then $a, b \in I_\alpha$. Since $I_\alpha \in \mathcal{I}(L)$, $a \vee b \in I_\alpha$. Since $\alpha \in A$ is arbitrary, we have that for each $\alpha \in A$, $a \vee b \in I_\alpha$. Hence $a \vee b \in I$.
      iii. Let $x \in L$, $a \in I$ and $\alpha \in A$. Then $a \in I_\alpha$. Since $I_\alpha \in \mathcal{I}(L)$, $x \in L$ and $a \in I_\alpha$, we have that $x \wedge a \in I_\alpha$. Since $\alpha \in A$ is arbitrary, we have that for each $\alpha \in A$, $x \wedge a \in I_\alpha$. Hence $x \wedge a \in I$. Since $x \in L$ and $a \in I$ are arbitrary, we have that for each $x \in L$ and $a \in I$, $x \wedge a \in I$.

      Thus $I \in \mathcal{I}(L)$.
   (b) Suppose that $A = \{1, 2\}$.
      i. Since $I_1, I_2 \in \mathcal{I}(L)$, $I_1 \neq \varnothing$ and $I_2 \neq \varnothing$. Thus there exist $x_1 \in I_1$ and $x_2 \in I_2$. Since $I_1 \in \mathcal{I}(L)$, $x_2 \in L$ and $x_1 \in I_1$, we have that $x_1 \wedge x_2 \in I_1$. Similarly, $x_1 \wedge x_2 \in I_2$. Hence $x_1 \wedge x_2 \in I_1 \cap I_2$ and $I_1 \cap I_2 \neq \varnothing$. Part 1($a$) implies that $I_1 \cap I_2 \in \mathcal{I}(L)$.
   (c) Suppose that L has a zero. Let $\alpha \in A$. Since $I_\alpha \in \mathcal{I}(L)$, a previous exercise implies that $0 \in I_\alpha$. Since $\alpha \in A$ is arbitrary, we have that for each $\alpha \in A$, $0 \in I_\alpha$. Hence $0 \in I$. Thus $I \neq \varnothing$. The previous part implies that $I \in \mathcal{I}(L)$.

2. Similar to (1).

□

Exercise 4.3.1.7. Let L be a lattice. Then

1. for each $I_1, I_2 \in \mathcal{I}(L)$, $\{x \in L : \text{there exist } a \in I_1 \text{ and } b \in I_2 \text{ such that } x \leq a \vee b\} \in \mathcal{I}(L)$.
2. for each $F_1, F_2 \in \mathcal{F}(L)$, $\{x \in L : \text{there exist } a \in F_1 \text{ and } b \in F_2 \text{ such that } a \vee b \leq x\} \in \mathcal{F}(L)$ (maybe FIX!!!).

Proof.

1. Let $I_1, I_2 \in \mathcal{I}(L)$. Set $I := \{x \in L : \text{there exist } a \in I_1 \text{ and } b \in I_2 \text{ such that } x \leq a \vee b\}$.
   (a) Since $I_1, I_2 \in \mathcal{I}(L)$, we have that $I_1 \neq \varnothing$ and $I_2 \neq \varnothing$. Hence there exist $x_1 \in I_1$ and $x_2 \in I_2$. Since $x_1 \leq x_1 \vee x_2$, we have that $x_1 \in I$. Thus $I \neq \varnothing$.
   (b) Let $a, b \in I$. Then there exist $x_a, x_b \in I_1$ and $y_a, y_b \in I_2$ such that $a \leq x_a \vee y_a$ and $b \leq x_b \vee y_b$. Define $z_1, z_2 \in L$ by $z_1 := x_a \vee x_b$ and $z_2 := y_a \vee y_b$. Since $I_1, I_2 \in \mathcal{I}(L)$, we have that $z_1 \in I_1$ and $z_2 \in I_2$. We note that
   $$\begin{aligned} a &\leq x_a \vee y_a \\ &\leq (x_a \vee y_a) \vee (x_b \vee y_b) \\ &= (x_a \vee x_b) \vee (y_a \vee y_b) \\ &= z_1 \vee z_2. \end{aligned}$$
   and similarly $b \leq z_1 \vee z_2$. Therefore $z_1 \vee z_2 \in \mathrm{ub}(\{a, b\}, \leq)$. Hence
   $$\begin{aligned} a \vee b &= \sup(\{a, b\}, \leq) \\ &\leq z_1 \vee z_2. \end{aligned}$$
   Thus $a \vee b \in I$. Since $a, b \in I$ are arbitrary, we have that for each $a, b \in I$, $a \vee b \in I$.
   (c) Let $x \in L$ and $y \in I$. Then there exists $a \in I_1$ and $b \in I_2$ such that $y \leq a \vee b$. Then
   $$\begin{aligned} x \wedge y &\leq y \\ &\leq a \vee b. \end{aligned}$$
   Therefore $x \wedge y \in I$. Since $x \in L$ and $y \in I$ are arbitrary, we have that for each $x \in L$ and $y \in I$, $x \wedge y \in I$.

   Therefore $I \in \mathcal{I}(L)$.
2. Similar to (1). FINISH!!!

□

Definition 4.3.1.8. Let L be a lattice. Define $\leq_{\mathcal{I}(L)} := \subset|_{\mathcal{I}(L)}$ (maybe standardize notation).

Exercise 4.3.1.9. Let L be a lattice.

1. Let $I_1, I_2 \in \mathcal{I}(L)$.
   (a) $\sup(\{I_1, I_2\}, \leq_{\mathcal{I}(L)}) = \{x \in L : \text{there exist } a \in I_1 \text{ and } b \in I_2 \text{ such that } x \leq a \vee b\}$.
   (b) $\inf(\{I_1, I_2\}, \leq_{\mathcal{I}(L)}) = I_1 \cap I_2$
2. Set $\leq_{\mathcal{F}(L)} := \subset|_{\mathcal{F}(L)}$ (maybe standardize notation). Let $F_1, F_2 \in \mathcal{F}(L)$. FIX!!!
   (a) $\sup(\{F_1, F_2\}, \leq_{\mathcal{I}(L)}) = \{x \in L : \text{there exist } a \in F_1 \text{ and } b \in F_2 \text{ such that } x \leq a \vee b\}$.
   (b) $\inf(\{F_1, F_2\}, \leq_{\mathcal{F}(L)}) = F_1 \cap F_2$

Proof.

1. (a) Set $I := \{x \in L : \text{there exist } a \in I_1 \text{ and } b \in I_2 \text{ such that } x \leq a \vee b\}$. (ref previous ex) implies that $I \in \mathcal{I}(L)$.

i. • Let $a \in I_1$. Since $I_2 \in \mathcal{I}(L)$, $I_2 \neq \varnothing$. Hence there exists $b \in I_2$. Then $a \leq a \vee b$. Since $a \in I_1$, $b \in I_2$ and $a \leq a \vee b$, we have that $a \in I$. Since $a \in I_1$ is arbitrary, we have that for each $a \in I_1$, $a \in I$. Thus $I_1 \subset I$.
   • Similarly, $I_2 \subset I$.

   Since $I_1, I_2 \subset I$, we have that $I \in \mathrm{ub}(\{I_1, I_2\}, \leq_{\mathcal{I}(L)})$.

ii. Let $K \in \mathrm{ub}(\{I_1, I_2\}, \leq_{\mathcal{I}(L)})$ and $x \in I$. Then $I_1, I_2 \subset K$ and there exist $a \in I_1$ and $b \in I_2$ such that $x \leq a \vee b$. Since $a \in I_1$, $b \in I_2$ and $I_1, I_2 \subset K$, we have that $a, b \in K$. Since $K \in \mathcal{I}(L)$, $a \vee b \in K$. Since $K \in \mathcal{I}(L)$ (ref a prev ex) implies that K is a lower set. Since K is a lower set, $a \vee b \in K$ and $x \leq a \vee b$, we have that $x \in K$. Since $x \in I$ is arbitrary, we have that for each $x \in I$, $x \in K$. Hence $I \subset K$. Since $K \in \mathrm{ub}(\{I_1, I_2\}, \leq_{\mathcal{I}(L)})$ is arbitrary, we have that for each $K \in \mathrm{ub}(\{I_1, I_2\}, \leq_{\mathcal{I}(L)})$, $I \leq_{\mathcal{I}(L)} K$.

Since

i. $I \in \mathrm{ub}(\{I_1, I_2\}, \leq_{\mathcal{I}(L)})$,

ii. for each $K \in \mathrm{ub}(\{I_1, I_2\}, \leq_{\mathcal{I}(L)})$, $I \leq_{\mathcal{I}(L)} K$,

we have that $I = \sup(\{I_1, I_2\}, \leq_{\mathcal{I}(L)})$.

(b) (ref previous ex) implies that $I_1 \cap I_2 \in \mathcal{I}(L)$. Since $I_1 \cap I_2 \subset I_1$ and $I_1 \cap I_2 \subset I_2$, we have that $I_1 \cap I_2 \in \mathrm{lb}(\{I_1, I_2\}, \leq_{\mathcal{I}(L)})$. Since $(\mathcal{I}(L), \leq_{\mathcal{I}(L)})$ is a subposet of $(\mathcal{P}(L), \subset)$, (define subproset/subposet and make exercise for the following fact) implies that

$$\begin{aligned}\inf(\{I_1, I_2\}, \leq_{\mathcal{I}(L)}) &\leq \inf(\{I_1, I_2\}, \subset) \\ &= I_1 \cap I_2 \\ &\leq \inf(\{I_1, I_2\}, \leq_{\mathcal{I}(L)}).\end{aligned}$$

Hence $I_1 \cap I_2 = \inf(\{I_1, I_2\}, \leq_{\mathcal{I}(L)})$.

2. Similar to (1). FINISH!!!

□

Definition 4.3.1.10. Let L be a lattice. Define $\vee_{\mathcal{I}(L)}, \wedge_{\mathcal{I}(L)} : \mathcal{I}(L) \times \mathcal{I}(L) \to \mathcal{I}(L)$ and $\vee_{\mathcal{F}(L)}, \wedge_{\mathcal{F}(L)} : \mathcal{F}(L) \times \mathcal{F}(L) \to \mathcal{F}(L)$ by

1. • $I_1 \vee_{\mathcal{I}(L)} I_2 := \{x \in L : \text{there exist } a \in I_1 \text{ and } b \in I_2 \text{ such that } x \leq a \vee b\}$
   • $I_1 \wedge_{\mathcal{I}(L)} I_2 := I_1 \cap I_2$.

2. FIX!!!

   • $F_1 \vee_{\mathcal{F}(L)} F_2 := \{x \in L : \text{there exist } a \in F_1 \text{ and } b \in F_2 \text{ such that } x \leq a \vee b\}$
   • $F_1 \wedge_{\mathcal{F}(L)} F_2 := F_1 \cap F_2$.

Exercise 4.3.1.11. Let L be a lattice. Then

1. $(\mathcal{I}(L), \vee_{\mathcal{I}(L)}, \wedge_{\mathcal{I}(L)})$ is a lattice

2. $(\mathcal{F}(L), \vee_{\mathcal{F}(L)}, \wedge_{\mathcal{F}(L)})$ is a lattice

Proof.

1. the previous exercise implies that $(\mathcal{I}(L), \leq_{\mathcal{I}(L)})$ is an ordered lattice. Exercise 4.1.6.6 then implies that $(\mathcal{I}(L), \vee_{\mathcal{I}(L)}, \wedge_{\mathcal{I}(L)})$ is a lattice.

2.

□

4.3.2 Properties of Ideals and Filters

Definition 4.3.2.1. Let L be a lattice.

- Let $I \in \mathcal{I}(L)$. Then I is said to be
  - **proper** if $I \neq L$.
  - **maximal** if I is maximal in $(\mathcal{I}(L) \setminus \{L\}, \subset)$.
  - **prime** if
    1. I is proper
    2. for each $a, b \in L$, $a \wedge b \in I$ implies that $a \in I$ or $b \in I$.
  - **principal** if there exists $a \in L$ such that $I = \downarrow a$.
- Let $F \in \mathcal{F}(L)$. Then F is said to be
  - **proper** if $F \neq L$.
  - **maximal** if F is maximal in $(\mathcal{F}(L) \setminus \{L\}, \subset)$.
  - **prime** if
    1. F is proper
    2. for each $a, b \in L$, $a \vee b \in F$ implies that $a \in F$ or $b \in F$.
  - **principal** if there exists $a \in L$ such that $F = \uparrow a$.

Exercise 4.3.2.2. Let L be a lattice.

1. For each $I \in \mathcal{I}(L)$.
   (a) if I is proper, then for each $x \in I$, x is not a one of L.
   (b) if I is prinicpal, then I is proper iff for each $x \in I$, x is not a one of L.
2. For each $F \in \mathcal{F}(L)$.
   (a) If F is proper, then for each $x \in F$, x is not a zero of L.
   (b) If F is prinicpal, then F is proper iff for each $x \in F$, x is not a zero of L.

Proof.

1. Let $I \in \mathcal{I}(L)$.
   (a) Suppose that I is proper. Let $x \in I$. For the sake of contradiction, suppose that x is a one of L. (previous ex) then implies that $(L, \leq) = \{x\}$. Let $y \in L$. Then $y \leq x$. Since I is a lower set, $x \in I$ and $y \leq x$, we have that $y \in I$. Since $y \in L$ is arbitrary, we have that for each $y \in L$, $y \in I$. Thus
   $$\begin{aligned} L &\subset I \\ &\subset L. \end{aligned}$$
   Thus $I = L$ and I is not proper. Therefore x is not a one of L. Since $x \in I$ is arbitrary, we have that for each $x \in I$, x is not a one of L.
   (b) Suppose that I is principal.
   - $(\implies)$:
     If I is proper, then $1(a)$ implies that for each $x \in I$, x is not a one of L.

- ($\Longleftarrow$):
  Suppose that I is not proper. Then $I = L$. Since I is principal, there exists $a \in I$ such that $I = \downarrow a$. Let $x \in L$. Since

$$\begin{aligned} x &\in L \\ &= I \\ &= \downarrow a, \end{aligned}$$

  we have that $x \leq a$. Hence

$$\begin{aligned} x \wedge a &= \inf\{x, a\} \\ &= x. \end{aligned}$$

  Since $x \in L$ is arbitrary, we have that for each $x \in L$, $x \wedge a = x$. Hence a is a one of L. Therefore I is not proper implies that there exists $a \in I$ such that a is a one of L. By contrapositive, we have that if for each $x \in I$, x is not a one of L, then I is proper.

2. FINISH!!

□

Exercise 4.3.2.3. Let L be a lattice and $x \in L$.

1. Then $\downarrow x \in \mathcal{I}_p(L)$ iff $x \in \text{prime}_\wedge(L)$.
2. Then $\uparrow x \in \mathcal{F}_p(L)$ iff $x \in \text{prime}_\vee(L)$.

Proof.

1. - ($\Longrightarrow$):
     Suppose that $\downarrow x \in \mathcal{I}_p(L)$. Then $\downarrow x$ is proper and for each $a, b \in L$, $a \wedge b \in \downarrow x$ implies that $a \in \downarrow x$ or $b \in \downarrow x$.
     - Since $\downarrow x$ is proper, previous exercise implies that x is not a one of L.
     - Let $a, b \in L$. Suppose that $a \wedge b \leq x$. Since $\downarrow x$ is a lower set, $x \in \downarrow x$ and $a \wedge b \leq x$, we have that $a \wedge b \in \downarrow x$. Therefore $a \in \downarrow x$ or $b \in \downarrow x$. Then $a \leq x$ or $b \leq x$. Since $a, b \in L$ with $a \wedge b \leq x$ are arbitrary, we have that for each $a, b \in L$, $a \wedge b \leq x$ implies that $a \leq x$ or $b \leq x$.

     Thus $x \in \text{prime}_\wedge(L)$.
   - ($\Longleftarrow$):
     Suppose that $x \in \text{prime}_\wedge(L)$.
     - For the sake of contradiction, suppose that $\downarrow x$ is not proper. Then $\downarrow x = L$. Thus $\text{ub}\, L = \{x\}$. A previous ex implies that x is a one of L. This is a contracition since $x \in \text{prime}_\wedge(L)$. Therefore $\downarrow x$ is proper.
     - Let $a, b \in L$. Suppose that $a \wedge b \in \downarrow x$. Then $a \wedge b \leq x$. Since $x \in \text{prime}_\wedge(L)$, $a \leq x$ or $b \leq x$. Hence $a \in \downarrow x$ or $b \in \downarrow x$. Since $a, b \in L$ with $a \wedge b \in \downarrow x$ are arbitrary, we have that for each $a, b \in L$, $a \wedge b \in \downarrow x$ implies that $a \in \downarrow x$ or $b \in \downarrow x$.

     Therefore $\downarrow x \in \mathcal{I}_p(L)$.
2. Similar to (1) FIX/CHECK/FINISH!!!

□

Exercise 4.3.2.4. Let L be a lattice.

1. For each $I \in \mathcal{I}(L)$, $I \in \mathcal{I}_p(L)$ iff $I^c \in \mathcal{F}(L)$.
2. For each $F \in \mathcal{F}(L)$, $F \in \mathcal{F}_p(L)$ iff $F^c \in \mathcal{I}(L)$.

Proof.

1. Let $I \in \mathcal{I}(L)$.

- ($\Longrightarrow$):
  Suppose that $I \in \mathcal{I}_p(L)$. Then I is proper and for each $a, b \in L$, $a \wedge b \in I$ implies that $a \in I$ or $b \in I$.
  (a) Since I is proper, $I \neq L$. Therefore $I^c \neq \varnothing$
  (b) Let $a, b \in I^c$. For the sake of contradiction, suppose that $a \wedge b \in I$. Since $I \in \mathcal{I}_p(L)$, $a \in I$ or $b \in I$. This is a contradiction since $a \in I^c$ and $b \in I^c$. Therefore $a \wedge b \notin I$. Hence $a \wedge b \in I^c$. Since $a, b \in I^c$ are arbitrary, we have that for each for each $a, b \in I^c$, $a \wedge b \in I^c$.
  (c) Since $I \in \mathcal{I}(L)$, a previous exercise implies that I is a lower set. an exercise in proset section implies that I^c is an upper set. Let $x \in L$ and $a \in I^c$. Since

$$\begin{aligned} x \vee a &\geq a \\ &\in I^c \end{aligned}$$

  and I^c is an upper set, we have that $x \vee a \in I^c$. Since $x \in L$ and $a \in I^c$ are arbitrary, we have that for each for each $x \in L$ and $a \in I^c$, $x \vee a \in I^c$.1
  Thus $I^c \in \mathcal{F}(L)$.
- ($\Longleftarrow$):
  Suppose that $I^c \in \mathcal{F}(L)$.
  (a) Since $I^c \in \mathcal{F}(L)$, $I^c \neq \varnothing$. Then $I \neq L$ and therefore I is proper.
  (b) Let $a, b \in L$. Suppose that $a \wedge b \in I$. For the sake of contradiction, suppose that $a \notin I$ and $b \notin I$. Then $a \in I^c$ and $b \in I^c$. Since $I^c \in \mathcal{F}(L)$, we have that $a \wedge b \in I^c$. This is a contradiction since $a \wedge b \in I$. Hence $a \in I$ or $b \in I$. Since $a, b \in L$ with $a \wedge b \in I$ are arbitrary, we have that for each $a, b \in L$, $a \wedge b \in I$ implies that $a \in I$ or $b \in I$.
  Thus $I \in \mathcal{I}_p(L)$.

2. use duality.

□

4.4 Contructions with Lattices

4.4.1 Sublattices

Definition 4.4.1.1. Let $(L, \vee_L, \wedge_L)$ and $(M, \vee_M, \wedge_M)$ be lattices. Then $(M, \vee_M, \wedge_M)$ is said to be a **sublattice of** $(L, \vee_L, \wedge_L)$ if

1. $M \subset L$
2. ι_M is a $((\vee_L, \wedge_L), (\vee_M, \wedge_M))$-homomorphism.

Exercise 4.4.1.2. Let $(L, \vee_L, \wedge_L)$ and $(M, \vee_M, \wedge_M)$ be lattices. Then $(M, \vee_M, \wedge_M)$ is a sublattice of L iff

1. $M \subset L$
2. $\vee_L|_{M^2} = \vee_M$ and $\wedge_L|_{M^2} = \wedge_M$.

Proof.
- ($\Longrightarrow$):
  Suppose that $(M, \vee_M, \wedge_M)$ is a sublattice of L.
  1. By definition, $M \subset L$.
  2. By definiton, ι_M is a $((\vee_L, \wedge_L), (\vee_M, \wedge_M))$-homomorphism. Then for each $a, b \in M$,

$$\begin{aligned} \vee_M(a, b) &= a \vee_M b \\ &= \iota_M(a \vee_M b) \\ &= \iota_M(a) \vee_L \iota_M(b) \\ &= a \vee_L b \\ &= \vee_L|_{M^2}(a, b). \end{aligned}$$

  Hence $\vee_L|_{M^2} = \vee_M$. Similarly, $\wedge_L|_{M^2} = \wedge_M$.

- ($\Longleftarrow$):
  Suppose that

  1. $M \subset L$
  2. $\vee_L|_{M^2} = \vee_M$ and $\wedge_L|_{M^2} = \wedge_M$.

  1. By assumption $M \subset L$.
  2. Since $\vee_L|_{M^2} = \vee_M$, we have that for each $a, b \in M$,

$$\begin{aligned}\iota_M(a \vee_M b) &= a \vee_M b \\ &= \vee_M(a, b) \\ &= \vee_L|_{M^2}(a, b) \\ &= \vee_L(a, b) \\ &= a \vee_L b.\end{aligned}$$

  Hence ι_M is a $(\vee_M, \vee_L)$-homomorphism. Similarly, ι_M is a $(\wedge_M, \wedge_L)$-homomorphism. Therefore ι_M is a $((\vee_L, \wedge_L), (\vee_M, \wedge_M)$ homomorphism.

  Thus $(M, \vee_M, \wedge_M)$ is a sublattice of L.

FINISH!!!! □

Definition 4.4.1.3. Let $(L, \vee_L, \wedge_L)$ and $(M, \vee_M, \wedge_M)$ be lattices. Then $(M, \vee_M, \wedge_M)$ is a sublattice of $(L, \vee_L, \wedge_L)$ if

1. $M \subset L$
2. $\vee_L|_{M^2} = \vee_M$ and $\wedge_L|_{M^2} = \wedge_M$.

Exercise 4.4.1.4. Let $(L, \vee, \wedge)$ be a lattice and $M \subset L$. Then $(M, \vee|_{M^2}, \wedge|_{M^2})$ is a sublattice of $(L, \vee, \wedge)$ iff

1. $M \neq \varnothing$
2. $\vee(M^2), \wedge(M^2) \subset M$.

Proof. Set $\vee_M := \vee|_{M^2}$ and $\wedge_M := \wedge|_{M^2}$

- ($\Longrightarrow$):
  Suppose that $(M, \vee_M, \wedge_M)$ is a sublattice of $(L, \vee, \wedge)$. Since $(M, \vee_M, \wedge_M)$ is a sublattice of $(L, \vee, \wedge)$, we have that $(M, \vee_M, \wedge_M)$ is a lattice.

  1. Since $(M, \vee_M, \wedge_M)$ is a lattice, $M \neq \varnothing$.
  2. Since $(M, \vee_M, \wedge_M)$ is a lattice, $\vee_M(M^2), \wedge_M(M^2) \subset M$.

- ($\Longleftarrow$):
  Suppose that

  1. $M \neq \varnothing$
  2. $\vee(M^2), \wedge(M^2) \subset M$.

  1. By assumption $M \neq \varnothing$,
  2. Since $\vee, \wedge$ are semilattice operators, $\vee, \wedge$ satisfy the lattice absorption identities and $\vee(M^2), \wedge(M^2) \subset M$, we have that $\vee_M, \wedge_M$ are semilattice operators and $\vee|_M, \wedge|_M$ satisfy the lattice absorption identities.

  Thus $(M, \vee_M, \wedge_M)$ is a lattice. By assumption, $M \subset L$ and $\vee_M = \vee|_{M^2}$, $\wedge_M = \wedge|_{M^2}$, we have that $(M, \vee_M, \wedge_M)$ is a sublattice of $(L, \vee, \wedge)$.

□

4.4.2 Product Lattices

4.4.3 Coproduct Lattices

4.4.4 Quotient Lattices

4.5 Modular and Distributive Lattices

4.5.1 Introduction

Definition 4.5.1.1. Let L be a lattice. Then L is said to be a **distributive lattice** if for each $a, b, c \in L$, $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$.

Exercise 4.5.1.2. Let L be a lattice. Then for each $a, b, c \in L$,

1. $(a \wedge b) \vee (a \wedge c) \leq a \wedge (b \vee c)$.
2. $a \wedge (b \vee c) \leq (a \vee b) \wedge (a \vee c)$

Proof. Let $a, b, c \in L$.

1.
   - Since $a \wedge b \leq a$ and $a \wedge c \leq a$, we have that $a \in \mathrm{ub}\{a \wedge b, a \wedge c\}$. Therefore
   $$\begin{aligned}(a \wedge b) \vee (a \wedge c) &= \sup\{a \wedge b, a \wedge c\} \\ &\leq a.\end{aligned}$$
   - Since
   $$\begin{aligned}a \wedge b &\leq b \\ &\leq b \vee c\end{aligned}$$
   and
   $$\begin{aligned}a \wedge c &\leq c \\ &\leq b \vee c,\end{aligned}$$
   we have that $b \vee c \in \mathrm{ub}\{a \wedge b, a \wedge c\}$. Therefore
   $$\begin{aligned}(a \wedge b) \vee (a \wedge c) &= \sup\{a \wedge b, a \wedge c\} \\ &\leq b \vee c.\end{aligned}$$

   Then $(a \wedge b) \vee (a \wedge c) \in \mathrm{lb}\{a, b \vee c\}$. Hence
   $$\begin{aligned}(a \wedge b) \vee (a \wedge c) &\leq \inf\{a, b \vee c\} \\ &= a \wedge (b \vee c).\end{aligned}$$
2. FINISH!!! (use duality)

□

Exercise 4.5.1.3. Let L be a lattice. Then for each $a, b, c \in L$,

1. $c \leq a$ implies that $(a \wedge b) \vee c \leq a \wedge (b \vee c)$
2. $a \leq c$ implies that $a \vee (b \wedge c) \leq (a \vee b) \wedge c$

Proof. Let $a, b, c \in L$.

1. Suppose that $c \leq a$. Then $a \wedge c = c$. The previous exercise then implies that

$$\begin{aligned}(a \wedge b) \vee c &= (a \wedge b) \vee (a \wedge c) \\ &\leq a \wedge (b \vee c).\end{aligned}$$

2. FINISH!!! use duality

□

Exercise 4.5.1.4. Let L be a lattice. Then for each $a, b, c \in L$,

$$(a \wedge b) \vee (b \wedge c) \vee (c \wedge a) \leq (a \vee b) \wedge (b \vee c) \wedge (c \vee a).$$

Proof. Let $a, b, c \in L$.

- We first note that

$$\begin{aligned}a \wedge b &\leq a \leq a \vee b, \\ b \wedge c &\leq b \leq a \vee b, \\ c \wedge a &\leq a \leq a \vee b.\end{aligned}$$

  Therefore $a \vee b \in \mathrm{ub}\{a \wedge b, b \wedge c, c \wedge a\}$ and

$$\begin{aligned}(a \wedge b) \vee (b \wedge c) \vee (c \wedge a) &= \sup\{a \wedge b, b \wedge c, c \wedge a\} \\ &\leq a \vee b.\end{aligned}$$

- Similarly,

$$\begin{aligned}a \wedge b &\leq b \leq b \vee c, \\ b \wedge c &\leq c \leq b \vee c, \\ c \wedge a &\leq c \leq b \vee c.\end{aligned}$$

  Therefore $b \vee c \in \mathrm{ub}\{a \wedge b, b \wedge c, c \wedge a\}$ and

$$\begin{aligned}(a \wedge b) \vee (b \wedge c) \vee (c \wedge a) &= \sup\{a \wedge b, b \wedge c, c \wedge a\} \\ &\leq b \vee c.\end{aligned}$$

- Finally, we have that

$$\begin{aligned}a \wedge b &\leq a \leq c \vee a, \\ b \wedge c &\leq c \leq c \vee a, \\ c \wedge a &\leq c \leq c \vee a.\end{aligned}$$

  Therefore $c \vee a \in \mathrm{ub}\{a \wedge b, b \wedge c, c \wedge a\}$ and

$$\begin{aligned}(a \wedge b) \vee (b \wedge c) \vee (c \wedge a) &= \sup\{a \wedge b, b \wedge c, c \wedge a\} \\ &\leq c \vee a.\end{aligned}$$

Hence $(a \wedge b) \vee (b \wedge c) \vee (c \wedge a) \in \mathrm{lb}\{a \vee b, b \vee c, c \vee a\}$ and

$$\begin{aligned}(a \wedge b) \vee (b \wedge c) \vee (c \wedge a) &\leq \inf\{a \vee b, b \vee c, c \vee a\} \\ &= (a \vee b) \wedge (b \vee c) \wedge (c \vee a).\end{aligned}$$

□

Exercise 4.5.1.5. Let L be a lattice. Then the following are equivalent:

1. For each $a, b, c \in L$, $c \leq a$ implies that $a \wedge (b \vee c) = (a \wedge b) \vee c$.
2. For each $a, b, c \in L$, $c \leq a$ implies that $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$.
3. For each $p, q, r \in L$, $p \wedge (q \vee (p \wedge r)) = (p \wedge q) \vee (p \wedge r)$.

Proof.

1. (1) $\implies$ (2):
   Suppose that for each $a, b, c \in L$, $c \leq a$ implies that $a \wedge (b \vee c) = (a \wedge b) \vee c$. Let $a, b, c \in L$. Suppose that $c \leq a$. Then $a \wedge c = c$. By assumption
   $$\begin{aligned} a \wedge (b \vee c) &= (a \wedge b) \vee c \\ &= (a \wedge b) \vee (a \wedge c) \end{aligned}$$
   Since $a, b, c \in L$ with $c \leq a$ are arbitrary, we have that for each $a, b, c \in L$, $c \leq a$ implies that $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$.
2. (2) $\implies$ (3):
   Suppose that for each $a, b, c \in L$, $c \leq a$ implies that $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$. Let $p, q, r \in L$. Define $a, b, c \in L$ by $a := p$, $b := q$ and $c := p \wedge r$. Then
   $$\begin{aligned} a \wedge c &= p \wedge (p \wedge r) \\ &= (p \wedge p) \wedge r \\ &= p \wedge r. \end{aligned}$$
   By assumption,
   $$\begin{aligned} p \wedge (q \vee (p \wedge r)) &= a \wedge (b \vee c) \\ &= (a \wedge b) \vee (a \wedge c) \\ &= (p \wedge q) \vee (p \wedge r). \end{aligned}$$
   Since $p, q, r \in L$ are arbitrary, we have that for each $p, q, r \in L$, $p \wedge (q \vee (p \wedge r)) = (p \wedge q) \vee (p \wedge r)$.
3. (3) $\implies$ (1):
   Suppose that for each $p, q, r \in L$, $p \wedge (q \vee (p \wedge r)) = (p \wedge q) \vee (p \wedge r)$. Let $a, b, c \in L$. Suppose that $c \leq a$. Define $p, q, r \in L$ by $p := a$, $q := b$ and $r := c$. Since $c \leq a$, we have that
   $$\begin{aligned} p \wedge r &= a \wedge c \\ &= c. \end{aligned}$$
   By assumption,
   $$\begin{aligned} a \wedge (b \vee c) &= p \wedge (q \vee (p \wedge r)) \\ &= (p \wedge q) \vee (p \wedge r) \\ &= (a \wedge b) \vee c. \end{aligned}$$
   Since $a, b, c \in L$ with $c \leq a$ are arbitrary, we have that for each $a, b, c \in L$, $c \leq a$ implies that $a \wedge (b \vee c) = (a \wedge b) \vee c$.

□

Definition 4.5.1.6. Let L be a lattice. Then L is said to be **modular** if for each $a, b, c \in L$,

$$c \leq a \text{ implies that } a \wedge (b \vee c) = (a \wedge b) \vee c.$$

Exercise 4.5.1.7. Let L be a lattice. Then the following are equivalent:

1. For each $a, b, c \in L$, $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$.

2. For each $p, q, r \in L$, $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$.

Proof.

1. (1) $\implies$ (2):
Suppose that for each $a, b, c \in L$, $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$. Let $p, q, r \in L$. Define $a, b, c \in L$ by $a := p \vee q$, $b := p$ and $c := r$. By absorption,

$$\begin{aligned} a \wedge b &= (p \vee q) \wedge p \\ &= p \wedge (p \vee q) \\ &= p. \end{aligned}$$

By assumption,

$$\begin{aligned} a \wedge c &= (p \vee q) \wedge r \\ &= r \wedge (p \vee q) \\ &= (r \wedge p) \vee (r \wedge q). \end{aligned}$$

Then by assumption and absorption,

$$\begin{aligned} (p \vee q) \wedge (p \vee r) &= a \wedge (b \vee c) \\ &= (a \wedge b) \vee (a \wedge c) \\ &= p \vee [(r \wedge p) \vee (r \wedge q)] \\ &= [p \vee (r \wedge p)] \vee (r \wedge q) \\ &= [p \vee (p \wedge r)] \vee (q \wedge r) \\ &= p \vee (q \wedge r). \end{aligned}$$

Since $p, q, r \in L$ are arbitrary, we have that for each $p, q, r \in L$, $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$.

2. (2) $\implies$ (1):
Suppose that for each $p, q, r \in L$, $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$. Let $a, b, c \in L$. Define $p, q, r \in L$ by $p := a \wedge b$, $q := a$ and $r := c$. By absorption,

$$\begin{aligned} p \vee q &= (a \wedge b) \vee a \\ &= a \vee (a \wedge b) \\ &= a. \end{aligned}$$

By assumption,

$$\begin{aligned} p \vee r &= (a \wedge b) \vee c \\ &= c \vee (a \wedge b) \\ &= (c \vee a) \wedge (c \vee b). \end{aligned}$$

Then by assumption and absorption,

$$\begin{aligned} (a \wedge b) \vee (a \wedge c) &= p \vee (q \wedge r) \\ &= (p \vee q) \wedge (p \vee r) \\ &= a \wedge [(c \vee a) \wedge (c \vee b)] \\ &= [a \wedge (c \vee a)] \wedge (c \vee b) \\ &= [a \wedge (a \vee c)] \wedge (b \vee c) \\ &= a \wedge (b \vee c). \end{aligned}$$

Since $a, b, c \in L$ are arbitrary, we have that for each $a, b, c \in L$, $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$.

□

Definition 4.5.1.8. Let L be a lattice. Then L is said to be **distributive** if for each $a, b, c \in L$,

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c).$$

Exercise 4.5.1.9. Let L be a lattice. If L is distributive, then L is modular.

Proof. Suppose that L is distributive. FINISH!!! □

Exercise 4.5.1.10. Let L be a lattice. Then for each $a, b, c \in L$, $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$ iff $a \wedge (b \vee c) = (a \wedge b) \vee (a \vee c)$.

Proof. Let $a, b, c \in L$.

- ($\Longrightarrow$):
  Suppose that $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$.
- ($\Longleftarrow$):

□

Chapter 5

Complete Lattices

5.1 Introduction

Exercise 5.1.0.1. Let $(X, \leq)$ be a poset. Then for each $A \subset X$, $\sup A$ exists iff for each $A \subset X$, $\inf A$ exists.

Proof.

- $(\implies)$:
  Suppose that for each $A \subset X$, $\sup A$ exists. Let $A \subset X$. Since $\operatorname{lb} A \subset X$, by assumption, $\sup(\operatorname{lb} A)$ exists. Since

$$\begin{aligned} \sup(\operatorname{lb} A) \text{ exists} &\iff \sup(\operatorname{lb} A) \neq \varnothing \quad \text{(by ref def here)} \\ &\iff \inf A \neq \varnothing \quad \text{(by Exercise 3.3.3.3)} \\ &\iff \inf A \text{ exists,} \end{aligned}$$

  we have that $\inf A$ exists.

- $(\impliedby)$:
  Similar to $(\implies)$.

$\square$

Definition 5.1.0.2. Let $(X, \leq)$ be a poset.

- Then $(X, \leq)$ is said to be a **complete lattice** if for each $A \subset X$, $\sup A$ exists.
- If $(X, \leq)$ is a complete lattice, we define the **arbitrary join and meet operators on** X, denoted $\bigvee_{\leq} : \mathcal{P}(X) \to X$ and $\bigwedge_{\leq} : \mathcal{P}(X) \to X$ respectively, by $\bigvee_{\leq} A := \sup(A, \leq)$ and $\bigwedge_{\leq} A := \inf(A, \leq)$.

Note 5.1.0.3. When the context is clear, we may write $\bigvee$ and $\bigwedge$ in place of $\bigvee_{\leq}$ and $\bigwedge_{\leq}$ respectively.

Exercise 5.1.0.4. Let $(X, \leq)$ be a complete lattice. Then $(X, \vee_{\leq}, \wedge_{\leq}, \inf(L, \leq), \sup(L, \leq),)$ is a bounded lattice.

Proof. FIX!!!

1. make previous exercise implies that $\operatorname{lb}(\varnothing, \leq) = X$. Since $(X, \leq)$ is a complete lattice and $\varnothing \subset X$, there exists $x_1 \in X$ such that $x_1 := \inf \varnothing$. In particular, $X \neq \varnothing$.
2.
   - make previous ex saying $\inf \varnothing = \sup X$ and $\sup \varnothing = \inf X$ with the following logic

$$\begin{aligned} &\inf \varnothing \\ &\qquad \in \operatorname{lb} \varnothing \cap \operatorname{ub}(\operatorname{lb} \varnothing) \\ &\qquad = X \cap \operatorname{lb}(X) \\ &\qquad = \operatorname{lb} X. \end{aligned}$$

     ref an ex implying that $\operatorname{lb} X = \inf X$. Hence X is a one of X.
   - similar x_0

Clearly $\bigvee X$ is a one of X and $\bigwedge X$ is a zero of X. $\square$

5.2 Complete Lattice Homomorphisms

Definition 5.2.0.1. Let $(X, \leq_X), (Y, \leq_Y)$ be complete lattices and $f : X \to Y$. Then f is said to

1. $(\leq_X, \leq_Y)$**-arbitrary join preserving** if for each $A \subset X$, $f(\sup(A, \leq_X)) = \sup(f(A), \leq_Y)$
2. $(\leq_X, \leq_Y)$**-arbitrary meet preserving** if for each $A \subset X$, $f(\inf(A, \leq_X)) = \inf(f(A), \leq_Y)$

Exercise 5.2.0.2. Let $(X, \leq_X), (Y, \leq_Y)$ be complete lattices, $f : X \to Y$ and $g : Y \to X$.

1. If f is arbitrary join preserving, then there exists $f_* : Y \to X$ such that g is a right adjoint of f.
2. If g is arbitrary meet preserving, then there exists $g^* : X \to Y$ such that g is a left adjoint of g.

Proof. 1. Suppose that

2.

□

Exercise 5.2.0.3. move to section on complete lattices Let $(X, \leq_X)$ and $(Y, \leq_Y)$ be posets, $f : X \to Y$. Then there exists $g : Y \to X$ such that (f, g) is a monotone galois connection iff f preserves arbitrary joins.

Proof. □

5.3 Properties of Ideals and Filters

5.3.1 Prime Ideals and Filters in Complete Lattices

Definition 5.3.1.1. Let L be a complete lattice.

- Let $I \in \mathcal{I}(L)$. Then I is said to be
  - **completely prime** if
    1. I is proper
    2. for each $A \subset L$, $\bigwedge A \in I$ implies that there exists $a \in A$ such that $a \in I$.
- Let $F \in \mathcal{F}(L)$. Then F is said to be
  - **completely prime** if
    1. F is proper
    2. for each $A \subset L$, $\bigvee A \in F$ implies that there exists $a \in A$ such that $a \in F$.

Exercise 5.3.1.2. Let L be a complete lattice.

1. For each $I \in \mathcal{I}(L)$, if I is completely prime, then $I \in \mathcal{I}_p(L)$.
2. For each $F \in \mathcal{F}(L)$, if F is completely prime, then $F \in \mathcal{F}_p(L)$.

Proof. Clear. fill out □

Exercise 5.3.1.3. Let L be a complete lattice.

1. For each $I \in \mathcal{I}_{cp}(L)$,
   (a) $\bigwedge I^c \in I^c$,
   (b) $\bigwedge I^c \in \text{prime}_\vee(L)$.
2. For each $F \in \mathcal{F}_{cp}(L)$,
   (a) $\bigvee F^c \in F^c$,

(b) $\bigvee F^c \in \text{prime}_\wedge(L)$.

Proof.

1. Let $I \in \mathcal{I}_{cp}(L)$.
   (a) For the sake of contradiction, suppose that $\bigwedge I^c \in I$. Since I is completely prime, there exists $a \in I^c$ such that $a \in I$. This is a contradiction. Hence $\bigwedge I^c \notin I$. Thus $\bigwedge I^c \in I^c$.
   (b) i. Since $I \in \mathcal{I}(L)$, a previous ex implies that $0 \in I$. Since $\bigwedge I^c \in I^c$, we have that $\bigwedge I^c \neq 0$.
   ii. Define $x \in L$ by $x := \bigwedge I^c$. Since $I \in \mathcal{I}(L)$, a prev ex implies that I is a lower set. Let $a, b \in L$. Suppose that $x \leq a \vee b$. For the sake of contradiction, suppose that $a \in I$ and $b \in I$. Since $I \in \mathcal{I}(L)$, we have that $a \vee b \in I$. Since I is a lower set and

$$\begin{aligned} x &\leq a \vee b \\ &\in I, \end{aligned}$$

   we have that $x \in I$. This is a contradiction since $x \in I^c$. Hence $a \notin I$ or $b \notin I$. Thus $a \in I^c$ or $b \in I^c$. Since $x = \bigwedge I^c$, we have that $x \leq a$ or $x \leq b$. Since $a, b \in L$ with $x \leq a \vee b$ are arbitrary, we have that for each $a, b \in L$, $x \leq a \vee b$ implies that $x \leq a$ or $b \leq b$.

   Hence $x \in \text{prime}_\vee(L)$.
2. use duality

□

Exercise 5.3.1.4. Let L be a complete lattice. Then

1. for each $x \in \text{prime}_\vee(L)$, $(\uparrow x)^c \in \mathcal{I}_{cp}(L)$.
2. for each $x \in \text{prime}_\wedge(L)$, $(\downarrow x)^c \in \mathcal{F}_{cp}(L)$.

Proof.

1. Let $x \in \text{prime}_\vee(L)$. a prev ex implies that $\uparrow x \in \mathcal{F}(L)$.
   (a) Since $x \in \text{prime}_\vee(L)$, $x \neq 0$. Hence $0 \in (\uparrow x)^c$. Thus $(\uparrow x)^c \neq \varnothing$.
   (b) Let $a, b \in (\uparrow x)^c$. For the sake of contradiction, suppose that $a \vee b \in \uparrow x$. Then $x \leq a \vee b$. Since $x \in \text{prime}_\vee(L)$, $x \leq a$ or $x \leq b$. Hence $a \in \uparrow x$ or $b \in \uparrow x$. This is a contraction since $a \in (\uparrow x)^c$ and $b \in (\uparrow x)^c$. Therefore $a \vee b \in (\uparrow x)^c$. Since $a, b \in (\uparrow x)^c$ are arbitrary, we have that for each $a, b \in (\uparrow x)^c$, $a \vee b \in (\uparrow x)^c$.
   (c) Let $y \in L$ and $a \in (\uparrow x)^c$. For the sake of contradiction, suppose that $y \wedge a \in \uparrow x$. Then

$$\begin{aligned} x &\leq y \wedge a \\ &\leq a. \end{aligned}$$

   Thus $a \in \uparrow x$, which is a contradiction. Hence $y \wedge a \in (\uparrow x)^c$. Since $y \in L$ and $a \in (\uparrow x)^c$ are arbitrary, we have that for each $y \in L$ and $a \in (\uparrow x)^c$, $y \wedge a \in (\uparrow x)^c$.

   Thus $(\uparrow x)^c \in \mathcal{I}(L)$.
   (a) Since $\uparrow x \neq \varnothing$, we have that $(\uparrow x)^c \neq L$. Thus $(\uparrow x)^c$ is proper.
   (b) Let $A \subset L$. Suppose that $\bigwedge A \in (\uparrow x)^c$. For the sake of contradiction, suppose that for each $a \in A$, $a \notin (\uparrow x)^c$. Then for each $a \in A$, $a \in \uparrow x$. Hence for each $a \in A$, $x \leq a$. Therefore $x \in \text{lb}\, A$. Thus

$$\begin{aligned} x &\leq \inf A \\ &= \bigwedge A. \end{aligned}$$

   Then $\bigwedge A \in \uparrow x$, which is a contradiction. Hence there exists $a \in A$ such that $a \in (\uparrow x)^c$. Since $A \subset L$ with $\bigwedge A \in (\uparrow x)^c$ is arbitrary, we have that for each $A \subset L$, $\bigwedge A \in (\uparrow x)^c$ implies that there exists $a \in A$ such that $a \in (\uparrow x)^c$.

Thus $(\uparrow x)^c \in \mathcal{I}_{cp}(L)$.

2. similar or duality

$\square$

maybe make ex in earlier section on ideals that $\downarrow x = \downarrow y$ iff $x = y$ and similar for $\uparrow x$. Then in this section, show that for complete lattice L, $\mathcal{F}_{cp}(L) \cong \text{prime}_{\wedge}(L)$ bijective.

5.4 Constructions with Complete Lattices

5.4.1 Complete Sublattices

Definition 5.4.1.1. define **complete sublattice** Let L, M be lattices

5.4.2 Products of Complete Lattices

5.4.3 Coproducts of Complete Lattices

5.4.4 Quotients of Complete Lattices

5.5 Fundamental Examples

5.5.1 Introduction

Exercise 5.5.1.1. Let L be a complete lattice.

5.5.2 Sets

5.5.3 Equivalence Relations

Proof. Let X be a set. Suppose that $X \neq \varnothing$. Then $(\text{Equiv}, \subset)$ is a complete lattice. □

5.5.4 Ideal and Filter Lattices

Exercise 5.5.4.1. Let L be a lattice. If L has a zero, then

1. $(\mathcal{I}(L), \leq_{\mathcal{I}(L)})$ is a complete lattice.
2. $(\mathcal{F}(L), \leq_{\mathcal{F}(L)})$ is a complete lattice.

5.5.5 Topological Spaces

5.6 Least Upper Bound Property

Definition 5.6.0.1. Let $(X, \leq)$ be a poset. Then $(X, \leq)$ is said to satisfy the

- **least upper bound (LUB) property** if for each $A \subset X$, if $A \neq \varnothing$ and A is bounded above, then there exists $x \in X$ such that $x = \sup A$
- **greatest lower bound (GLB) property** if for each $A \subset X$, if $A \neq \varnothing$ and A is bounded below, then there exists $x \in X$ such that $x = \inf A$.

Exercise 5.6.0.2. LUB iff GLB

Proof. FINISH!!!! □

Definition 5.6.0.3. Suplattice:
Let $(L, \leq)$ be a poset. Then $(L, \leq)$ is said to be a **suplattice** if for each $A \subset L$, there exists $x \in L$ such that $x = \sup A$.

5.6.1 Complete Lattice Homomorphisms

Chapter 6

Frames and Locales

move to after universal algebra, because we doing the same things

6.1 Introduction

6.1.1 Frames

Definition 6.1.1.1. Let L be a complete lattice. Then L is said to be a **frame** if for each $a \in L$ and $(b_\alpha)_{\alpha \in A} \subset L$, $a \wedge \left(\bigvee_{\alpha \in A} b_\alpha \right) = \bigvee_{\alpha \in A} a \wedge b_\alpha$.

Definition 6.1.1.2. Let L, M be frames and $f : L \to M$. Then f is said to be a **frame homomorphism** if

1. for each $(x_\alpha)_{\alpha \in A} \subset L$, $f\left(\bigvee_{\alpha \in A} x_\alpha \right) = \sup_{\alpha \in A} f(x_\alpha)$.
2. for each $a, b \in L$, $f(a \wedge b) = f(a) \wedge f(b)$

maybe reword this with some vocab to make shorter, like "preserves arbitrary joins" and "preserves meets"

Definition 6.1.1.3. (check notation consistent with category theory notes) We define the **category of frames**, denoted **Frm**, by

- $\mathrm{Obj}(\mathbf{Frm}) := \{L : L \text{ is a frame}\}$
- $\mathrm{Hom}_{\mathbf{Frm}}(L, M) := \{f : L \to M : f \text{ is a frame homomorphism}\}$

Exercise 6.1.1.4. We have that **Frm** is a category

Proof. FINISH!!! □

6.1.2 Locales

Definition 6.1.2.1. We define the **category of locales**, denoted **Loc** by $\mathbf{Loc} := \mathbf{Frm}^{\mathrm{op}}$.

6.2 More Lattice Stuff to Come

- talk about join and meet irriducibility
- talk about join and meet primality
- talk about maximality.
- the goal is to get all the background for sober topological/measure spaces, locale theory for constructive topology and universal algebra

Chapter 7

Some Chapter

7.1 Closure Operators

Definition 7.1.0.1. Let A be a set and $C : \mathcal{P}(A) \to \mathcal{P}(A)$. Then C is said to be a **closure operator on** A if for each $X, Y \in \mathcal{P}(A)$,

1. $X \subset C(X)$,
2. $C^2(X) = C(X)$,
3. $X \subset Y$ implies that $C(X) \subset C(Y)$.

Exercise 7.1.0.2. Let A be a set and $C : \mathcal{P}(A) \to \mathcal{P}(A)$. Suppose that C is a closure operator on A. Then for each $(E_j)_{j \in J} \subset \mathcal{P}(A)$,

1. $C\left(\bigcap\limits_{j \in J} E_j\right) \subset \bigcap\limits_{k \in J} C(E_k)$,
2. $\bigcup\limits_{k \in J} C(E_k) \subset C\left(\bigcup\limits_{j \in J} E_j\right)$.

Proof. Let $(E_j)_{j \in J} \subset \mathcal{P}(A)$.

1. Let $k \in J$. Then $\bigcap\limits_{j \in J} E_j \subset E_k$. So $C\left(\bigcap\limits_{j \in J} E_j\right) \subset C(E_k)$. Since $k \in J$ is arbitrary, we have that
$$C\left(\bigcap_{j \in J} E_j\right) \subset \bigcap_{k \in J} C(E_k).$$
2. Let $k \in J$. Then $E_k \subset \bigcup_{j \in J} E_j$. Hence $C(E_k) \subset C\left(\bigcup\limits_{j \in J} E_j\right)$. Since $k \in J$ is arbitrary, we have that
$$\bigcup_{k \in J} C(E_k) \subset C\left(\bigcup_{j \in J} E_j\right)$$

$\square$

Definition 7.1.0.3. Let A be a set, $C : \mathcal{P}(A) \to \mathcal{P}(A)$ and $X \subset A$. Suppose that C is a closure operator on A. Then X is said to be C-closed if $C(X) = X$.

Definition 7.1.0.4. Let A be a set and $C : \mathcal{P}(A) \to \mathcal{P}(A)$. Suppose that C is a closure operator on A. We define the **lattice of C-closed subsets of** A, denoted $L_C(A) \subset \mathcal{P}(A)$, by

$$L_C(A) := \{X \subset A : X \text{ is } C\text{-closed}\}$$

.

Exercise 7.1.0.5. Let A be a set and $C : \mathcal{P}(A) \to \mathcal{P}(A)$. Suppose that C is a closure operator on A. Then

1. for each $(E_j)_{j\in J} \subset L_C(A)$, $\bigcap_{j\in J} E_j \in L_C(A)$ and $\bigcup_{j\in J} E_j \in L_C(A)$.
2. $(L_C(A), \subset)$ is a complete lattice define complete lattice

$$C\left(\bigcap_{j\in J} E_j\right) = \bigcap_{j\in J} E_j$$

and

$$C\left(\bigcup_{j\in J} E_j\right) = \bigcup_{j\in J} E_j.$$

Proof.

1. Let $(E_j)_{j\in J} \subset L_C(A)$.
   - A previous exercise Exercise B.0.0.3 implies that

$$\begin{aligned} C\left(\bigcap_{j\in J} E_j\right) &\subset \bigcap_{k\in J} C(E_k) \\ &= \bigcap_{k\in J} E_k \\ &\subset C\left(\bigcap_{k\in J} E_k\right). \end{aligned}$$

   Hence $C\left(\bigcap_{j\in J} E_j\right) = \bigcap_{k\in J} E_k$.
   - A previous exercise Exercise B.0.0.3 implies that

$$\begin{aligned} \bigcup_{k\in J} E_k &= \bigcup_{k\in J} C(E_k) \\ &\subset C\left(\bigcup_{j\in J} E_j\right) \\ &\subset \bigcap_{k\in J} C(E_k) \\ &= \bigcap_{k\in J} E_k \\ &\subset C\left(\bigcap_{k\in J} E_k\right). \end{aligned}$$

   Hence $C\left(\bigcap_{j\in J} E_j\right) = \bigcap_{k\in J} E_k$.
2.

FINISH!!!, don't need to show second part, □

Definition 7.1.0.6. then is said to be an **algebraic closure operator on** A if

Chapter 8

Universal Algebra

8.1 Introduction

Definition 8.1.0.1. Let $A, J \in \mathrm{Obj}(\mathbf{Set})$ be a set and $f \in \mathcal{F}^*(A)^J$. Then (A, f) is said to be an **algebra** if $A \neq \varnothing$ and $J \neq \varnothing$.

Definition 8.1.0.2. Let $\mathcal{A}$ be an algebra. Write $\mathcal{A} = (A, f)$. Set $J := \operatorname{dom} f$.

- We define the **universe of** $\mathcal{A}$, denoted $\operatorname{Uni} \mathcal{A}$, by $\operatorname{Uni} \mathcal{A} := A$.
- We define the **operations of** $\mathcal{A}$, denoted $\operatorname{Oper} \mathcal{A}$, by $\operatorname{Oper} \mathcal{A} := f$.
- We define the **type of** $\mathcal{A}$, denoted $\operatorname{Type} \mathcal{A} : J \to \mathbb{N}_0$, by $\operatorname{Type} \mathcal{A}(j) := \operatorname{arity} f_j$.

Definition 8.1.0.3. Let $\mathcal{A}$, $\mathcal{B}$ be algebras. Then $\mathcal{A}$ and $\mathcal{B}$ are said to be **type-similar**, denoted $\mathcal{A} \sim_{\mathrm{Type}} \mathcal{B}$, if $\operatorname{Type} \mathcal{A} = \operatorname{Type} \mathcal{B}$.

Definition 8.1.0.4. Let $(A, f), (B, g)$ be algebras and $\alpha : A \to B$. Suppose that $(A, f) \sim_{\mathrm{Type}} (B, g)$. Set $J := \operatorname{dom} f$ and $\rho := \operatorname{Type}(A, f)$. Write $f = (f_j)_{j \in J}$ and $g = (g_j)_{j \in J}$. Then α is said to be an $((A, f), (B, g))$**-homomorphism** if for each $j \in J$,

$$\alpha \circ f_j = g_j \circ \alpha^{\rho(j)}.$$

Exercise 8.1.0.5. Let $(A, f), (B, g), (C, h)$ be algebras and $\alpha : A \to B$, $\beta : B \to C$. Suppose that $(A, f) \sim_{\mathrm{Type}} (B, g), (C, h)$. Set $J := \operatorname{dom} f$ and $\rho := \operatorname{Type}(A, f)$. Write $f = (f_j)_{j \in J}$, $g = (g_j)_{j \in J}$ and $h = (h_j)_{j \in J}$. If α is a $((A, f), (B, g))$-homomorphism and β is a $((B, g), (C, h))$-homomorphism, then $\beta \circ \alpha$ is a $((A, f), (C, h))$-homomorphism.

Proof. Suppose that α is a $((A, f), (B, g))$-homomorphism and β is a $((B, g), (C, h))$-homomorphism. Let $j \in J$. Since α is a $((A, f), (B, g))$-homomorphism, $\alpha \circ f_j = g_j \circ \alpha^{\rho(j)}$. Since β is a $((B, g), (C, h))$-homomorphism, $\beta \circ g_j = h_j \circ \beta^{\rho(j)}$. Therefore

$$\begin{aligned}
(\beta \circ \alpha) \circ f_j &= \beta \circ (\alpha \circ f_j) \\
&= \beta \circ (g_j \circ \alpha^{\rho(j)}) \\
&= (\beta \circ g_j) \circ \alpha^{\rho(j)} \\
&= (h_j \circ \beta^{\rho(j)}) \circ \alpha^{\rho(j)} \\
&= h_j \circ (\beta^{\rho(j)} \circ \alpha^{\rho(j)}) \\
&= h_j \circ ([\beta \circ \alpha]^{\rho(j)}).
\end{aligned}$$

Since $j \in J$ is arbitrary, we have that for each $j \in J$ and $(\beta \circ \alpha) \circ f_j = h_j \circ ([\beta \circ \alpha]^{\rho(j)})$. Hence $\beta \circ \alpha$ is a $((A, f), (C, h))$-homomorphism. $\square$

Exercise 8.1.0.6. Let (A, f) be and algebra. Set $J := \operatorname{dom} f$. Write $f = (f_j)_{j \in J}$. Then id_A is a $((A, f), (A, f))$-homomorphism.

Proof. Since for each $j \in J$,

$$\begin{aligned} \mathrm{id}_A \circ f_j &= f_j \\ &= f_j \circ \mathrm{id}_{A^{\rho(j)}} \\ &= f_j \circ \mathrm{id}_A^{\rho(j)}, \end{aligned}$$

we have that id_A is a $((A, f), (A, f))$-homomorphism. □

Definition 8.1.0.7.

- Let J be a set and $\rho : J \to \mathbb{N}_0$. Then ρ is said to be a **type function**. We define Types $:= \{\rho : \rho \text{ is a type function}\}$.
- Let $\rho \in$ Types. We define the **category of algebras with type** ρ, denoted $\mathbf{Alg}_\rho$, by
  - $\mathrm{Obj}(\mathbf{Alg}_\rho) := \{\mathcal{A} : \mathcal{A} \text{ is an algebra and } \mathrm{Type}\,\mathcal{A} = \rho\}$
  - For $(A, f), (B, g) \in \mathrm{Obj}(\mathbf{Alg}_\rho)$, $\mathrm{Hom}_{\mathbf{Alg}_\rho}(\mathcal{A}, \mathcal{B}) := \{\alpha \in B^A : \alpha \text{ is a } (\mathcal{A}, \mathcal{B})\text{-homomorphism.}\}$

Exercise 8.1.0.8. Let $\rho \in$ Types. Then $\mathbf{Alg}_\rho$ is a category.

Proof. FINISH!!! □

8.2 Subalgebras

Definition 8.2.0.1. Let $\rho \in \text{Types}$ and $\mathcal{A}, \mathcal{B} \in \text{Obj}(\mathbf{Alg}_\rho)$. Write $\mathcal{A} = (A, f)$ and $\mathcal{B} = (B, g)$. Set $J := \operatorname{dom} \rho$. Write $f = (f_j)_{j \in J}$ and $g = (g_j)_{j \in J}$.

- Then $\mathcal{B}$ is said to be a **subalgebra of** $\mathcal{A}$ if
  1. $B \subset A$
  2. $\iota_B \in \text{Hom}_{\mathbf{Alg}_\rho}(\mathcal{B}, \mathcal{A})$.
- We define $\text{SubAlg}(\mathcal{A}) := \{\mathcal{B} \in \text{Obj}(\mathbf{Alg}_\rho) : \mathcal{A} \sim_{\text{Type}} \mathcal{B} \text{ and } \mathcal{B} \text{ is a subalgebra of } \mathcal{A}\}$.

Exercise 8.2.0.2. Let $\rho \in \text{Types}$ and $\mathcal{A}, \mathcal{B} \in \text{Obj}(\mathbf{Alg}_\rho)$. Write $\mathcal{A} = (A, f)$ and $\mathcal{B} = (B, g)$. Set $J := \operatorname{dom} \rho$. Write $f = (f_j)_{j \in J}$ and $g = (g_j)_{j \in J}$. Then $(B, g) \in \text{SubAlg}(A, f)$ is a subalgebra of (A, f) iff

1. $B \subset A$
2. for each $j \in J$, $f_j|_{B^{\rho(j)}} = g_j$.

Proof.

- ($\Longrightarrow$):
  Suppose that $\mathcal{B}$ is a subalgebra of $\mathcal{A}$.
  1. By definition, $B \subset A$.
  2. Let $j \in J$. Since $\mathcal{B}$ is a subalgebra of $\mathcal{A}$, $\iota_B \in \text{Hom}_{\mathbf{Alg}_\rho}(\mathcal{B}, \mathcal{A})$. Then
  $$\begin{aligned} f_j|_{B^{\rho(j)}} &= f_j \circ \iota_{B^{\rho(j)}} \\ &= f_j \circ \iota_B^{\rho(j)} \\ &= \iota_B \circ g_j \\ &= g_j. \end{aligned}$$
  Since $j \in J$ is arbitrary, we have that for each $j \in J$, $f_j|_{B^{\rho(j)}} = g_j$.
- ($\Longleftarrow$):
  Suppose that
  1. $B \subset A$
  2. for each $j \in J$, $f_j|_{B^{\rho(j)}} = g_j$.

  Let $j \in J$. Then
  $$\begin{aligned} \iota_B \circ g_j &= g_j \\ &= f_j|_{B^{\rho(j)}} \\ &= f_j \circ \iota_{B^{\rho(j)}} \\ &= f_j \circ \iota_B^{\rho(j)}. \end{aligned}$$
  Since $j \in J$ is arbitary, we have that for each $j \in J$, $\iota_B \circ g_j = f_j \circ \iota_B^{\rho(j)}$. Thus $\iota_B \in \text{Hom}_{\mathbf{Alg}_\rho}(\mathcal{B}, \mathcal{A})$.
  1. $B \subset A$
  2. Let $j \in J$ and $a \in B^{\rho(j)}$. Then
  $$\begin{aligned} f_j(\iota_B^{\rho(j)}(a)) &= f_j(\iota_{B^{\rho(j)}}(a)) \\ &= f_j \circ \iota_{B^{\rho(j)}}(a) \\ &= f_j|_{B^{\rho(j)}}(a) \\ &= g_j(a) \\ &= \iota_B(g_j(a)). \end{aligned}$$

□

Definition 8.2.0.3. Let $\rho \in \text{Types}$ and $(A, f), (B, g) \in \text{Obj}(\mathbf{Alg}_\rho)$. Set $J := \operatorname{dom} \rho$. Write $f = (f_j)_{j \in J}$ and $g = (g_j)_{j \in J}$.

- Then (B, g) is said to be a **subalgebra of** (A, f) if
  1. $B \subset A$
  2. for each $j \in J$, $f_j|_{B^{\rho(j)}} = g_j$.
- We define $\text{SubAlg}(\mathcal{A}) := \{\mathcal{B} \in \text{Obj}(\mathbf{Alg}_\rho) : \mathcal{A} \sim_{\text{Type}} \mathcal{B} \text{ and } \mathcal{B} \text{ is a subalgebra of } \mathcal{A}\}$.

Definition 8.2.0.4. Let $\rho \in \text{Types}$, $(A, f) \in \text{Obj}(\mathbf{Alg}_\rho)$ and $B \in \text{Obj}(\mathbf{Set})$.

- Then B is said to be a **subuniverse of** (A, f) if
  1. $B \subset A$,
  2. B is $\operatorname{Im} f$-closed.
- We define $\text{SubUni}(\mathcal{A}) := \{B \in \text{Obj}(\mathbf{Set}) : B \text{ is a subuniverse of } \mathcal{A}\}$.

Exercise 8.2.0.5. Let $\rho \in \text{Types}$ and $(A, f), (B, g) \in \text{Obj}(\mathbf{Alg}_\rho)$. If (B, g) is a subalgebra of (A, f), then B is a subuniverse of (A, f).

Proof. Set $J := \operatorname{dom} \rho$ and $\rho := \text{Type}(A, f)$. Suppose that (B, g) is a subalgebra of (A, f).

1. Since (B, g) is a subalgebra of (A, f), we have that $B \subset A$.
2. Let $j \in J$. Then for each $a \in B^\rho(j)$,

$$\begin{aligned} f_j(a_1, \ldots, a_{\rho(j)}) &= f_j|_{B^{\rho(j)}}(a_1, \ldots, a_{\rho(j)}) \\ &= g_j(a_1, \ldots, a_{\rho(j)}) \\ &\in B. \end{aligned}$$

   Since $j \in J$ is arbitrary, we have that B is $\operatorname{Im} f$-closed. Thus B is a subuniverse of A.

□

Definition 8.2.0.6. Let $\mathcal{A}$ be an algebra and B a subuniverse of $\mathcal{A}$. Set $\mathcal{S}(B, \mathcal{A}) := \{S \subset A : S \text{ is a subuniverse of } \mathcal{A} \text{ and } B \subset S\}$. We define the **subuniverse of $\mathcal{A}$ generated by** B, denoted $\text{Sg}(B, \mathcal{A})$, by

$$\text{Sg}(B, \mathcal{A}) := \bigcap_{S \in \mathcal{S}(B, \mathcal{A})} S$$

show $\mathcal{S} \neq \varnothing$ and intersection of subiniverses is subuniverse

Exercise 8.2.0.7. Let (A, f) be an algebra and $B \subset A$. Then

1. $\text{Sg}(B, f)$ is a subuniverse of A
2. $B \subset \text{Sg}(B, f)$.

.

Proof.

1. Set $\mathcal{S} := \{S \subset A : S \text{ is an } f\text{-subuniverse of } A\}$. By construction, for each $S \in S$, S is f-closed. Since $\text{Sg}(B, f) = \bigcap_{S \in \mathcal{S}} S$, Exercise B.0.0.3 A previous exercise in the set theory section implies that $\text{Sg}(B, f)$ is f-closed. Hence $\text{Sg}(B, f)$ is an f-subuniverse of A.

2. By construction, for each $S \in \mathcal{S}$, $B \subset S$. Thus

$$\begin{aligned} B &\subset \bigcap_{S \in \mathcal{S}} S \\ &= \mathrm{Sg}(B, f). \end{aligned}$$

□

Exercise 8.2.0.8. Let (A, f) be an algebra. Then $\mathrm{Sg}(\cdot, f)$ is an algebraic closure operator on A.

Proof. □

8.3 Congruences

Definition 8.3.0.1. Let $\rho \in \mathrm{Types}$ and $(A, f) \in \mathrm{Obj}(\mathbf{Alg}_\rho)$. Set $J := \mathrm{dom}\,\rho$. Write $f = (f_j)_{j \in J}$.

- Let $\theta \in \mathrm{Equiv}(A)$. Then θ is said to be a **congruence on** (A, f) if for each $j \in J$ and $a, b \in A^{\rho(j)}$, $a\theta^{\times \rho(j)}b$ implies that $f_j(a)\theta f_j(b)$.
- We define $\mathrm{Cong}(A, f) := \{\theta \in \mathrm{Equiv}(A) : \theta \text{ is a congruence on } (A, f)\}$.

Note 8.3.0.2. Question: θ is congruence iff $f_j^{\times 2}(\theta^{\times \rho(j)}) \subset \theta$, is there a generalization of equivalence relation to $\theta \subset A^n$ that is reflexive ($\Delta_{A,n} \subset \theta$) symmetric (invariant under S_n), and transitive (idk what the generalization of composition should be), then define congruence by

$$f_j^{\times n}(\theta^{\times \rho(j)}) \subset \theta$$

when $n = 3$, the question becomes what meaning could capture the statement that a set of three objects 'has sameness', but no two element subset of them 'has sameness' in the same way.

Exercise 8.3.0.3. Let $\rho \in \mathrm{Types}$, $(A, f) \in \mathrm{Obj}(\mathbf{Alg}_\rho)$ and $\theta \in \mathrm{Cong}(A, f)$.

Proof. □

Definition 8.3.0.4. Let $\mathcal{A} \in \mathrm{Obj}(\mathbf{Alg}_\rho)$ and $\theta \in \mathrm{Cong}(\mathcal{A})$. Write $\mathcal{A} = (A, f)$, set $J := \mathrm{dom}\,\rho$ and write $f = (f_j)_{j \in J}$. We define

- $f^\theta \in \mathcal{F}^*(A/\theta)^J$ by $f_j^\theta := \mathrm{Factor}_{\theta^{\rho(j)}, \theta}(f_j)$.
- **quotient algebra of $\mathcal{A}$ by** θ, denoted $\mathcal{A}/\theta$, by $\mathcal{A}/\theta := (A/\theta, f^\theta)$.

introduce factor notation here and apply it to analysis and other notes, aka if we fix $\pi_X : X \to X/\theta_X$, $\pi_Y : Y \to Y/\theta_Y$ and consider $f : X \to Y$ which is (θ_X, θ_Y)-invariant, then there exists a unique $\tilde{f} : X/\theta_X \to Y/\theta_Y$ such that $\pi_Y \circ f = \tilde{f} \circ \pi_{\theta_X}$, we define $\mathrm{Factor}_{\theta_X, \theta_Y}(f) := \tilde{f}$.

I guess we can define a Factor map or something similar in other contexts, i.e. for other universal properties/limits/colimits since they all have some unique map involved as a factor of some expression related to the original map.

need to go over equiv relations in set theory notes

Exercise 8.3.0.5. Let $\rho \in \mathrm{Type}$, $\mathcal{A} \in \mathrm{Obj}(\mathbf{Alg}_\rho)$ and $\theta \in \mathrm{Cong}(\mathcal{A})$. Then $\mathcal{A}/\theta \in \mathrm{Obj}(\mathbf{Alg}_\rho)$.

Proof. Write $\mathcal{A} = (A, f)$. Set $J := \mathrm{dom}\,\rho$. By assumption, $A, J \neq \varnothing$. Hence $(A/\theta, f^\theta)$ is an algebra. By definition, for each $j \in J$, $f_j \in \mathcal{F}^j(A)$ iff $f_j^\theta \in \mathcal{F}^j(A/\theta)$. Hence

$$\begin{aligned} \mathrm{Type}(A/\theta, f^\theta) &= \mathrm{Type}(A, f) \\ &= \rho. \end{aligned}$$

Thus $(A/\theta, f^\theta) \in \mathrm{Obj}(\mathbf{Alg}_\rho)$. □

Chapter 9

Groups

9.0.1 Direct Products

Definition 9.0.1.1. Let G, H be groups. Define a product $* : (G \times H) \times (G \times H) \to G \times H$ by

$$(x_1, y_1) * (x_2, y_2) = (x_1 x_2, y_1 y_2)$$

Then $(G \times H, *)$ is called the **direct product of G and H**.

Exercise 9.0.1.2. Let G, H be groups. Then the direct product $G \times H$ is a group.

Proof. Clear. □

Definition 9.0.1.3. Let G, H be groups. Define $\pi_G : G \times H \to G$ and $\pi_H : G \times H \to H$ by $\pi_G(x, y) = x$ and $\pi_H(x, y) = y$. Then π_G and π_H are respectively called the **projection maps onto G and H**.

Exercise 9.0.1.4. Let G, H be groups. Then

1. $\pi_G : G \times H \to G$ and $\pi_H : G \times H \to H$ are homomorphisms
2. $\ker \pi_G \cong H$ and $\ker \pi_H \cong G$

Proof.

1. Clear
2. Define $\iota_G : G \to \ker \pi_H$ by
$$\iota_G(x) = (x, e_H)$$
Then ι_G is an isomorphism. Similarly, we can define $\iota_H : H \to \ker \pi_G$ and show that it is an isomorphism.

□

Definition 9.0.1.5. Let G, H, K be groups, $\phi \in \mathrm{Hom}(G, K)$ and $\psi \in \mathrm{Hom}(H, K)$. We define $\phi \times \psi : G \times H \to K$ by $\phi \times \psi(x, y) = \phi(x)\psi(y)$

Exercise 9.0.1.6. Let G, H, K be groups, $\phi \in \mathrm{Hom}(G, K)$ and $\psi \in \mathrm{Hom}(H, K)$. If K is abelian, then $\phi \times \psi \in Hom(G \times H, K)$.

Proof. Let $x_1, x_2 \in G$ and $y_1, y_2 \in H$. Then

$$\begin{aligned}
\phi \times \psi[(x_1, y_1)(x_2, y_2)] &= \phi \times \psi(x_1 x_2, y_1 y_2) \\
&= \phi(x_1 x_2)\psi(y_1 y_2) \\
&= \phi(x_1)\phi(x_2)\psi(y_1)\psi(y_2) \\
&= \phi(x_1)\psi(y_1)\phi(x_2)\psi(y_2) \\
&= [\phi \times \psi(x_1, y_1)][\phi \times \psi(x_2, y_2)]
\end{aligned}$$

□

Exercise 9.0.1.7. Let G, H, K be groups and $\phi \in \mathrm{Hom}(G \times H, K)$. Then there exist $\phi_G \in \mathrm{Hom}(G, K)$, $\phi_H \in \mathrm{Hom}(H, K)$ such that $\phi_G \times \phi_H = \phi$.

Proof. Suppose that K is abelian. Define $\iota_G \in \mathrm{Hom}(G, \ker \pi_H)$ and $\iota_H \in \mathrm{Hom}(H, \ker \pi_G)$ as in part (2) of Exercise 9.0.1.4 Define $\phi_G \in \mathrm{Hom}(G, K)$ and $\phi_H \in \mathrm{Hom}(H, K)$ by $\phi_G = \phi \circ \iota_G$ and $\phi_H = \phi \circ \iota_H$. Let $(x, y) \in G \times H$. Then

$$\begin{aligned}
\phi_G \times \phi_H(x, y) &= \phi_G(x)\phi_H(y) \\
&= \phi \circ \iota_G(x)\phi \circ \iota_H(y) \\
&= \phi(x, e_H)\phi(e_G, y) \\
&= \phi(x, y)
\end{aligned}$$

So $\phi = \phi_G \times \phi_H$ $\square$

9.1 Rings

Definition 9.1.0.1. Let R be a set and $+, * : R \times R \to R$ (we write $a+b$ and ab in place of $+(a,b)$ and $*(a,b)$ respectively). Then R is said to be a **ring** if for each $a, b, c \in R$,

1. R is an abelian group with respect to $+$. The identity element with respect to $+$ is denoted by 0.
2. R is a monoid with respect to $*$. The identity element of R with respect to $*$ is denoted 1.
3. R is commutative with respect to $*$.
4. $*$ distributes over $+$.

Definition 9.1.0.2. Let R be a ring and $I \subset R$. Then I is said to be an **ideal** of R if for each $a \in R$ and $x, y \in I$,

1. $x + y \in I$
2. $ax \in I$

Definition 9.1.0.3. Let R be a ring and $A, B \subset R$. We define the **product** of A and B, denoted AB, to be

$$AB = \left\{ \sum_{i=1}^{n} a_i b_i : a_i \in A, b_i \in B, n \in \mathbb{N} \right\}$$

Exercise 9.1.0.4. Let R be a ring and $I \subset R$. Then I is an ideal of R iff $RI \subset I$.

Proof. Suppose that $RI \subset I$. Let $a \in R$ and $x, y \in I$. Then by assumption $x + y = 1x + 1y \in I$ and $ax \in I$. So I is an ideal of R
Conversely, suppose that I is an ideal of R. Let $a_1, \cdots, a_n \in R$ and $x_1, \cdots, x_n \in I$. Then by assumption, for each $i = 1, \cdots, n$, $a_i x_i \in I$ and therefore $\sum_{i=1}^{n} a_i b_i \in I$. Hence $RI \subset I$. $\square$

9.2 Modules

9.2.1 Introduction

Definition 9.2.1.1. Let R be a ring, M a set, $+ : M \times M \to M$ and $* : R \times M \to M$ (we write rx in place of $*(r, x)$). Then M is said to be an R**-module** if

1. M is an abelian group with respect to $+$. The identity element of M with respect to $+$ is denoted by 0.
2. for each $r \in R$, $*(r, \cdot)$ is a group endomorphism of M
3. for each $x \in M$, $*(\cdot, x)$ is a group homomorphism from R to M
4. $*$ is a monoid action of R on M

Note 9.2.1.2. For the remainder of this section, we assume that R is a commutative ring.

Exercise 9.2.1.3. Let M be an R-module. Then for each $r \in R$ and $x \in M$,

1. $r0 = 0$
2. $0x = 0$
3. $(-1)x = -x$

Proof. Let $r \in R$ and $x \in M$. Then

1. $$\begin{aligned} r0 &= r(0+0) \\ &= r0 + r0 \end{aligned}$$

   which implies that $r0 = 0$.

2. $$\begin{aligned} 0x &= (0+0)x \\ &= 0x + 0x \end{aligned}$$

   which implies that $0x = 0$.

3. $$\begin{aligned} (-1)x + x &= (-1)x + 1x \\ &= (-1+1)x \\ &= 0x \\ &= 0 \end{aligned}$$

   which implies that $(-1)x = -x$.

□

Definition 9.2.1.4. Let M an R-module and $N \subset M$. Then N is said to be a **submodule** of M if for each $r \in R$ and $x, y \in N$, we have that $rx \in N$ and $x + y \in N$.

Definition 9.2.1.5. Let M be an R-module. We define $\mathcal{S}(M) = \{N \subset M : N \text{ is a submodule of } M\}$.

Exercise 9.2.1.6. Let M be an R-module and $N \in \mathcal{S}(M)$. Then N is a subgroup of M.

Proof. Let $x, y \in M$. Then $x - y = 1x + (-1)y \in N$. So N is a subgroup of M. □

Definition 9.2.1.7. Let M be an R-module and $N \in \mathcal{S}(M)$. We define

1. $M/N = \{x + N : x \in M\}$
2. $+ : M/N \times M/N \to M/N$ by
$$(x + N) + (y + N) = (x + y) + N$$
3. $* : R \times M/N \to M/N$ by
$$r(x + N) = (rx) + N$$

Under these operations (see next exercise), M/N is an R-module known as the **quotient module** of M by N.

Exercise 9.2.1.8. Let M be an R-module and $N \in \mathcal{S}(M)$. Then

1. the monoid action defined above is well defined
2. the quotient M/N is an R-module

Proof.

1. Let $r \in R$ and $x + N, y + N \in M/N$. Recall from group theory that $x + N = y + N$ iff $x - y \in N$. Suppose that $x + N = y + N$. Then $x - y \in N$ and there exists $n \in N$ such that $x - y = n$. Therefore
$$\begin{aligned} rx - ry &= r(x - y) \\ &= rn \\ &\in N \end{aligned}$$
So $rx + N = ry + N$.
2. Properties (1) - (4) in the definition of a module are easily shown to be satisfied for M/N since they are true for M.

□

Definition 9.2.1.9. Let M and N be R-modules and $\phi : M \to N$. Then ϕ is said to be a **module homomorphism** if for each $r \in R$ and $x, y \in M$

1. $\phi(rx) = r\phi(x)$
2. $\phi(x + y) = \phi(x) + \phi(y)$

Exercise 9.2.1.10. Let M and N be R-modules and $\phi : M \to N$. Then ϕ is a iff for each $r \in R$ and $x, y \in M$, $\phi(x + ry) = \phi(x) + r\phi(y)$.

Proof. Clear. □

Exercise 9.2.1.11. Let M and N be R-modules and $\phi : M \to N$ a homomorphism. Then

1. $\ker \phi$ is a submodule of M
2. $\operatorname{Im} \phi$ is a submodule of N

Proof. Let $r \in R$, $x, y \in \ker \phi$ and $w, z \in \operatorname{Im} \phi$. Then

1. $$\begin{aligned} \phi(rx) &= r\phi(x) \\ &= r0 \\ &= 0 \end{aligned}$$
So $rx \in \ker \phi$. Group theory tells us that $\ker \phi$ is a subgroup of M, so $x + y \in \ker \phi$. Hence $\ker \phi$ is a submodule of M.
2. Similar.

□

Definition 9.2.1.12. Let M be an R-module and $A \subset M$. We define the **submodule of M generated by** A, denoted $\operatorname{span}(A)$, to be

$$\operatorname{span}(A) = \bigcap_{N \in \mathcal{S}(M)} N$$

Exercise 9.2.1.13. Let M be an R-module and $A \subset M$. Then $\operatorname{span}(A) \in \mathcal{S}(M)$

Proof. Let $r \in R$ and $x, y \in \operatorname{span}(A)$. Basic group theory tells us that $\operatorname{span}(A)$ is a subgroup of M. So $x + y \in \operatorname{span}(A)$. For $N \in \mathcal{S}(M)$, by definition we have $x \in N$ and therefore $rx \in N$. So $rx \in \operatorname{span}(A)$. Hence $\operatorname{span}(A)$ is a submodule of M. □

Exercise 9.2.1.14. Let M be an R-module and $A \subset M$. If $A \neq \varnothing$, then

$$\operatorname{span}(A) = \left\{ \sum_{i=1}^{n} r_i a_i : r_i \in R, a_i \in A, n \in \mathbb{N} \right\}$$

Proof. Clearly □

Definition 9.2.1.15. Let M

9.3 Fields

9.4 Vector Spaces

9.5 Appendix

9.5.1 Monoids

Definition 9.5.1.1. Let G be a set and $* : G \times G \to G$ (we write ab in place of $*(a, b)$). Then

1. $*$ is called a **binary operation** on G
2. $*$ is said to be **associative** if for each $x, y, z \in G$, $(xy)z = x(yz)$
3. $*$ is said to be **commutative** if for each $x, y \in G$, $xy = yx$

Definition 9.5.1.2. Let G be a set, $* : G \times G \to G$, $e, x, y \in G$. Then e is said to be an **identity element** if for each $x \in G$, $ex = xe = x$.

Definition 9.5.1.3. Let G be a set and $* : G \times G \to G$. Then G is said to be a **monoid** if

1. $*$ is associative
2. there exits $e \in G$ such that e is an identity element.

Exercise 9.5.1.4. Let G be a monoid. Then the identity element is unique.

Proof. Let $e, f \in G$. Suppose that e and f are identity elements. Then $e = ef = f$. □

Note 9.5.1.5. Unless otherwise specified, we will denote the identity element of a monoid by e.

Definition 9.5.1.6. Let G be a monoid, X a set and $* : G \times X \to X$ (we write gx in place of $*(g, x)$). Then $*$ is said to be a **monoid action** of G on X if for each $g, h \in G$ and $x \in X$,

1. $(gh)x = g(hx)$
2. $ex = x$

Appendix A

Summation

Definition A.0.0.1. Let $f : X \to [0, \infty)$, Then we define

$$\sum_{x \in X} f(x) := \sup_{\substack{F \subset X \\ F \text{ finite}}} \sum_{x \in F} f(x)$$

This definition coincides with the usual notion of summation when X is countable. For $f : X \to \mathbb{C}$, we can write $f = g + ih$ where $g, h : X \to \mathbb{R}$. If

$$\sum_{x \in X} |f(x)| < \infty,$$

then the same is true for g^+, g^-, h^+, h^-. In this case, we may define

$$\sum_{x \in X} f(x)$$

in the obvious way.

The following note justifies the notation $\sum_{x \in X} f(x)$ where $f : X \to \mathbb{C}$.

Note A.0.0.2. Let $f : X \to \mathbb{C}$ and $\alpha : X \to X$ a bijection. If $\sum_{x \in X} |f(x)| < \infty$, then $\sum_{x \in X} f(\alpha(x)) = \sum_{x \in X} f(x)$.

Appendix B

Asymptotic Notation

Definition B.0.0.1. Let X be a topological space, Y, Z be normed vector spaces, $f : X \to Y$, $g : X \to Z$ and $x_0 \in X \cup \{\infty\}$. Then we write

$$f = o(g) \qquad \text{as } x \to x_0$$

if for each $\epsilon > 0$, there exists $U \in \mathcal{N}(x_0)$ such that for each $x \in U$,

$$\|f(x)\| \leq \epsilon \|g(x)\|$$

Exercise B.0.0.2. Let X be a topological space, Y, Z be normed vector spaces, $f : X \to Y$, $g : X \to Z$ and $x_0 \in X \cup \{\infty\}$. If there exists $U \in \mathcal{N}(x_0)$ such that for each $x \in U \setminus \{x_0\}$, $g(x) > 0$, then

$$f = o(g) \text{ as } x \to x_0 \quad \text{iff} \quad \lim_{x \to x_0} \frac{\|f(x)\|}{\|g(x)\|} = 0$$

Exercise B.0.0.3. Let X and Y a be normed vector spaces, $A \subset X$ open and $f : A \to Y$. Suppose that $0 \in A$. If $f(h) = o(\|h\|)$ as $h \to 0$, then for each $h \in X$, $f(th) = o(|t|)$ as $t \to 0$.

Proof. Suppose that $f(h) = o(\|h\|)$ as $h \to 0$. Let $h \in X$ and $\epsilon > 0$. Choose $\delta' > 0$ such that for each $h' \in B(0, \delta')$, $h' \in A$ and

$$\|f(h')\| \leq \frac{\epsilon}{\|h\| + 1} \|h'\|$$

Choose $\delta > 0$ such that for each $t \in B(0, \delta)$, $th \in B(0, \delta')$. Let $t \in B(0, \delta)$. Then

$$\begin{aligned} \|f(th)\| &\leq \frac{\epsilon}{\|h\| + 1} |t| \|h\| \\ &< \epsilon |t| \end{aligned}$$

So $f(th) = o(|t|)$ as $t \to 0$. $\square$

Definition B.0.0.4. Let X be a topological space, Y, Z be normed vector spaces, $f : X \to Y$, $g : X \to Z$ and $x_0 \in X \cup \{\infty\}$. Then we write

$$f = O(g) \qquad \text{as } x \to x_0$$

if there exists $U \in \mathcal{N}(x_0)$ and $M \geq 0$ such that for each $x \in U$,

$$\|f(x)\| \leq M \|g(x)\|$$

Appendix C

Categories

move to notation?

Definition C.0.0.1. We define the category of topological measure spaces, denoted $\mathbf{TopMsr}_+$, by

- $\mathrm{Obj}(\mathbf{TopMsr}_+) := \{(X, \mu) : X \in \mathrm{Obj}(\mathbf{Top}) \text{ and } \mu \in M(X)\}$
- $\mathrm{Hom}_{\mathbf{TopMsr}_+}((X, \mu), (Y, \nu)) := \mathrm{Hom}_{\mathbf{Top}}(X, Y) \cap \mathrm{Hom}_{\mathbf{Msr}_+}((X, \mathcal{B}(X), \mu), (Y, \mathcal{B}(Y), \nu))$

Appendix D

Vector Spaces

it might be better to cover some category theory and write everything in terms of $\mathrm{Hom}_{\mathbf{Vect}_{\mathbb{K}}}$ and $\mathrm{Obj}(\mathbf{Vect}_{\mathbb{K}})$

D.1 Introduction

Definition D.1.0.1. Let X be a set, $\mathbb{K}$ a field, $+ : X \times X \to X$ and $\cdot : \mathbb{K} \times X \to X$. Then $(X, +, \cdot)$ is said to be a **$\mathbb{K}$-vector space** if

1. $(X, +)$ is an abelian group
2.

Definition D.1.0.2. Let $(X, +_X, \cdot_X)$ and $(E, +_E, \cdot_E)$ be vector spaces. Suppose that $E \subset X$. Then $(E, +_E, \cdot_E)$ is said to be a subspace of X if

1. $+_E = +_X|_{E \times E}$
2. $\cdot_E = \cdot_X|_{\mathbb{K} \times E}$

Exercise D.1.0.3. Let $(X, +_X, \cdot_X)$ and $(E, +_E, \cdot_E)$ be vector spaces. Suppose that $E \subset X$.

Exercise D.1.0.4. Let $(X, +, \cdot)$ be a vector space and $E \subset X$. Then E is a subspace of X

Definition D.1.0.5. Let X be a vector space and $(E_j)_{j \in J}$ a collection of subspaces of X. Then $\bigcap_{j \in J} E_j$ is a subspace of X.

Proof. Set $E := \bigcap_{j \in J} E_j$. Let $x, y \in E$ and $\lambda \in \mathbb{K}$. Then for each $j \in J$, $x, y \in E_j$. Since for each $j \in J$, E_j is a subspace of X, we have that for each $j \in J$, $x + \lambda y \in E_j$. Thus $x + \lambda y \in E$. Since $x, y \in E$ and $\lambda \in \mathbb{K}$ are arbitrary, (cite exercise here) we have that E is a subspace of X. □

Definition D.1.0.6. Let X, Y be vector spaces and $T : X \to Y$. Then T is said to be **linear** if for each $x_1, x_2 \in X$ and $\lambda \in \Lambda$,

1. $T(x_1 + x_2) = T(x_1) + T(x_2)$,
2. $T(\lambda x_1) = \lambda T(x_1)$.

We define $L(X; Y) := \{T : X \to Y : \ T \text{ is linear}\}$.

Exercise D.1.0.7. Let X, Y be vector spaces and $T : X \to Y$. Then T is linear iff for each $x_1, x_2 \in X$ and $\lambda \in \Lambda$,

$$T(x_1 + \lambda x_2) = T(x_1) + \lambda T(x_2)$$

Proof. Clear. (add details) □

Definition D.1.0.8. define addition/scalar multiplication of linear maps

Exercise D.1.0.9. Let X, Y be vector spaces. Then $L(X;Y)$ is a $\mathbb{K}$-vector space.

Proof. Clear □

Definition D.1.0.10. Let X be a vector space over $\mathbb{K}$ and $T : X \to \mathbb{K}$. Then T is said to be a **linear functional on** X if T is linear. We define the **dual space of** X, denoted X^*, by $X^* := \{T : X \to \mathbb{K} : T \text{ is linear}\}$.

Exercise D.1.0.11. Let X be a vector space. Then X^* is a vector space.

Proof. Clear. □

D.2 Bases

Definition D.2.0.1. Let X be a vector space and $(e_\alpha)_{\alpha \in A} \subset X$. Then $(e_\alpha)_{\alpha \in A}$ is said to be

- **linearly independent** if for each $(\alpha_j)_{j=1}^n \subset A$, $(\lambda_j)_{j=1}^n \subset \mathbb{K}$, $\sum_{j=1}^n \lambda_j e_{\alpha_j} = 0$ implies that for each $j \in [n]$, $\lambda_j = 0$.
- a **Hamel basis for** X if $(e_\alpha)_{\alpha \in A}$ is linearly independent and $\operatorname{span}(e_\alpha)_{\alpha \in A} = X$.

Exercise D.2.0.2. every vector space has a Hamel basis

Proof. □

Exercise D.2.0.3.

Exercise D.2.0.4. Let X be a $\mathbb{K}$-vector space and $x \in X$. Then $x = 0$ iff for each $\phi \in X^*$, $\phi(x) = 0$.

Proof.

- $(\implies)$:
  Suppose that $x = 0$. Linearity implies that for each $\phi \in X^*$ $\phi(x) = 0$.
- $(\impliedby)$:
  Conversely, suppose that $x \neq 0$. Define $\epsilon_x : \operatorname{span}(x) \to \mathbb{K}$ by $\epsilon_x(\lambda x) := \lambda$. Let $u, v \in \operatorname{span}(x)$. Then there exists $\lambda_u, \lambda_v \in \mathbb{K}$ such that $u = \lambda_u x$ and $v = \lambda_v x$. Suppose that $u = v$. Then

  $$\begin{aligned}(\lambda_u - \lambda_v)x &= \lambda_u x - \lambda_v x \\ &= u - v \\ &= 0\end{aligned}$$

  Since $x \neq 0$, we have that $\lambda_u - \lambda_v = 0$ and therefore $\lambda_u = \lambda_v$. Hence

  $$\begin{aligned}\lambda_u &= \epsilon_x(u) \\ &= \epsilon_x(v) \\ &= \lambda_v.\end{aligned}$$

  Thus ϵ_x is well defined.

□

D.3 Multilinear Maps

Definition D.3.0.1. Let $X_1, \cdots, X_n, Y$ be vector spaces and $T : \prod_{j=1}^{n} X_j \to \mathbb{K}$. Then T is said to be **multilinear** if for each $j_0 \in [n]$ and $(x_j)_{j=1}^n \in \prod_{j=1}^{n} X_j$, $T(x_1, \ldots, x_{j_0-1}, \cdot, x_{j_0+1})$ is linear.

$$L^n(X_1, \ldots, X_n; Y) = \left\{ T : \prod_{j=1}^{n} X_j \to Y : T \text{ is multilinear} \right\}$$

If $X_1 = \cdots = X_n = X$, we write $L^n(X; Y)$ in place of $L^n(X, \ldots, X; Y)$.

Definition D.3.0.2. define addition and scalar mult of multilinear maps

Exercise D.3.0.3. Let $X_1, \cdots, X_n, Y$ be vector spaces. Then $L^n(X_1, \ldots, X_n; Y)$ is a $\mathbb{K}$-vector space.

Proof. content... □

Exercise D.3.0.4. Let $X_1, \cdots, X_n, Y, Z$ be $\mathbb{K}$-vector spaces, $\alpha \in L^n(X_1, \ldots, X_n; Y)$ and $\phi \in L^1(Y; Z)$. Then $\phi \circ \alpha \in L^n(X_1, \ldots, X_n; Z)$.

Proof. Let $(x_j)_{j=1}^n \in \prod_{j=1}^{n} X_j$ and $j_0 \in [n]$. Define $f : X_{j_0} \to Y$ by

$$f(a) := \alpha(x_1, \ldots, x_{j_0-1}, a, x_{j_0+1}, \ldots, x_n)$$

Since $\alpha \in L^n(X_1, \ldots, X_n; Y)$, f is linear. Since ϕ is linear, and $\phi \circ f$ is linear. Since $(x_j)_{j=1}^n \in \prod_{j=1}^{n} X_j$ and $j_0 \in [n]$ are arbitrary, we have that $\phi \circ \alpha \in L^n(X_1, \ldots, X_n; Y)$. □

D.4 Tensor Products

Definition D.4.0.1. Let X, Y and T be vector spaces over $\mathbb{K}$ and $\alpha \in L^2(X, Y; T)$. Then (T, α) is said to be a **tensor product of** X **and** Y if for each vector space Z and $\beta \in L^2(X, Y; Z)$, there exists a unique $\phi \in L^1(T; Z)$ such that $\phi \circ \alpha = \beta$, i.e. the following diagram commutes:

$$\begin{tikzcd} X \times Y \arrow[r, "\alpha"] \arrow[dr, "\beta"'] & T \arrow[d, dashed, "\phi"] \\ & Z \end{tikzcd}$$

Exercise D.4.0.2. Let X, Y, S, T be vector spaces, $\alpha \in L^2(X, Y; S)$ and $\beta \in L^2(X, Y; T)$. Suppose that (S, α) and (T, β) are tensor products of X and Y. Then S and T are isomorphic.

Proof. Since (T, β) is a tensor product of X and Y, $\beta \in L^2(X, Y; T)$ there exists a unique $f \in L^1(T; T)$ such that $f\ circ\beta = \beta$, i.e. the following diagram commutes:

$$\begin{tikzcd} & T \arrow[dd, dashed, "f"] \\ X \times Y \arrow[ur, "\beta"] \arrow[dr, "\beta"'] & \\ & T \end{tikzcd}$$

Since $\mathrm{id}_T \in L^1(T; T)$ and $\mathrm{id}_T \circ \beta = \beta$, we have that $f = \mathrm{id}_T$. Since (S, α) is a tensor product of X and Y, there exists a unique $\phi : S \to T$ such that $\phi \circ \alpha = \beta$, i.e. the following diagram commutes:

$$\begin{tikzcd} X \times Y \arrow[r, "\alpha"] \arrow[dr, "\beta"'] & S \arrow[d, dashed, "\phi"] \\ & T \end{tikzcd}$$

Similarly, since (T, β) is a tensor product of X and Y, there exists a unique $\psi : T \to S$ such that $\psi \circ \beta = \alpha$, i.e. the following diagram commutes:

$$\begin{tikzcd} X \times Y \arrow[r, "\beta"] \arrow[dr, "\alpha"'] & T \arrow[d, dashed, "\psi"] \\ & S \end{tikzcd}$$

Therefore

$$\begin{aligned} (\phi \circ \psi) \circ \beta &= \phi \circ (\psi \circ \beta) \\ &= \phi \circ \alpha \\ &= \beta, \end{aligned}$$

i.e. the following diagram commutes:

$$\begin{tikzcd} & T \arrow[d, dashed, "\psi"] & \\ X \times Y \arrow[ur, "\beta"] \arrow[r, "\alpha"] \arrow[dr, "\beta"'] & S \arrow[d, dashed, "\phi"] & \\ & T & \end{tikzcd} \implies \begin{tikzcd} & T \arrow[dd, dashed, "\phi\circ\psi"] \\ X \times Y \arrow[ur, "\beta"] \arrow[dr, "\beta"'] & \\ & T \end{tikzcd}$$

By uniqueness of $f \in L^1(T; T)$, we have that

$$\begin{aligned} \mathrm{id}_T &= f \\ &= \phi \circ \psi \end{aligned}$$

A similar argument implies that $\psi \circ \phi = \mathrm{id}_S$. Hence ϕ and ψ are isomorphisms with $\phi^{-1} = \psi$. Hence S and T are isomorphic. $\square$

Definition D.4.0.3. Let X, Y be vector spaces, $x \in X$ and $y \in Y$. We define $x \otimes y : X^* \times Y^* \to \mathbb{K}$ by $x \otimes y(\phi, \psi) := \phi(x)\psi(y)$.

Exercise D.4.0.4. Let X, Y be vector spaces, $x \in X$ and $y \in Y$. Then $x \otimes y \in L^2(X^*, Y^*; \mathbb{K})$.

Proof. Let $\phi_1, \phi_2 \in X^*$, $\psi \in Y^*$ and $\lambda \in \mathbb{K}$. Then

$$\begin{aligned} x \otimes y(\phi_1 + \lambda\phi_2, \psi) &= [\phi_1 + \lambda\phi_2(x)]\psi(y) \\ &= \phi_1(x)\psi(y) + \lambda\phi_2(x)\psi(y) \\ &= x \otimes y(\phi_1, \psi) + \lambda x \otimes y(\phi_2, \psi) \end{aligned}$$

Since $\phi_1, \phi_2 \in X^*$, $\psi \in Y^*$ and $\lambda \in \mathbb{K}$ are arbitrary, we have that for each $\psi \in Y^*$, $x \otimes y(\cdot, \psi)$ is linear. Similarly for each $\phi \in X^*$, $x \otimes y(\phi, \cdot)$ is linear. Hence $x \otimes y$ is bilinear and $x \otimes y \in L^2(X^*, Y^*; \mathbb{K})$. □

Definition D.4.0.5. Let X, Y be vector spaces. We define

- the **tensor product of X and Y**, denoted $X \otimes Y \subset L^2(X^*, Y^*; \mathbb{K})$, by
$$X \otimes Y := \operatorname{span}(x \otimes y : x \in X \text{ and } y \in Y),$$
- the **tensor map**, denoted $\otimes : X \times Y \to X \otimes Y$, by $\otimes(x, y) := x \otimes y$.

Exercise D.4.0.6. Let X, Y be vector spaces, $(x_j)_{j=1}^n \subset X$ and $(y_j)_{j=1}^n \subset Y$. The following are equivalent:

1. $\sum_{j=1}^n x_j \otimes y_j = 0$
2. for each $\phi \in X^*$ and $\psi \in Y^*$, $\sum_{j=1}^n \phi(x_j)\psi(y_j) = 0$
3. for each $\phi \in X^*$, $\sum_{j=1}^n \phi(x_j)y_j = 0$
4. for each $\psi \in Y^*$, $\sum_{j=1}^n \psi(y_j)x_j = 0$

Proof.

1. $(1) \implies (2)$:
   Suppose that $\sum_{j=1}^n x_j \otimes y_j = 0$. Let $\phi \in X^*$ and $\psi \in Y^*$. Then
$$\begin{aligned} \sum_{j=1}^n \phi(x_j)\psi(y_j) &= \phi\left(\sum_{j=1}^n \psi(y_j)x_j\right) \\ &= \end{aligned}$$
2.
3.

□

Exercise D.4.0.7. Let X, Y be vector spaces. Then $(X \otimes Y, \otimes)$ is a tensor product of X and Y.

Proof. Let Z be a vector space and $\alpha \in L^2(X, Y; Z)$. Define $\phi : X \otimes Y \to Z$ by $\phi\left(\sum_{j=1}^n \lambda_j x_j \otimes y_j\right) := \sum_{j=1}^n \lambda_j \alpha(x_j, y_j)$.

- **(well defined):**
  Let $u \in X \otimes Y$. Then there exist $(\lambda_j)_{j=1}^n \subset \mathbb{K}$, $(x_j)_{j=1}^n \subset X$, $(y_j)_{j=1}^n \subset Y$ such that $u = \sum_{j=1}^n \lambda_j x_j \otimes y_j$. Suppose that $u = 0$. Let $\phi \in Z^*$. Then $\phi \circ \alpha \in L^2(X, Y; Z)$.

□

Bibliography

[1] Introduction to Algebra

[2] Introduction to Analysis

[3] Introduction to Fourier Analysis

[4] Introduction to Measure and Integration